

GINKGO BIOWORKS (NYSE: DNA)

A Snake Oil Salesman And Some Hedge Funds Partner Up To Pimp The Latest “Synthetic Biology” Scam -- As Phantom Revenue, A Hocus-Pocus Business Model, Rampant Related-Party Games, And A Decade of Colossal Failure Get Shoveled Into Yet Another Garbage SPAC.

\$23B market cap | \$11.98/share | ADV 8.8MM shares | Short interest 22mm shares *10/5 per Capital IQ/Markit

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Part 1: Ginkgo Bioworks is a colossal scam, a Frankenstein mash-up of the worst frauds of the last 20 years. At \$23B market cap, it is rare to see a related-party scheme on Ginkgo's scale in the US markets – it is, quite simply, the US version of the “China Hustle.”

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1. <u>Introduction to the real Ginkgo Bioworks.</u> We conducted an intensive investigation into Ginkgo's business model and practices, with a particular focus on the related-party entities that drive the bulk of its revenue. We completed 21 research interviews, encompassing a broad sample of former employees and executives of Ginkgo, as well as individuals who are currently employed at its related-party “customers.” Our research leads us to conclude that Ginkgo is a house of cards – in our opinion, one of the most brazen frauds of the last 20 years.	7-34
2. <u>Ginkgo's business model is based on a dubious shell game.</u> The majority of its foundry revenue, an absurd 72% in 2020, and essentially 100% of its deferred revenue are derived from related-party “customers” it created, funded, controls, or influences via its ownership position and board seats. Investments into these entities by Ginkgo and its largest investors are recycled back to Ginkgo and recorded as deferred or current revenue. The scheme reflects its woeful, decade-long failure to derive real revenue from third-party customers, forcing it to cover it up with a ploy that we believe to be enabled by its largest holders.	35-46
3. <u>We believe that Ginkgo is concealing the true extent of its dependence on related party revenue</u> and that it is far greater than it reports. We have uncovered a smoking gun that indicates that essentially ALL of its foundry revenue is derived from related parties, suggesting that Ginkgo has engaged in a brazen effort to misclassify and misreport related party revenue and deceive investors with phony accounting. Our interviews with ex-employees indicated fear inside Ginkgo that its related-party model would fail regulatory scrutiny and derail an IPO, which we believe has prompted a cover-up of its magnitude.	47-53
4. <u>We believe that not only is ~100% of reported foundry revenue from related-parties, but that the actual truth is far more nefarious and suggests that it is a hoax for the ages.</u> Ginkgo is opaque about the percent of its revenue and deferred revenue which is cash versus non-cash. Based on interviews with its related-party “customers,” <u>we believe that at least half of Ginkgo's reported foundry revenue is phantom – that is, non-cash and pure accounting hocus-pocus.</u> We spoke with one of its largest such customers, from whom Ginkgo reported \$38MM and \$30MM of deferred revenue in 2019 and 2020. A senior employee there stated unequivocally that they have never paid Ginkgo cash for foundry services and are merely using “free” R&D credits following investments by Ginkgo and Viking. Giving away the essence of the scam, the customer stated they would not use Ginkgo if they had to pay cash, dismissing them as a failed contract research organization (CRO) that they neither trust to deliver nor could justify except under a round-tripping deal.	54-67

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Part 1: Ginkgo is a colossal scam (cont'd)

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5. <u>Former employees we interviewed indicated a widespread awareness of and concern inside Ginkgo about its related-party business model</u> : worry inside the company that it may not be able to go public or could “be slammed by regulators”; staff indicated “confusion” at what Ginkgo counts as revenue and questioned if “money actually passed hands”; and suggested that concern around the circular flows was widespread internally.	68-72
6. <u>We spoke with several former employees who indicated that they had been terminated for refusing to play along with the scheme, and who stated they knew of others pushed out for similar reasons</u> . The first worked at a key related-party “customer” and alleged he was fired because he resisted Ginkgo’s pressure to round-trip R&D spend from the entity, stating he was asked to do “some unethical things” that made him uncomfortable. A second ex-employee in a senior role at another related-party “customer” indicated he “got axed” for pushing back on Ginkgo’s attempt to exploit the entity for fake work it didn’t need: “there was real pressure to produce, to do work in their foundry that would have no real value”; Ginkgo had a “certain timeline” and “certain revenue expectations” from the “customer.” A third ex-employee was afraid to bring up concerns about circular revenue flows; suggested a colleague may have been fired for doing so; and indicated a personal awareness of three individuals who were allegedly terminated for pushing back.	73-76
7. <u>The related party “customers” are mostly fronts for Ginkgo and its revenue is therefore an elaborate fiction</u> . <u>The entities are generally small, failed projects that share the following features</u> : 1) based inside Ginkgo’s headquarters; 2) have vaporous, glossy, Instagram-like websites that appear to be operated by Ginkgo; 3) have too few employees on LinkedIn to drive the “revenue” that Ginkgo books from them, and in some cases appear to have no employees at all; 4) appear to be controlled and run by Ginkgo, as the entities typically lack finance, treasury, or HR employees, indicating they are neither designed to nor actually operate independently; and 5) appear to be further controlled via “G&A Services Agreements” that suggest Ginkgo runs them.	77-94
8. <u>Ex-Ginkgo employees as well as current employees of these related-party “customers” corroborated our findings that they are located at Ginkgo’s headquarters and are anything but arms-length vendor-customer relationships</u> : “overlap and intermingling”; “a close association”; “specifically propped up” by Ginkgo; murky situations where Ginkgo still paid staff even though they “worked” for the related-party “customer.” Employees at one key related-party described the day-to-operation as more like a subsidiary of Ginkgo or a captive team within the company.	95-103

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Part 1: Ginkgo is a colossal scam (cont'd)	Pages
9. In the last two months, timed with its SPAC listing, Ginkgo has announced a <u>flurry of new R&D partners, a dog whistle that the scam is about to hit overdrive. We believe these “partners” are undisclosed related parties and fronts.</u> Without proliferating new round-tripping arrangements, Ginkgo can’t show “growth” in Q3 and coming quarters. We present several smoking guns.	104-113
10. <u>Aside from its “customers” being related parties and effectively fake, we believe that Ginkgo takes the scam into even more aggressive territory by booking revenue from them that is simply fictitious, overstated and/or based on overcharging.</u> First, the revenue that Ginkgo books from these entities bears no plausible relationship to their size, suggesting they are fraudulent. Second, the timing of when Ginkgo books revenue and/or deferred revenue is equally implausible, sometimes before the entities even appear to have any employees or are functional, which further indicates they are simply sham transactions.	114-121
11. <u>Current employees at Ginkgo’s related-party “customers,” as well as former employees of Ginkgo, corroborated our findings that their model is based on overcharging captive entities:</u> foundry fees are “prohibitively expensive; “obligated” to pay whatever Ginkgo dictates. Employees at these entities provided specific examples of Ginkgo charging them rates 50-250% above market. An ex-employee suggested that overcharging and padding statements of work was part of Ginkgo’s culture: “We do it all the time. They’re never going to know”; “I just got the sense it happened all the time”; alleged personally witnessing an attempt to overcharge and defraud the US government on a DARPA project; “I saw them actually make it. I saw the spreadsheet.”	122-128
12. <u>Related-party “customers” appear to be staffed with what we believe to be loyal, longtime Ginkgo plants in critical positions, who we think rotate from entity to entity to ensure that the entities play ball by round-tripping revenue back to Ginkgo via foundry R&D spend.</u> An ex-employee who alleged being terminated for pushing back on improper related-party payments back to Ginkgo detailed steps allegedly taken by a Ginkgo co-founder to impose control over payment flows after staff attempted to act as fiduciaries on behalf of the related party entity.	129-132

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Part 2: Behind the synthetic biology and “Foundry” hype, Ginkgo is a glorified contract research organization that does commodity yeast strain engineering – with a decade of spectacular failure in delivering anything of value for customers.

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13. <u>No unique technology, intellectual property, or differentiated capabilities.</u> Ex-employees indicated the foundry is nothing special and simply a collection of automated liquid handling machines and fermenters from the same two suppliers used by the rest of the industry, including its closest competitor Zymergen. Ginkgo is simply a failed CRO that does commodity strain engineering, with yeast as its platform of choice, similar to everyone else and as has been standard practice for 50 years.	133-145
14. <u>Ginkgo’s related-party “customers” slammed Ginkgo’s foundry services.</u> We find it stunning that even “customers” who are captive and being funded by Ginkgo and its investors have little positive to say. We note these customers are intimately familiar with the foundry as they are co-located inside Ginkgo’s headquarters: “I can’t understand the basis for the hype”; “hard for me to point to something that they do that is totally different than anybody else in the industry.” “a kind of a glorified CRO; similar strain engineering services are widely available from other vendors.	146-153
15. <u>Long history of failure with little to nothing delivered in the way of successful strains and products.</u> Ginkgo promotes 70+ third party collaboration projects, but our research indicates that the historical failure rate has been 90-100%. It is self-evident why Ginkgo had to resort to an elaborate round-tripping scheme where it purchases “R&D Services” from itself.	154-161
16. <u>Each of Ginkgo’s key related-party collaborations has been a flop. We present five case studies: 1) Synlogic; 2) Genomatica; 3) Joyn Bio; 4) Motif; and 5) Allonnia.</u> Interviews with current employees and executives at these entities indicate Ginkgo’s failure to deliver anything of value. Synlogic had to turn to Zymergen after Ginkgo failed. Genomatica stated they would not use Ginkgo if they had to pay cash. A Joyn executive called Ginkgo a “glorified CRO.” Motif “did not have a good experience” and the “CEO of Motif is barely on speaking terms with the Ginkgo folks.” Allonnia is “extremely unhappy.” We note that Ginkgo and/or its investors funded these entities and/or sit on their boards – yet they still slam it.	162-170
17. <u>Ginkgo is virtually identical to its closest synthetic biology competitor Zymergen (ticker: ZY), which fell >70% in a single day two months ago. We believe a similar outcome is more likely than not at Ginkgo.</u> One ex-employee after another indicated that Zymergen and Ginkgo are the same animal, even using the same suppliers to equip their labs; that the Ginkgo CEO’s “biggest fear was always Zymergen”; and that after Zymergen’s implosion, Ginkgo “announced all these deals to try to be able to show that they’re different.”	171-175

Speculative fads on Wall Street often end in a climactic, defining moment of excess – a merger, IPO, bankruptcy, or fraud on a scale so massive that boundaries are pushed, speed limits get smashed, and the whole rig collapses of its own weight. We have no idea whether Ginkgo will be that signature moment for the SPAC era, but at \$23B market cap – conjured from thin air by hedge funds pushing paper around related-parties – we think it'll be a contender for the Hall of Shame, if not a 6am raid by the Feds and some perp walks. Ginkgo is, we believe, a colossal fraud, a Frankenstein mash-up of the worst scams of the last 20 years: Enron's abuse of related parties and special purpose entities; Intrexon, the father of synthetic biology scams, and its circular "revenue" charade; the vaporous grandiosity of WeWork and Theranos; and Ginkgo's Siamese twin Zymergen, which fell 75% in a single day in August.

Market capitalization

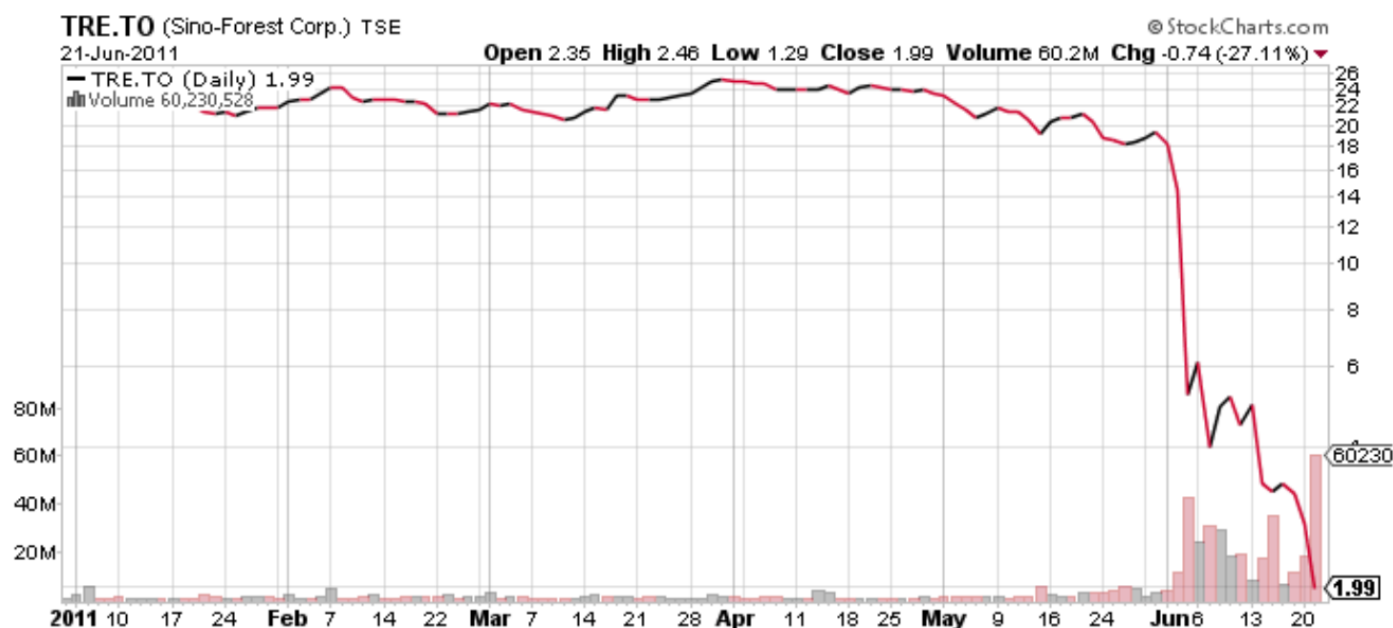
in millions USD	
Share Price, as of 10/5	11.98
Shares Outstanding	1,957
Market Capitalization	23,450
- Cash & Short Term Investments	236
+ Total Debt	16
+ Total Minority Interest	7
= Total Enterprise Value	23,237

Ginkgo stock chart



At \$23B market cap, it is rare to see a related-party racket on Ginkgo's scale in the US markets. Related parties are such an alarming red flag and self-evident mechanism for abuse that If most or all of your “revenue” comes “customers” you created, funded, co-locate in your offices, and who you control or influence, you are presumed to be a rig until proven otherwise. One has to turn to offshore, Wild-West jurisdictions to witness related-party games as flagrant as Ginkgo. **It is the US version of the “China Hustle,”** when Chinese frauds – generally of the related-party genre - listed in the US a decade ago, with a recent revival. **The defining moment of the China Hustle era was Sino-Forest,** which fell 82% the day it was exposed by Muddy Waters.

China Hustle trailer: <https://www.youtube.com/watch?v=55892jT06aI>



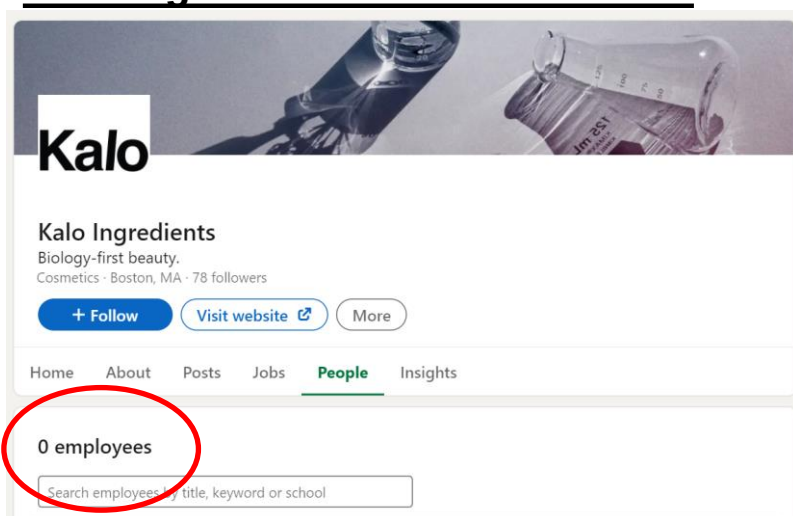
We investigated each of Ginkgo's related party "customers," and as a preview of what a hard-core China Hustle-type scam it is, we note Kalo Ingredients, which has no employees listed on LinkedIn, yet Ginkgo booked an \$11.9MM deferred revenue balance as of 3/31/2021. We find this particularly brazen, as Kalo was only formed 2 weeks earlier as an entity, according to its Massachusetts corporate registry filing, which further indicates that the entity's "Principal Offices" are Ginkgo's headquarters, and its 3 'Managers' are Ginkgo's CEO and two Ginkgo cronies. Kalo's site is a front, and it appears to be hosted by Ginkgo.

SPAC Prospectus Aug 13, 2021

Balances as of March 31, 2021

	Joyn	Motif	Genomatica	Allonnia	Synlogic	Kalo	Total
Accounts receivable, net	\$ —	\$ 3,391	\$ 1,500	\$ 2,167	\$ —	\$ —	\$ 7,058
Deferred revenue, current and non-current	\$8,281	\$53,299	\$ 26,680	\$37,959	\$ 66	\$11,897	\$138,182

Kalo Ingredients LinkedIn search



MA corporate registry filing

Corporations Division
Business Entity Summary
 ID Number: 001501343
 Summary for: KALO INGREDIENTS, LLC
 The exact name of the Foreign Limited Liability Company (LLC): KALO INGREDIENTS, LLC
 Entity type: Foreign Limited Liability Company (LLC)
 Identification Number: 001501343
 Date of Registration in Massachusetts: 04-14-2021
 Last date certain:
 Organized under the laws of: State: DE Country: USA on: 03-15-2021
 The location of the Principal Office:
 Address: C/O GINKGO BIOWORKS, INC. 27 DRYDOCK AVE., FLOOR 8
 City or town, State, Zip code, Country: BOSTON, MA 02210 USA
 The location of the Massachusetts office, if any:
 Address: C/O GINKGO BIOWORKS, INC. 27 DRYDOCK AVE., FLOOR 8
 City or town, State, Zip code, Country: BOSTON, MA 02210 USA
 The name and address of the Resident Agent:
 Name: GINKGO BIOWORKS, INC.
 Address: 27 DRYDOCK AVE., FLOOR 8
 City or town, State, Zip code, Country: BOSTON, MA 02210 USA
 The name and business address of each Manager:

Title	Individual name	Address
MANAGER	JASMINA AGANOVIC	C/O GINKGO BIOWORKS, INC. 27 DRYDOCK AVE., FLOOR 8 BOSTON, MA 02210 USA
MANAGER	JASON KELLY	C/O GINKGO BIOWORKS, INC. 27 DRYDOCK AVE., FLOOR 8 BOSTON, MA 02210 USA
MANAGER	JASON KAKOYIANIS	C/O GINKGO BIOWORKS, INC. 27 DRYDOCK AVE., FLOOR 8 BOSTON, MA 02210 USA

Organized as a Delaware entity on 3/15/2021

Principal Office listed as Ginkgo's corporate HQ: C/O Ginkgo BIOWORKS INC. 27 DRYDOCK AVE, FLOOR 8 BOSTON, MA 02201

"Managers" listed are CEO Jason Kelly, Jason Kakoyiannis, and Jasmina Aganovic

We dispatched private investigators to Ginkgo's headquarters, where they took photographs. Kalo isn't even listed in the tenant directory. Related-party "customers" Joyn Bio and Allonnia are, but they share the same phone number as Ginkgo. Motif, another "customer," has its own number, but is still in the same facility. As we discuss in a later section, we think that related party "customers" like Kalo Ingredients are fronts for Ginkgo and its revenue is therefore an elaborate fiction.

Tenant Contact Directory	
JLL Management Office :	617-428-0003
Northeast Security :	857-337-5200
Allonnia:	877-422-5362
Biobuilder:	Please call your contact person
Dana Farber Security Office:	617-632-3131
Emulate:	Please call your contact person
Ginkgo Bioworks / Joyn Bio LLC :	877-422-5362
Invicro :	617-904-2100
Motif Foodworks :	617-610-2549
Seaport Diagnostics :	857-233-7172

Allonnia shares the same phone number with Ginkgo and Joyn Bio, another related-party "customer"

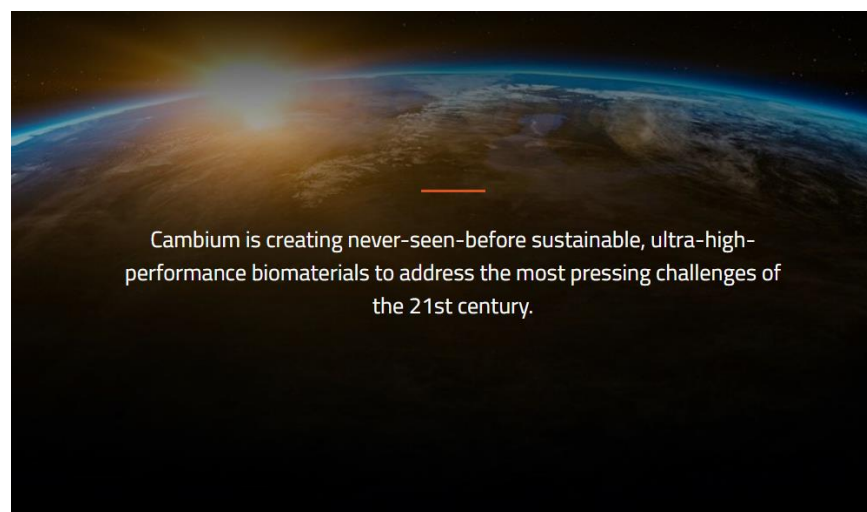
Motif Foodworks is in the same facility

Kalo, as we shall detail, is only the tip of the iceberg. In the last two months, timed with its SPAC listing, Ginkgo has announced a flurry of new R&D partners, a dog whistle that the scam is about to hit overdrive – without proliferating new round-tripping arrangements, it can't show “growth” in Q3 and coming quarters. The press releases are silent as to whether Ginkgo or its investors are backing these “customers” – that is, whether they are undisclosed related parties and fronts. An ex-employee who was alarmed told us they are, and we were stunned by what we found. As a preview, we note Cambium Biomaterials. It lists 3 employees on LinkedIn, only one of whom appears to be full-time while another is a documentary film-maker. The address is a co-working space. The website is a single page of vaporware. We will share a smoking gun....

Ginkgo press release on Sep 2, 2021

Cambium Biomaterials Announces Partnership with Ginkgo Bioworks to Accelerate Industrial Adoption of High-Performance Biomaterials

Cambium site is a single page of vaporware



Ex-employees who alerted us to round-tripping deals with undisclosed related parties declined to share their names. However, we have deduced that Cambium is one, among others we expose later. On Sep 17, 2021, Cambium filed a Form D stating that it sold \$8MM of equity on Sep 2, 2021, as part of a \$28MM round – the same day as Ginkgo’s misleading press release. The filing didn’t name the investor, so we spoke to an individual closely involved with Cambium - who stated that Ginkgo invested \$5MM as part of an \$11MM financing in conjunction with an R&D services agreement. The individual described a long-standing personal relationship between principals at both companies.

Excerpts from Cambium’s SEC Form D filing

Notice of Exempt Offering of Securities

1. Issuer's Identity

CIK (Filer ID Number)
0001883516

Name of Issuer
Cambium Biomaterials, Inc.

Jurisdiction of Incorporation/Organization
DELAWARE

Year of Incorporation/Organization
☐ Over Five Years Ago
☒ Within Last Five Years (Specify Year) 2019
☐ Yet to Be Formed

Previous Names ☒ None

Entity Type
☒ Corporation
☐ Limited Partnership
☐ Limited Liability Company
☐ General Partnership
☐ Business Trust
☐ Other (Specify)

**Cambium Biomaterials Inc,
626 Bancroft Way Suite A,
Berkeley, CA 94710**

2. Principal Place of Business and Contact Information

Name of Issuer
Cambium Biomaterials, Inc.

Street Address 1
626 BANCROFT WAY, SUITE A

City
BERKELEY

State/Province/Country
CALIFORNIA

Street Address 2

ZIP/Postal Code
94710

Phone Number of Issuer
(415) 878-3752

7. Type of Filing

☒ New Notice **Date of First Sale 2021-09-02** ☐ First Sale Yet to Occur
☐ Amendment

8. Duration of Offering

Does the Issuer intend this offering to last more than one year? ☐ Yes ☒ No

9. Type(s) of Securities Offered (select all that apply)

☒ Equity
☐ Debt

13. Offering and Sales Amounts

Total Offering Amount \$28,014,379 USD or ☐ Indefinite
Total Amount Sold \$8,012,877 USD
Total Remaining to be Sold \$20,001,502 USD or ☐ Indefinite

Date of First Sale: 2021-09-02

\$28MM offering, \$8MM already sold

Given its preposterous valuation and multiple, the market appears utterly oblivious that Ginkgo is a house of cards where the rot runs deep. A Capital IQ search of all public companies in the world with >\$50MM LTM revenue yields ~322,000 in total, and Ginkgo is the single most expensive at 166X revenue. As we think the company's revenue is fictitious – accounting hocus-pocus where investments in related-parties are round-tripped back as “revenue” - the actual price to sales multiple is infinity.

Company Name	TEV/LTM Total Revenues
Ginkgo Bioworks Holdings Inc. (NYSE:DNA)	166
Snowflake Inc. (NYSE:SNOW)	98
Carnival Corporation & plc (NYSE:CCL)	81
Liberty Broadband Corporation (NasdaqGS:LBRD.K)	67
Cloudflare, Inc. (NYSE:NET)	67
Alpha Dhabi Holding PJSC (ADX:ALPHADHABI)	66
Datadog, Inc. (NasdaqGS:DDOG)	55
Zscaler, Inc. (NasdaqGS:ZS)	51
WuXi Biologics (Cayman) Inc. (SEHK:2269)	49
Tianqi Lithium Corporation (SZSE:002466)	49
Prosus N.V. (ENXTAM:PRX)	49
CrowdStrike Holdings, Inc. (NasdaqGS:CRWD)	47
Atlassian Corporation Plc (NasdaqGS:TEAM)	46
Samsung Biologics Co.,Ltd. (KOSE:A207940)	44
Afterpay Limited (ASX:APT)	42
MongoDB, Inc. (NasdaqGM:MDB)	41
Shopify Inc. (NYSE:SHOP)	41
Yunnan Energy New Material Co., Ltd. (SZSE:002812)	41

The largest holder in Sino-Forest was billionaire John Paulson, illustrating a key feature of the largest frauds: a famous, respected investor - the “smart money” - whose presence lends credibility to the scam. Without the legitimacy they provide, a fraud can’t scale in the first place. JP Morgan stood behind Madoff; Bill Ackman bet the farm on Valeant; Theranos had Walgreens; Nikola had GM. The largest holder in Ginkgo - the “smart money” - is hedge fund Viking Global. Viking is a “Tiger cub” – a term for funds started by alumni of Julian Robertson’s Tiger Management. Viking is run by Andreas Halvorsen and was the largest holder with a 24% stake prior to the offering, currently worth \$3.5B per Capital IQ, followed by a hedge fund hotel of similar players.

Latest Holders			
Holder	Common Stock Equivalent Held	% Of CSO	Market Value (USD in mm)
Viking Global Investors LP	302,942,974	15.5	3,487
Cascade Investment, L.L.C.	151,865,481	7.8	1,748
General Atlantic Service Company, L.P.	102,559,867	5.2	1,181
Baillie Gifford & Co.	89,497,288	4.6	1,030
Che, Austin (Founder, Head of Strategy & Treasurer)	83,020,866	4.2	956
Kelly, Jason (Co-Founder, CEO & Director)	82,581,106	4.2	951
Canton, Bartholomew (Founder, CTO & Secretary)	81,729,071	4.2	941
Shetty, Reshma P. (Co-Founder, President, COO & Director)	81,729,071	4.2	941
Senator Investment Group LP	80,153,273	4.1	923
Anchorage Capital Group, L.L.C.	74,929,312	3.8	862
Eagle Equity Partners, LLC	31,590,948	1.6	364
Novalis LifeSciences Investments LLC	5,780,364	0.3	67
The Adhira S. Canton Irrevocable Trust	1,291,794	0.1	15
The Asha S. Canton Irrevocable Trust	1,291,794	0.1	15
Dekkers Ph.D., Marijn E. (Chairman of the Board)	374,960	0.0	4
Dmytruk, Mark E. (Chief Financial Officer)	314,097	0.0	4
Henry M.B.A., Christian O. (Director)	234,548	0.0	3
Sankar, Shyam (Director)	234,548	0.0	3
Fallon, Marie E	58,894	0.0	1

We admire a number of funds that spun out of Tiger, but for those who think that having a ~\$35B AUM Tiger cub as the largest holder is automatic indicia of legitimacy, judgment, or perfunctory due diligence, we caution that one of the largest holders in Luckin Coffee, a Chinese fraud that imploded, was Lone Pine – a prominent ~\$30B Tiger cub, a fund we otherwise respect. We further note the case of Bill Hwang, who ran Tiger Asia and Archegos, and whose history with regulators and recent troubles are well-documented.

Luckin Coffee stock chart



Bill Hwang media coverage

"Tiger Cub" manager pleads guilty in insider trading case

SEC Opens Probe Into Archegos Trades That Triggered Rout

Bill Hwang Invested Millions in Other Tiger Cubs, Including

Unlike Paulson, Ackman, and other famous investors who fell for a fraud, Viking's role in Ginkgo is different, and we believe, more troubling. Halvorsen is not a stupid man. He has likely been shorting stocks for a quarter century, and he and his crackerjack long/short analysts can probably smell an old-fashioned related-party fraud a mile away. Yet, at one related-party "customer" of Ginkgo's after another that we investigated, we encountered Viking – playing a central role or lurking in the background. In a number of related-parties, such as the shell Kalo Ingredients, Ginkgo is coy and states merely that they were financed by "certain of our investors," which we find unusual as some of these appear to be ~\$80-120MM rounds – what is there to hide?

Representative examples of Viking's investments in Ginkgo's related-party "customers"

Genomatica announced \$90 equity raise – Viking is largest shareholder, Ginkgo investor Casdin Capital led new round
"Bioengineering export Genomatica has announced it has issued \$90 million in its latest equity offering, a move which is said to allow the company to 'fully execute on its business plan and fuel new growth opportunities', which include sustainable technologies. Contributors to the funding include Casdin Capital, the new lead investor, **Viking Global Investors, the company's largest shareholder**, and organism engineering partner Ginkgo Bioworks." Source: <https://www.globalcosmeticsnews.com/genomatica-closes-90-million-financing-to-grow-sustainable-technologies/>

Verb Biotics and Ayana Bio – both related-parties funded by Viking and Ginkgo investor Cascade
"Simultaneously with the arrangements between Ginkgo and each of Ayana Bio, LLC and Verb Biotics, LLC, **Cascade Investment L.L.C. and entities affiliated with Viking Global** Opportunities Illiquid Investments Sub-Master LP, each of which hold more than 5% of Ginkgo's outstanding capital stock, invested approximately \$7.5 million and \$22.5 million, respectively, in each company.." Source: Soaring Eagle Acquisition Corp prospectus 8/13/2021

Kalo Ingredients

"Kalo was capitalized through a Series A preferred unit financing that raised approximately \$77 million in gross proceeds from **an investor group which included certain of our investors** and industry strategic investors." Source: Soaring Eagle Acquisition Corp prospectus 8/13/2021

We are particularly troubled that Ginkgo's largest holders are making massive investments in related-party "customers" that are just shells, which appear to have no employees or which otherwise strain credibility as ones they would fund if they were not recycling "revenue" back. The game that Ginkgo's backers are playing is the same that Softbank tried with WeWork. Softbank reportedly offered to anchor WeWork's IPO – offering more money to get a mark-up on its reportedly previous investments of \$18.5B. We think Ginkgo's holders saw it for what it was – a failing company unable to attract real customers – and began to fund related-parties in what strikes us as an attempt to mark up their holdings. The math is genius and vaporous: invest \$1 in a related-party and recycle it as \$1 of "sales" – which gets marked up to \$166 given its price to sales multiple of 166X.

Ayana Bio and Verb Biotics are two shells Ginkgo created in Sept 2021 – our reading of the SPAC disclosures leads us to conclude that Viking and Cascade – two of Ginkgo's largest holders – recently invested \$60MM total into both shells. Both have half-page websites that look like a cut and paste and appear to be hosted by Ginkgo.

Verb Biotics and Ayana Bio websites are half a page

verb**bi**otics

We identify and design new strains of probiotic bacteria for nutrition, health and wellness.

Get More Information →

Name	Email Address
Submit	



AYANA**BIO**

Harnessing nature's bioactives for use as complementary medicine to support human health and wellness.

We are coming soon. Get More Information →

Name	Email Address
Submit	



WSJ article on Softbank/WeWork IPO

"Among the ideas considered was **SoftBank serving as an anchor investor in the IPO** by buying a significant portion of the roughly \$3 billion to \$4 billion the company is expected to raise."

A key mechanism by which Ginkgo's largest investors appear to prop up its related-party "customers" is the Ferment Fund, which doesn't even appear to be registered as a fund or file like one. Ginkgo announced it in 2019 as a \$350MM vehicle for Viking, General Atlantic, and Cascade to finance its "spinout companies." An ex-employee characterized it as a "slush fund," a sentiment we heard across other interviews. The "Fund" appears to have various identities, such as the "Ferment Consortium" or simply "company formation studio." The SPAC prospectus indicates \$90MM invested, of which \$58MM was entities affiliated with Viking.

Ginkgo press release Oct 2019

BOSTON, Oct. 10, 2019 /PRNewswire/ -- Today at [Ginkgo Bioworks'](#) second annual Ferment meeting, the organism company unveiled the Ferment Consortium, a private investment vehicle for funding Ginkgo spinout companies with transformative potential in established industries. Companies backed by Ginkgo's Ferment Consortium will have the advantage of launching with full access to Ginkgo's platform for cell programming and the market expertise of leading strategic partners. The fund is supported by a \$350 million commitment from Ginkgo's current major investors, including Viking Global Investors, General Atlantic, and Cascade Investment.

SPAC prospectus

Agreements in Connection with Platform Ventures
Initial investments in new companies launched by Ginkgo in connection with the agreement were approximately **\$12.9 million from Cascade Investment L.L.C., \$19.5 million from entities affiliated with General Atlantic (GK), L.P., and \$57.8 million from entities affiliated with Viking Global Opportunities Illiquid Investments Sub-Master LP.**

Former director-level employee called it a "slush fund"

"The whole Ferment Consortium, this \$300 million slush fund to start child companies of Ginkgo, I would say for sure Ginkgo controls all of those almost certainly...Ferment is basically Viking plus a few other investors. It's a way for Ginkgo to - in an ideal world, people will just pay Ginkgo a fair market rate for their services. Ginkgo can't get enough business to support their valuation just inside their boundary. So, they have this slush fund, that they can basically use to invest in companies, to start companies, and in some cases, it's a small company that has raised very little capital." - Former director-level employee

The “fund” appears to be run by Jason Kakoyiannis, Ginkgo’s Head of Ventures. He appears to be a central, enabling player in the web of related-party “customers,” popping up repeatedly during our investigation. The website for the fund is a one page brochure. We question whether the fund has ever invested in a company the purpose of which was not to recycle R&D “revenue” back to Ginkgo.

LinkedIn profile - excerpt

Jason Kakoyiannis · 2nd

Executive. Advisor. Investor. New Company creation at Ferment. Head of Ventures at Ginkgo Bioworks. "The future will be grown."

Los Angeles Metropolitan Area · [Contact info](#)

Experience



Head of Ventures

Ginkgo Bioworks, Inc.

Apr 2015 – Present · 6 yrs 7 mos

As a member of Ginkgo Bioworks' business team since 2015, I'm responsible for a \$400 million new company formation studio, dubbed the Ferment Consortium ("Ferment" for short--see www.ferment.co). Together, Ferment and Ginkgo concept and stand up companies that use Ginkgo's cell programming platform to advance products in food, agriculture, nutrition, waste remediation, personal care and beyond. We work with strategic investors such as Bayer, Battelle, Fonterra, Louis Dreyfus Co, and capital partners like Viking Global Investors, Cascade, General Atlantic, and Breakthrough Energy Ventures. Partner companies formed include Motif Foodworks, Allonnia, Kalo, and Joyn Bio. ...see more



Board Member

Allonnia

Jan 2020 – Present · 1 yr 10 mos

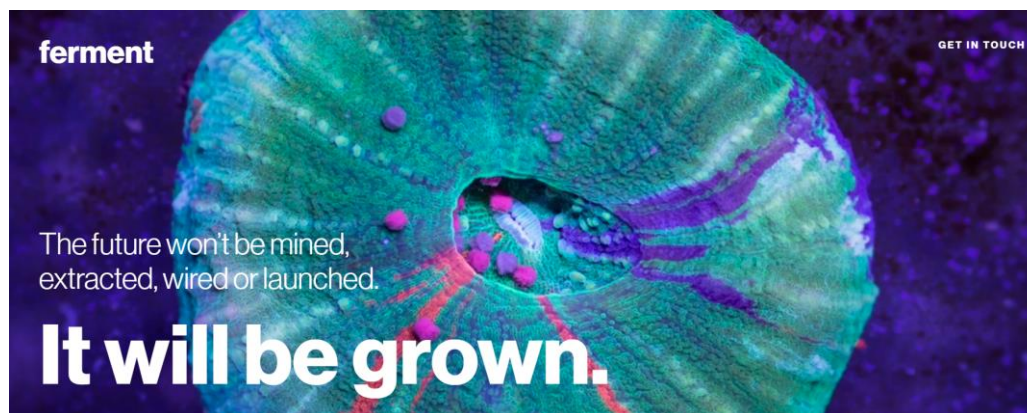


Board Member

Motif Ingredients

Jul 2018 – Present · 3 yrs 4 mos
Greater Boston Area

Excerpted sections from Ferment Fund website



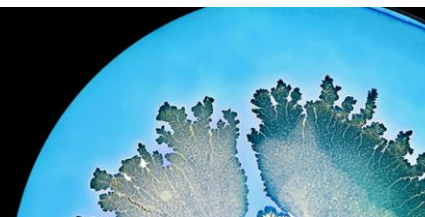
Programmable biology is

The future of food, pharma, farming.
Of design and manufacturing.
Of everything.



"The interesting thing to program in the 21st century isn't computers — it's **biology.**"
Spencer H. Ginkgo Bioworks

New ideas don't have to wait. Projects that used to require massive technical resources, infrastructure, and timelines can now launch almost overnight, with Ginkgo as the technical backend.



We further note a new venture capital entity called First Bight Ventures, which we have not seen Ginkgo disclose. It appears to have a single employee and a one page site, with a focus on synthetic biology. Ginkgo's name is absent from the site and the founder's LinkedIn bio. We spoke with an entrepreneur who recently engaged with the firm, and who believed that "it's the VC arm of Ginkgo, or that's how they frame it." We share the entrepreneur's interpretation, but caution that we can't attest to its accuracy: "They basically say, we would like to make an investment, and we expect you to spend that investment on Ginkgo's services. They don't explicitly say 100%, but I imagine they expect most of it."

Synthetic biology entrepreneur notes interactions with First Bight

Q: "So, you've spoken to Ginkgo's investment arm?"

A: **"I have spoken to their investment arm."**

Q: "And this is when? This is recent? Or a few years ago?"

A: "In the last few months, yeah."

Q: "Who runs it?"

A: "It is Veronica Wu, and **the name of the firm is First Bight.**"

Q: "What is First Bight?"

A: **"It's the VC arm of Ginkgo, or that's how they frame it.** There is some independence. I don't know exactly how the legality of it works, but **I know Ginkgo is basically their funding mechanism**...Ginkgo made the introduction to us, and they said First Bight, their goal is to invest in companies that will use Ginkgo's services. So, **a lot of their money or investment comes back to Ginkgo**...I think they're independent but Ginkgo is their sole funding source. So, I think they're probably a separate legal entity, but I imagine Ginkgo is their limited partner. **They basically say, we would like to make an investment, and we expect you to spend that investment on Ginkgo's services. They don't explicitly say 100%, but I imagine they expect most of it.** From their standpoint, it does seem fairly clever for them. I don't know how well it benefits the company that they're investing in. I think they're still raising their fund—they still haven't closed their fund. They haven't made any investments yet, I believe." – Synthetic biology entrepreneur

LinkedIn bio of First Bight founder - excerpt

Veronica Wu (She/Her) · 2nd

Founder/Managing Partner at First Bight Ventures

Houston, Texas, United States · [Contact info](#)



Founder/Managing Partner

First Bight Ventures · Full-time

Apr 2021 – Present · 7 mos

Houston, Texas, United States

First Bight Ventures is the first early stage fund dedicated to advancing and accelerating synthetic biology companies. We look to leverage our extensive industry network to broaden applications of synthetic biology into agriculture, medicine, industries and consumer goods. We believe the expansion of synthetic biology will allow us to have a better and more sustainable world.

After Viking, the second and third largest holders of Ginkgo are Cascade and General Atlantic. Cascade is a Bill Gates vehicle, though we suspect he's barely aware of Ginkgo as the CIO is Michael Larson, whose conduct was the subject of a recent New York Times investigation. General Atlantic's investment was sponsored by Michelle Dipp, an individual we have followed for years and who is a gift that keeps on giving to those who short scams. She was previously CEO of Ovascience, which fell 99% and remains the most revolting situation we have encountered, for reasons that are sordid and still only known by few. GA CEO Bill Ford trumpeted her hiring as a Managing Director in 2017. We recall chuckling at their due diligence. In 2020 Dipp settled fraud charges with the SEC and quietly departed GA along the way. She remains a huge Ginkgo fan.

SEC press release on fraud settlement with Dipp

SEC Charges Former
Biotechnology Company CEO for
Misleading Investors

ADMINISTRATIVE PROCEEDING

File No. 3-19843

June 30, 2020 - The Securities and Exchange Commission today announced that Michelle Dipp, a co-founder and former CEO of the now defunct OvaScience, Inc., agreed to settle fraud charges for misleading investors about the availability and commercial prospects for the company's fertility treatments that were in development.

Source: <https://www.sec.gov/enforce/33-10794-s>; <https://twitter.com/dipp/status/1392173411766575111>

Dipp is active on Twitter re Ginkgo



Michelle Dipp
@dipp

...

Congrats @Ginkgo @jrkelly @reshmapshetty @bcanton @TomKnightSynBio Austin Che - what a testament to your brilliance + hard work + teamwork. This is just the beginning... #synbio



Jennifer Lum @lum · May 11

Congratulations to the @Ginkgo team on bringing #synbio to industrial scale. @jrkelly @reshmapshetty

cc @dipp twitter.com/jrkelly/status...

1:43 PM · May 11, 2021 · Twitter for iPhone

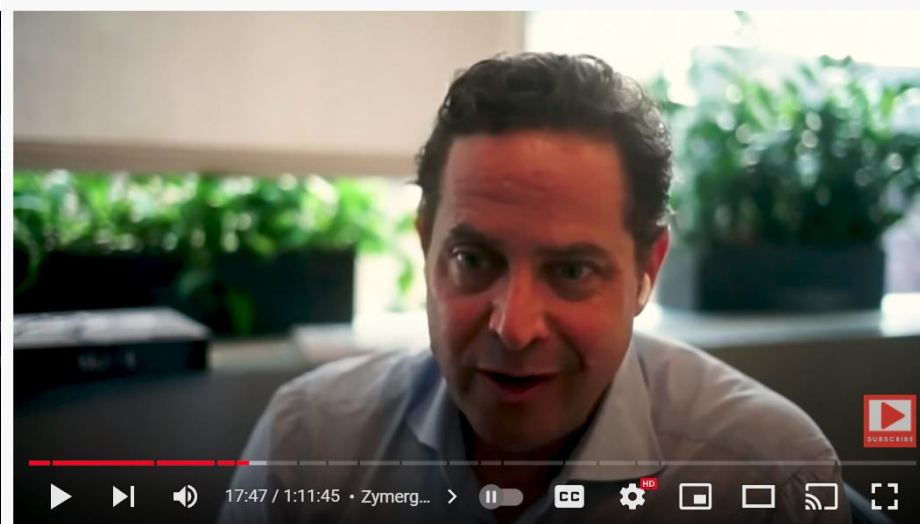
Ginkgo was founded in 2009 by MIT alums. It is a “synthetic biology” company, though we think of it less as a company than a series of pithy and slick taglines masquerading as one - “Biology by design,” for example. The “Letter from our Founders” appears to be written by a 10-year old, opening with “Biology is special” and becoming more trite and grandiose as it dribbles down the bib. We recall how WeWork’s S-1 began: “Our mission is to elevate the world’s consciousness.” The enduring, seductive tagline that has raised billions is that we “program cells as easily as we can program computers.” A recent interview with Ginkgo’s CEO reminded us of performance art by his Siamese synthetic biology twin from Zymergen, right before it self-detonated.

Ginkgo CEO Jason Kelly



<https://twitter.com/SquawkStreet/status/1438894791601840129?s=20>

Ex-CEO of Zymergen who just left as it collapsed



<https://www.youtube.com/watch?v=FYFca7tfjeg>

Similar to WeWork, Ginkgo at its core is a vanity project by “visionary” founders with little to no prior experience, sort of a fake-it-til-you-make-it Emily In Paris adventure chronicled on Instagram. It even publishes its own glossy print magazine, called Grow.

Grow is a magazine that tells the unfolding story of synthetic biology.



The Serendipity of Llama Biocapital

Can llamas heal the world?

The Highbrow Neanderthal

Long before we arrived on the scene, Neanderthals were making artworks, developing elaborate rituals, and having adventurous sex.



DIALOGUE | CHRISTINA AGAPAKIS

The New Natural

Artist Ani Liu and entrepreneur Jasmina Aganovic discuss how science and technology will shape the future of aesthetics.

The Beauty Issue

Beauty is often considered a superficial quality, but it has tremendous power over us. An expression of our ever-evolving and surprising diversity, beauty can inspire great joy and creativity, but it can also become a tool of oppression. In Grow's second print issue, we explore biology's most mysterious trick in all its facets.

Like most synthetic biology scams, Ginkgo bends over backwards to obscure what it actually does, flinging buzzwords at the wall with abandon - Apple App Store, Amazon's AWS, cloud, SaaS, machine learning - along with bespoke phrases like "We're automating organism design." The "foundation" of their platform is the "Foundry" and the "Codebase" – always capitalized: "Our Foundry leverages proprietary software, automation, and data analytics to reduce the cost of cell programming" while "Our Codebase consists of reusable biological assets that helps [sic] accelerate the engineering process."

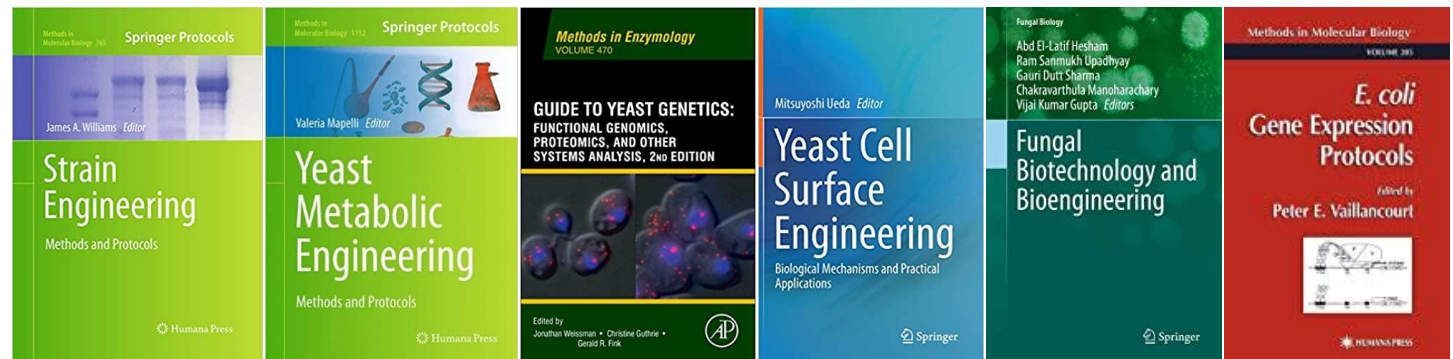
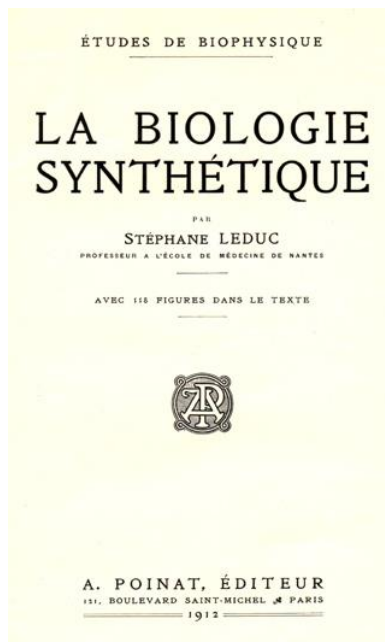
"Foundry" description from prospectus

"**Our Foundry wraps proprietary software and automation around core cell engineering workflows**—designing DNA, writing DNA, inserting that DNA into cells, testing to measure cell performance—and leverages data analytics and data science to inform each iteration of design."

"Codebase" description from prospectus

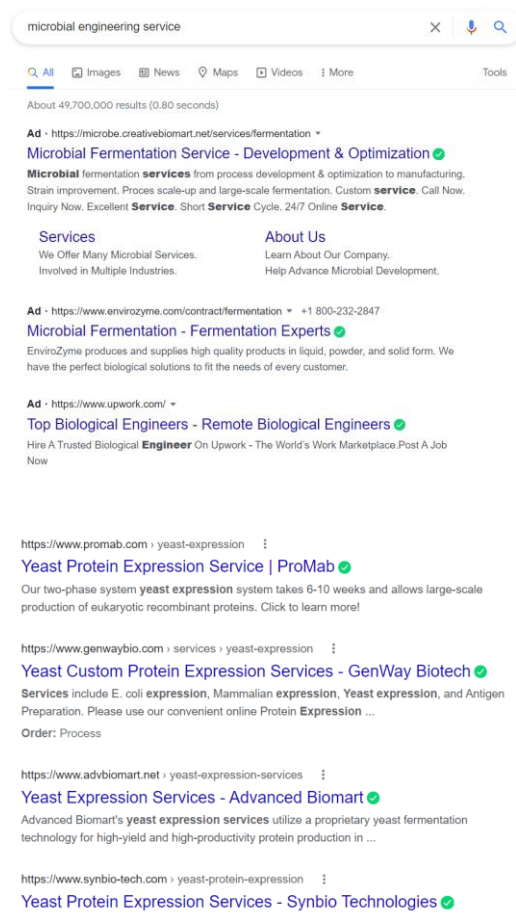
"**Our Codebase includes both our physical (engineered cells and genetic parts) and digital (genetic sequences and performance data) biological assets**, and accumulates as we execute more cell programs on the platform. Every program, whether successful or not, generates valuable Codebase and helps inform future experimental designs and provides reusable genetic parts, making our cell program designs more efficient."

In our recent report on Berkeley Lights (BLI) - a key Ginkgo partner - we stated that synthetic biology is a meaningless term and may as well be a synonym for publicly-traded scam. Allow us to offer a more precise definition: using widespread 50-year old genetic engineering tools to tweak yeast cells to excrete mostly low-value industrial inputs like flavors, fragrances, and drug ingredients. This is the essence of the mysterious foundry – Ginkgo is nothing more than a commodity strain engineering CRO, and a crappy one at that according to its own related-party “customers.” Before stock promoters claimed the term “synthetic biology,” it was and is still known as microbial or strain engineering. Humans have tweaked yeast for thousands of years for bread, beer, and other fermented products. The first synthetic biology book is from 1912.



A few Google searches for strain or microbial engineering, yeast expression services, microbial fermentation CRO, fermentation CDMO, etc. is all one needs to grasp the commodity, crowded nature of Ginkgo's business – where large players like Lonza and Abbvie, and bottom-feeder CRO's proliferating Google ads, compete on price.

Google searches for microbial fermentation service and yeast expression service



Source: Google searches

Abbvie

MICROBIAL FERMENTATION

MORE INFO

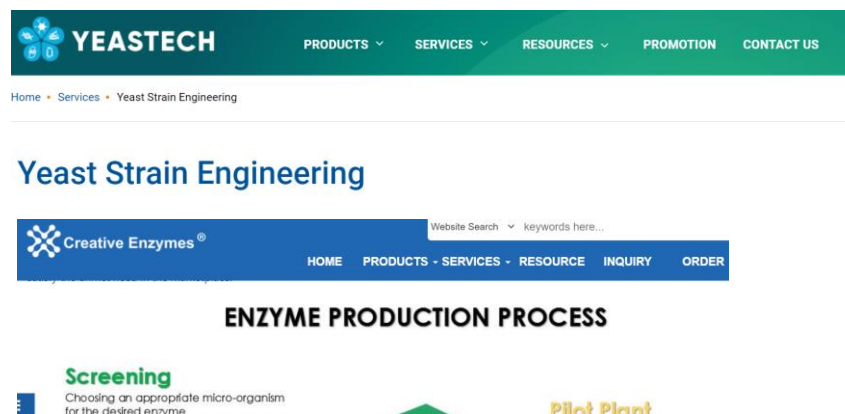
AbbVie Contract Manufacturing offers world-class fermentation capabilities based on more than 60 years' leadership in fermentation. Your project will benefit from AbbVie's scientific expertise as a leading development and manufacturer of fermentation processes at our 180-acre facility in Chicago, Illinois. We bring the same degree of expertise and passion to your project as we do with our own, supporting you from cell bank development through to large-scale manufacturing.

Lonza

Microbial fermentation excellence

As a leading development and manufacturing service provider, we strive to match your requirements with a complete line of service offerings from strain development through to large scale manufacturing. We have over 30 years of experience in microbial fermentation for biopharmaceuticals, including a proven track record of bringing products to BLA or sBLA.

Countless smaller CRO's



Modified yeast strains are used to produce countless products, and PubMed yields thousands of papers on the subject. As we discuss in Part 2, Ginkgo's only "innovation" is blowing hundreds of millions to outfit its "Foundry" with fermenters and liquid handling machines from the same vendors as everyone else. Its "Codebase" is just a sequencing database, eclipsed by far larger, widely-available libraries online. It is no wonder that ex-employees as well as its related-party "customers" described Ginkgo as a "glorified CRO" doing standard small molecule screens; a "schtick"; "doesn't have any unique equipment"; "I don't think it has any really unique IP"; "hard for me to point to something that they do that is totally different than anybody else in the industry."

Senior employee at related-party "customer" Joyn Bio calls them "a kind of a glorified CRO"

"They can engineer a strain and test them with very high throughput. **But how we are working with them now is as a kind of a glorified CRO.**" – Senior employee at Joyn

Ex-employee in a managerial role states Ginkgo has no unique equipment or "any really unique IP"

"On a high level, **I don't think they're unique and special at all...it doesn't have any unique equipment. I don't think it has any really unique IP.**" - Former employee in a managerial role

Ex-executive stated foundry is "not unique"; just standard small molecule screens people have done for years

"The foundry itself is not unique. People for many years, whether that's Amyris or Intrexon or name your favorite pharma company, **people have been doing small molecule screens and high throughput and doing similar types of work for many years.**" - Former executive

Second Joyn Bio source, an executive, can't point to anything different at Ginkgo's foundry

Q: "The foundry services that Ginkgo provides, are these are basically services you can get from a lot of other people?"

A: "Yeah, I would say there's probably—it would be **hard for me to point to something that they do that is totally different than anybody else in the industry.**"

Q: "That doesn't bode well in the valuation."

A: "No comment." – Current Joyn executive

It is therefore not surprising that Ginkgo's financials are a self-evident catastrophe – the ultimate greater fool SPAC. In our last report, we described Berkeley Lights as a dumpster fire, but perhaps we should have reserved the term, as BLI looks like Berkshire Hathaway in comparison. We note that Ginkgo is opaque about its gross margin and cost of goods – highly unusual, in our experience, and indicative of a broken business model with fatal unit level economics. It is rare – even among the frauds, promotes, and detritus that we short – to see losses and cash flow characteristics as dreadful as Ginkgo's, much less driving a \$23B valuation.

<i>in millions USD</i>	2019	2020	LTM 6/30/21
Revenue	54.2	76.7	133.1
Research & development	96.3	159.8	208.9
Operating income	(71.6)	(137.0)	(207.4)
Net income	(119.3)	(126.6)	(200.5)
Operating cash flow	(44.7)	(135.8)	(167.7)
Capital expenditure	(22.2)	(57.8)	(94.0)
Free cash flow	(66.9)	(193.6)	(261.7)
EBITDA	(60.8)	(123.2)	(187.0)
R&D as % of revenue	177.7%	208.4%	156.9%
Operating margin	-132.1%	-178.7%	-155.8%

Ginkgo's inability to execute on a single initiative except fundraising is stunning. In 2020, the company made a random pivot into COVID testing kits, a commodity business with a short half-life as the pandemic fades. Ginkgo of course applies a fancier label – “Biosecurity revenue,” which comprised 50% of its 1H 2021 revenue, with foundry R&D services as the other half. Even for a simple cost-of-goods business, Ginkgo reported \$41MM of what appear to be “Biosecurity” cost of goods – suggesting a pitiful 6% gross margin on \$44MM of 1H revenue for the segment, with free cash flow that we suspect is another fiasco, given undisclosed capital expenditure to kick it off. At its 1H business update on Aug 18th, Ginkgo already missed its COVID revenue projection as testing fades, leaving a massive comps headwind for next year, to the extent YoY comparisons with a fictitious P&L are even relevant. We note below that ex-employees mocked the company's foray into low-margin testing.

“In 2020, we incurred significant operating and capital expenditures in launching our biosecurity offering, which includes COVID-19 testing products...” – SPAC prospectus

Covid-related revenue is unsustainable and testing is a commodity

*“Covid testing was also just a **ploy to get into pharma** because for years they've tried to get into pharma and **nobody wanted to work with them. It's still just like a commodity business** that they're in, sort of NGS of viral samples; it's like the Quest Diagnostics and all these other labs, **it's a race to the bottom for pricing** and things like that. They've got **some inflated revenues or short-term revenues** but I don't see that as building long-term value .” –Former director-level employee*

“Nothing special”

*“I don't even know why Ginkgo is doing Covid-19 testing. PCR testing, you can do that at any university lab, really at any company. It just requires a couple of DNA sequencers, I mean, Illumina instruments, **nothing special**.” – Former employee in a managerial role*

Ginkgo's trajectory is typical of the synthetic biology space, which is littered with scams and blows ups, whether Zymergen two months ago or Intrexon, which we shorted successfully for years. Intrexon pioneered the round-tripping hustle in the space, using the term "Exclusive Channel Collaboration" for related-parties, while Ginkgo calls them "Platform Ventures." **An ex-employee of Ginkgo who worked at Intrexon stated that Ginkgo copied their playbook; says he's "been around the block enough times to understand" Ginkgo's related-party games; and noted it's the same model where money is "going from the left hand to the right."**

"More than one chapter in Ginkgo's playbook copied from Intrexon"

"If you're a student of the history of synthetic biology, there's more than one chapter in Ginkgo's playbook copied from Intrexon because what RJ Kirk was doing, he basically invested in distressed assets or companies that needed some sort of assistance, and that revenue came back to Intrexon and it looked like proper revenue for the company, it helped build the valuation of the company that allowed them to go out and do big acquisitions and things like that. **You know how Intrexon ended."**

Q: "Is this basically **Intrexon 2.0**, just to be blunt?"

A: "Yeah, I would say so." – Former senior employee at Ginkgo

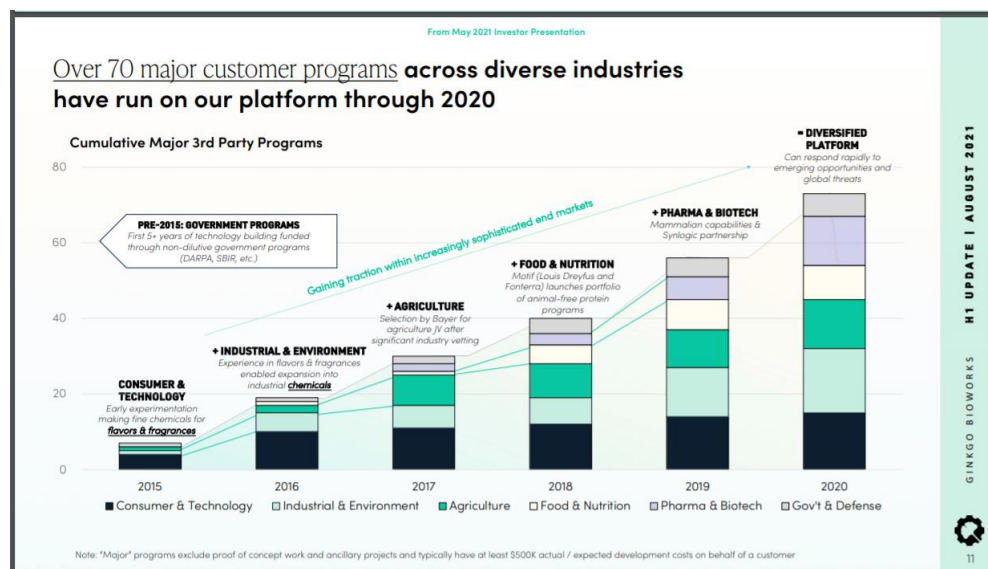
Same model of pushing to control related-parties to capture and round-trip their "revenue"

"From a business perspective, they want all the revenue, and they want to control the entity, for sure. It reminds me a lot of RJ because, in the short term, everything looked amazing, but they never got a single project over the finish line. I know short-term, now that they're a public company, they have revenue expectations and things like that. But it's kind of interesting because the whole Ferment Consortium, this \$300 million slush fund to start child companies of Ginkgo, I would say for sure Ginkgo controls all of those almost certainly." – Former senior employee at Ginkgo

Same playbook with money "going from the left hand to the right"

"I think the related parties, going back to my comment about Intrexon... I know enough about what's going on there and have been around the block enough times to understand...Having come from Intrexon and having seen that playbook played out before...going from the left hand to the right hand, so to speak. They can do that because they can raise money like no one else I've ever seen before." – Former senior employee at Ginkgo

Ginkgo promotes “over 70 major customer programs” to date, but our research indicates a colossal, decade-long failure to deliver strains that work – a devastating indictment of the hype around its “Foundry.” An ex-business development executive stated that his team did 20 collaborations and “zero percent panned out.” A former strain engineer stated that “9 out of 10” projects didn’t succeed. Others confirmed the color. We find it revealing that Ginkgo tends to lead its customer case studies with Cronos – a shady Canadian pot company with \$57MM LTM revenue and \$194MM in losses. Ginkgo’s ticker is DNA – one with a hallowed past given it belonged to Genentech - yet it has now fallen to a failed CRO where Cronos is the best it can do.



Ex Ginkgo director-level employee: “There’s no real business there”; “They don’t have those products”
“I think the big hiccup or red flag right now is you could say, **where are those products? For a company that’s been around for 13 years**, it seems to me, if I was going public, I’d have this long list of amazing products—they don’t all have to be billion-dollar products, but **something that looks tangible** that says, we have succeeded and provided value to our partners and provided a good return on investment. **And they don’t have that story. They don’t have those products.** And so, do you believe that they are going to build those products and are going to have some success? Or do you believe that it’s just a smokescreen? **There’s no real business there.”** - Former director-level employee

Ginkgo's product failure is identical to Zymergen's, which claimed to have a product but didn't, triggering a 70% plunge two months ago. The companies are identical - both use yeast as their platform of choice; the same suppliers to outfit their facilities; and so many similarities that multiple ex-Ginkgo employees, including one who also worked at Zymergen, told us they expect the same implosion. We preview their comments, covered in more detail in Part 2: "can't tell what's different"; both founders "have their little magical thing they spin to get deals"; can't figure out the "secret sauce" at either company; both "were always looking jealously at each other"; "Wow, it's like Zymergen too here. All the same stuff is happening, except it's happening even faster at Ginkgo."

Current executive at related-party "customer" Joyn Bio "can't tell what's different"

*"Zach Serber, the CSO, was the same kind of thing as Jason. They each have their little magical thing that they spin to get deals. You could never really get to what was the secret sauce. **I can't tell what's different about them.**" – Current Joyn executive*

Ginkgo and Zymergen "were always looking jealously at each other"; not much differentiation between the two

"I think both companies were always looking jealously at each other or trying to one-up each other. But I don't think there's a large differentiation between the two companies *per se besides that historical philosophy. I think they've worked on very similar projects and ideas. Like, for Joyn Bio, Zymergen was a runner-up, and you can see in their S1. They kept working on a nitrogen-fixing microbe because they thought it was such a valuable idea or useful thing to do. There are a lot of parallels; they're **more similar to each other than they are different.**" - Former director level employee*

Ginkgo employee who worked at Zymergen stated it's "all the same" but the decline is even faster at Ginkgo

*"[A colleague who worked at Zymergen] was telling me, **"Wow, it's like Zymergen too here. All the same stuff is happening, except it's happening even faster at Ginkgo.**" The decline in the culture." There was a very fast decline when it got really close to IPO; the culture became terrible, I would say, in a matter of a few months."* –Former Ginkgo employee

Ex-employees familiar with Zymergen, whether by having worked there or having contacts, indicated the culture at Ginkgo was similar and frequently used the word “toxic,” a common feature we’ve noted in this genre of shorts: “Absolutely, it was toxic”; “The culture is pretty toxic. There are some really terrible things going on”; “There were things that went on, and I’m not going to go into the details.” An ex-executive stated a member of his team had to leave because of “horrible diet problems that developed because of working at Ginkgo,” and added that the founders are inexperienced and have alienated staff.

“Absolutely, it was toxic”

“Absolutely, it was toxic. Absolutely. It was toxic. I had one individual who worked for me. She had to leave because she had **horrible diet problems that developed because of working at Ginkgo.** Physically, it was just killing people. It was not because people were working hard—because we all worked hard—it's a mental anguish that was put on people...**It was abusive,** and part of it was that it was frustrating because this is Jason's first job.” – Former Ginkgo executive

“The culture is pretty toxic”

“The culture is very toxic...We used to call Zymergen Theranos because it was lying so much about the Hyaline film, and all the stuff that came out about Hyaline being not good for customers and stuff, we all knew that for a whole year, we just couldn't talk about it. **I just think that it's lying about stuff that's creating an extremely toxic culture, silencing...The culture is pretty toxic. There are some really terrible things going on.”** –Former Ginkgo employee

Founders are inexperienced and have alienated staff; “things that went on”

“Ginkgo is not an easy place to work. The biggest issue is that the founders and the management are one and the same. This is their first job out of school, and they really don't know how to build employees and retain good people, and they don't care if they piss people off with their actions. **There were things that went on, and I'm not going to go into the details,** but there are a number of things that went on in the company my last year or so.” – Former Ginkgo employee

In August, MIT Technology Review published a damning investigation into Ginkgo. We have long followed the author's prescient exposes on questionable biotech companies. A senior employee of Joyn Bio, a key related-party "customer" based inside Ginkgo's headquarters, asked us if we had seen the piece: "we were afraid that they had bugged our Slack channel"; the article "picked up on all of the things that we were concerned about Ginkgo, about their grandiose claims" and "versus the reality of what we see day to day."

Is Ginkgo's synthetic-biology story worth \$15 billion?

"Jason Kelly has promised a manufacturing revolution with DNA. Just don't ask him to make any products."

Jason Kelly has promised a manufacturing revolution with DNA.

Just don't ask him to make any products.

<https://www.technologyreview.com/2021/08/24/1032308/is-ginkgos-synthetic-biology-story-worth-15-billion/>

"We were afraid that they had bugged our Slack channel"

Q: "There was a recent article in the MIT Technology Review, which I'm sure you can find."

Q: "I read it. It was very skeptical. Is that your perspective?"

A: **"I do. There were several of us here who said we were afraid that they had bugged our Slack channel."**

Q: "What were you guys saying and thinking internally that the article captured and echoed?"

A: "[Chuckles] **They talk a good game, and then six months later, they're still saying the same things. Jason Kelly's been giving the same pitch for eight years**..." – Senior employee at Joyn Bio

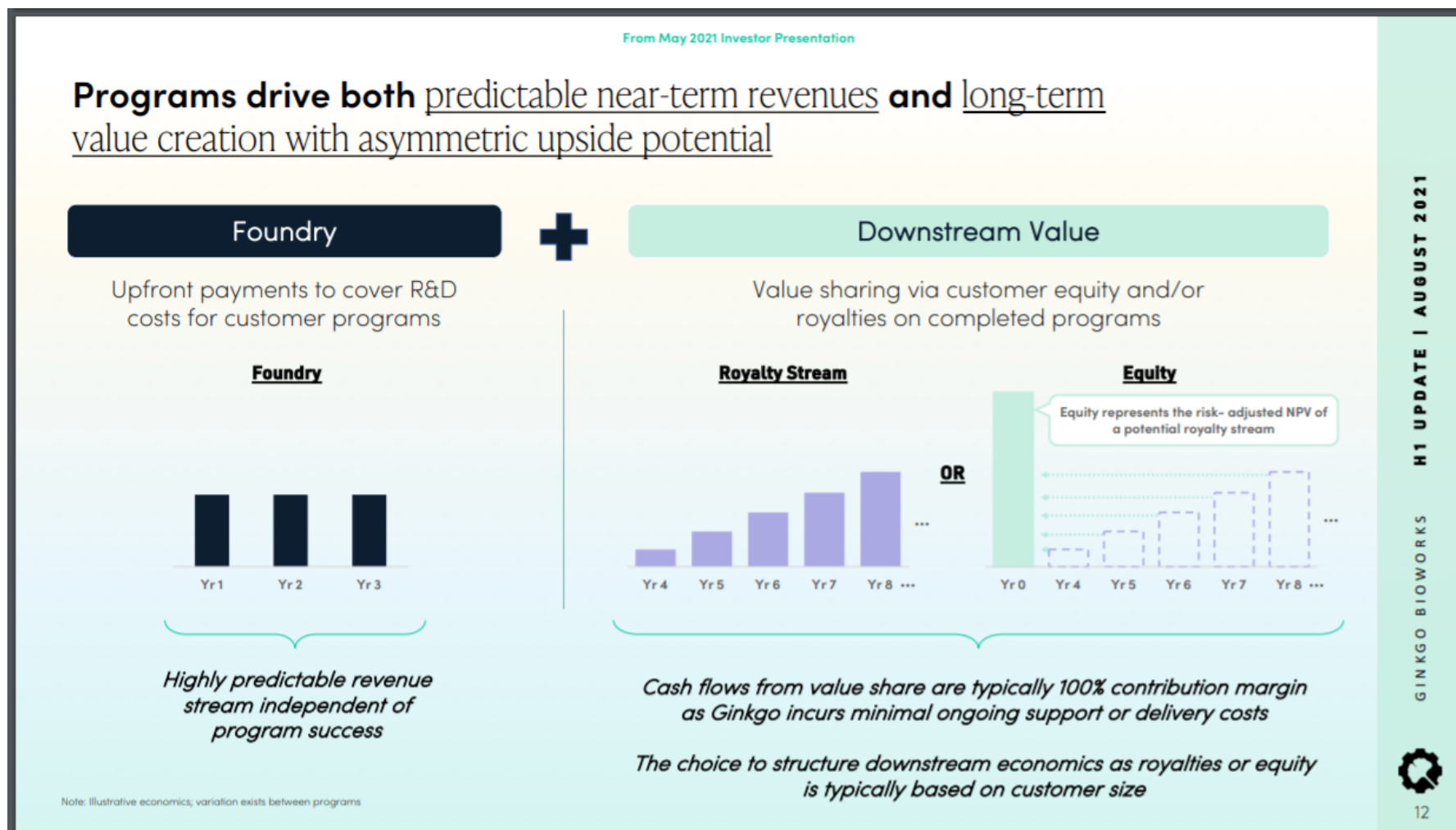
Article picked up on concerns internally regarding the reality vs. the hype

"It was really interesting that they sort of picked up on all of the things that we were concerned about Ginkgo, about their grandiose claims about what they're going to accomplish versus the reality of what we see day-to-day, Recently Ginkgo also came out with a projection of the number of new ventures that they're going to start. And I think if you look at the number and the timeframe, it announced one new venture every three days. That's the kind of thing. I don't think that's possible for a giant multinational to do that, much less a company the size of Ginkgo." – Senior employee at Joyn Bio

2. Ginkgo's business model is based on a dubious shell game

Ginkgo's business model is built upon selling R&D services ("Foundry revenue" or "Foundry usage fees") and also taking royalties or equity stakes in customer programs ("Downstream value").

Ginkgo corporate presentation, Aug 18, 2021



Given that “Customer programs” are at the heart of the model, one would assume that Ginkgo is providing R&D services to third parties with whom it has typical vendor/customer relationships. In reality, the programs are related parties that Ginkgo calls either “Platform Ventures” – de novo entities it creates and funds which then become its customers – or “Structured Partnerships,” which are pre-existing entities that Ginkgo and/or its own investors and other related parties invest in.

Excerpts from Prospectus, Sept 17, 2021

Platform Ventures

Platform Ventures allow leading multinationals to **partner with Ginkgo and financial investors to form new ventures** in identified market segments with potential to benefit from synthetic biology. **In exchange for an equity position in the venture**, we contribute license rights to our proprietary cell programming technology and intellectual property, while our partners contribute relevant industry expertise, other resources and venture funding. **We also provide R&D services for which we receive cash payments for our costs incurred, plus a margin.**

Structured Partnerships

Structured Partnerships **allow Ginkgo to partner with existing entities** with complementary assets for high potential synthetic biology applications

Ginkgo discloses roughly 7 platform ventures and structured partnerships in total that comprise most of its foundry revenue. Each has a similar structure: Ginkgo and its investors create and fund the entity, which then simply returns and round-trips back cash or “non-cash consideration” to Ginkgo under the guise of R&D “prepayments” or services, which it books as Foundry revenue or deferred revenue. The disclosure for Joyn Bio LLC, a “Platform Venture” formed in 2017 to focus the agriculture vertical, lays out the template: an equity investment commensurate with a Foundry Services Agreement with Ginkgo.

Excerpt from Prospectus, Sept 17, 2021

Joyn Bio, LLC

Founded in 2017, Joyn Bio, LLC (“Joyn”) was formed to focus on engineered microbes for use in agricultural applications. **Along with certain of our investors, we formed Cooksonia, LLC (“Cooksonia”) which holds a 50% equity interest in Joyn [...]**

Ginkgo also entered into a Foundry Services Agreement (“Joyn FSA”) with Joyn under which we provide R&D services [...]

The Company contributed \$5.0 million in cash and certain intellectual property to Cooksonia in exchange for a 70% equity interest in Cooksonia (“Class A Units”). **Cooksonia received \$20.0 million in cash from another investor, who is a related party of the Company**, for a 20% equity interest in Cooksonia (“Class B Units”).

Bayer contributed \$20.0 million in cash funding plus specified intellectual property. In addition, Bayer committed to contribute up to an additional \$60.0 million to be paid subject to certain funding procedures.

Ginkgo's business model is little more than a shell game where what's taken from the left hand is given to the right. The vast majority of Ginkgo's foundry revenue comes from transactions with its "Platform Ventures" and "Structured Partnerships. These related party entities comprised a shocking 72% of Foundry revenue in 2020. The disclosure is buried in an index on pages 470 and 502 of a 725 page prospectus, under Note 21 in section F-95 and Note 17 in section F-129. We summarize the information from these notes in a table.

<i>in millions USD</i>		2019	2020	Q1 2021
Foundry revenue by related party				
	Joyn	9,349	7,273	1,598
	Motif	18,986	20,798	5,492
	Genomatica	6,248	9,431	3,298
	Allonnia		4,960	2,266
	Synlogic	17	73	6
	Glycosyn	668	-	-
	Kalo	-	-	-
	Total	35,268	42,535	12,660
Total foundry revenue		54,184	59,221	22,504
Related parties as % of total foundry revenue		65%	72%	56%

An even greater percentage – essentially 100% - of Ginkgo’s deferred revenue also comes from transactions with these related party entities – 94% to be exact, as of March 31, 2021. Given that Ginkgo steadily draws down the deferred revenue account on its balance to create revenue on its income statement, the prospectus is signaling that related parties will continue to be the source of “growth” in sales.

<i>in millions USD</i>				
Deferred revenue by related party		12/31/2019	12/31/2020	3/31/2021
	Joyn	17,135	9,862	8,281
	Motif	62,513	53,952	53,299
	Genomatica	38,059	30,128	26,680
	Allonnia	24,480	26,064	37,959
	Synlogic	144	72	66
	Kalo	-	-	11,897
	Total	142,331	120,078	138,182
Total deferred revenue		147,898	128,475	147,408
Related parties as % of total deferred revenue		96%	93%	94%

We note a particularly alarming and aggressive feature of the structure: immediate “prepayments” from related parties back to Ginkgo. In Joyn’s case, the filings disclose \$35MM of prepayments, which Ginkgo presumably booked as “Foundry revenue” or deferred “revenue.” The prepayments are similar in amount to the \$25MM contribution in Joyn by Ginkgo and Ginkgo’s related party investors – a rather elegant mathematical outcome where what’s contributed is what’s spiffed back.

Excerpt from Prospectus, Sept 17, 2021

Joyn paid us **a non-refundable \$20.0 million prepayment for services to be provided** under the Joyn FSA and made **an additional \$15.0 million prepayment for services** during the year ended December 31, 2019.

The haste with which related parties recycle proceeds back to Ginkgo is startling. In virtually every case, Ginkgo and its investors inject cash into the related party, followed by Ginkgo immediately **booking large chunks of revenue and deferred revenue.** We believe this serves several **purposes, beyond just conjuring revenue from thin air: it creates a deferred revenue slush fund** that enables Ginkgo to create quarterly accounting revenue at will, versus being dependent on the timing of the entity's R&D spend; it gets Ginkgo's hands back on its cash instantly, lest the related party ever have second thoughts about using Ginkgo's foundry services (a recurring source of tension with some related parties, which we discuss later); and it enables Ginkgo to create a false investor narrative around a deferred revenue backlog.

Ginkgo invested \$80MM in its related party Synlogic, but took \$30MM out at closing – a prepayment for five years of R&D spend. In a legitimate vendor-customer transaction, no customer would ever pre-pay five years of R&D up front. We wonder if the \$30MM was even included in the wire to Synlogic and then wired back to Ginkgo – or if Ginkgo just kept it and dispensed with any such housekeeping formalities altogether.

Synlogic press release announcing Ginkgo's equity investment, June 12, 2019

Ginkgo has purchased 6,340,771 shares of Synlogic common stock as well as pre-funded warrants to purchase up to 2,548,117 shares of Synlogic common stock, both at a price of \$9.00 per share. Gross proceeds to Synlogic are approximately \$80 million. The transactions were executed and closed on June 11, 2019. **At the closing, under the foundry services agreement, Synlogic paid \$30.0 million to Ginkgo for synthetic biology services to be provided over an initial period of five years which can be extended.** Synlogic has exclusive rights to any Synthetic Biotic medicines that it develops as part of the collaboration and to intellectual property covering such products.

The last two pages illustrated Joyn Bio and Synlogic as examples where Ginkgo recorded large prepayments from related parties in coordination with financings by it and/or its own investors. We looked at each disclosed related party with a meaningful deferred revenue balance, and the modus operandi is identical. Allonnia is another related party, where the initial Series A Preferred tranche appears to be for \$33MM in Dec 2019. By Dec 31, 2019, Ginkgo already booked \$24.5MM in deferred revenue from the entity. As the only private related party with financials disclosed in Ginkgo's prospectus, one can see that Allonnia simultaneously recorded \$24.5MM in R&D expenses.

Allonnia release announcing Ginkgo investment, Oct 2020 (includes follow-on Series A >\$33MM)

Allonnia Launches from Ginkgo Bioworks'
Ferment Consortium with \$40M Series A to
Address Critical Problems in Waste
Remediation with Synthetic Biology, Appoints
Nicole Richards as CEO

Initial Series A tranche appears to be \$33MM in Dec 2019

Pursuant to the Series A-1 Unit Purchase Agreement, as amended, the Company issued 2,970,000 and 1,664,911 units of Series A-1 at a purchase price of \$11.111111 per unit in 2019 and 2020, respectively, for the total gross proceeds of \$51,499,011.

Allonnia financials buried at end of SPAC prospectus show \$24.5MM of R&D expenses in 2019, simultaneous with Ginkgo booking \$24.5MM of deferred revenue from the entity

	2020	2019
Operating expenses:		
Research and development	\$ 3,966,936	\$ 24,481,519
General and administrative	2,106,245	162,978
Total operating expenses	6,073,181	24,644,497
Net operating loss	(6,073,181)	(24,644,497)
Interest income	61,663	—
Net loss	\$ (6,011,518)	\$ (24,644,497)

in millions USD	2019	2020
Revenue Ginkgo recognized from Allonnia		4,960
Deferred revenue balance at period end with Allonnia	24,480	26,064
Total booked as revenue and deferred revenue	24,480	31,024

Motif Ingredients, now re-branded at Motif Foodworks, exhibits the same playbook: a \$90MM investment into the related party announced in Feb 2019, simultaneous with Ginkgo booking \$81.5MM of revenue and deferred in 2019.

Press release announcing Ginkgo's equity investment, Feb 26, 2019

Ginkgo Bioworks Launches Motif Ingredients with \$90 Million Series A Financing

Ginkgo immediately booked \$81.5mm of revenue and deferred revenue

<i>in millions USD</i>	2019
Revenue Ginkgo recognized from Motif	18,986
Deferred revenue balance at period end with Motif	62,513
Total booked as revenue and deferred revenue	81,499

Kalo Ingredients is a newly-formed related party, founded in 2021. By the end of Q1 2021, it appears to have completed a \$77MM Series A investment involving Ginkgo's investors with Ginkgo simultaneously booking \$11.9MM of deferred revenue. We expect Ginkgo to sharply escalate the revenue and deferred revenue it books from Kalo, as the entire purpose of these related parties appears to be to create the illusion of "growth."

Prospectus indicates \$77MM financing including "certain of our investors," which appears to be in Q1 2021

Kalo Ingredients, LLC

Founded in 2021, Kalo Ingredients, LLC ("Kalo") was formed to focus on the application of synthetic biology in the beauty and personal care products industry. In March 2021, we entered into an intellectual property contribution agreement that granted Kalo rights to our intellectual property, subject to mutually agreed upon technical development plans. In return for our contribution of intellectual property and access to our platform, we received common units in Kalo with a right to additional units subject to additional closings of Kalo's Series A preferred units. The initial fair value of our common units received in Kalo was \$11.9 million which has subsequently been reduced to a carrying value of zero as a result of the allocation of losses under our accounting for equity method investments. Kalo was capitalized through a Series A preferred unit financing that raised approximately \$77 million in gross proceeds from an investor group which included certain of our investors and industry strategic investors. Ginkgo also entered into a Technical Development Agreement with Kalo under which we will provide R&D services in return for cash consideration on a cost-plus fixed margin basis.

Ginkgo immediately booked \$11.9MM of deferred revenue, as of 3/3/21

<i>in millions USD</i>	3/31/2021
Revenue Ginkgo recognized from Kalo	-
Deferred revenue balance at period end with Kalo	11,897
Total booked as revenue and deferred revenue	11,897

Genomatica is a related party that Ginkgo calls a “Structured Partnership” – its term for pre-existing entities in which it invests, as opposed to new entities it creates itself. Genomatica appears to have been equitized in 2016 and 2018 with a Series A and Series B Preferred financing, respectively. The timing is opaque given Ginkgo’s lack of disclosure. Nonetheless, Ginkgo discloses a carrying value at cost of \$55MM in Genomatica preferred stock, which we suspect closed in 2018. By 2019, Ginkgo already booked \$44MM of revenue and deferred revenue.

SPAC prospectus indicates \$55MM Ginkgo investment in Genomatica

	As of June 30, 2021	As of December 31, 2020
Investments:		
Genomatica, Inc. preferred stock	\$55,000	\$ 55,000
Synlogic, Inc. warrant	9,912	5,504
Total	<u>\$64,912</u>	<u>\$ 60,504</u>

SPAC prospectus indicates \$44MM of revenue and deferred revenue booked in 2019

<i>in millions USD</i>	2019	2020
Revenue Ginkgo recognized from Genomatica	6,248	9,431
Deferred revenue balance at period end ith Genomatica	38,059	30,128
Total booked as revenue and deferred revenue	44,307	39,559

3. We believe that Ginkgo is concealing the true extent of its dependence on related-party revenue

We noted earlier that the majority of Ginkgo’s foundry revenue comes from related party entities: 65% in 2019, 72% in 2020, and 56% in 2021. We believe that the actual figure is essentially 100%, and think that Ginkgo is misclassifying and underreporting related party revenue. While the reported 60-70% percentage is bad enough and indicative of a broken, hocus-pocus business model, it’s still better than 100%. Ginkgo can claim that at least 30-40% of foundry revenue comes from third parties, and that its reliance decreased in Q1 2021 to “only” 56%. Ex-employees indicated fear inside Ginkgo that its related party model doesn’t stand up to regulatory scrutiny and could prevent it from going public – hence, we believe, an effort to fake the reported figure.

(in thousands)	Year Ended December 31,	
	2020	2019
Foundry revenue (includes related party revenue of \$42,535 and \$35,268, respectively)	\$ 59,221	\$ 54,184

<i>in millions USD</i>	2019	2020	Q1 2021
Foundry revenue by related party			
Joyn	9,349	7,273	1,598
Motif	18,986	20,798	5,492
Genomatica	6,248	9,431	3,298
Allonnia		4,960	2,266
Synlogic	17	73	6
Glycosyn	668	-	-
Kalo	-	-	-
Total	35,268	42,535	12,660
Total foundry revenue	54,184	59,221	22,504
Related parties as % of total foundry revenue	65%	72%	56%

***Reported
related party
foundry
revenues
totaled 65% in
2019 and 72% in
2020***

As we examined Ginkgo's related party disclosures, we noted a major anomaly: Ginkgo invested \$80MM in Synlogic in 2018, but disclosed almost no revenue from it in 2019 and 2020 in the footnotes. Similarly, it reported a minimal deferred revenue balance with Synlogic, in Note 21 in the SPAC prospectus. In every other case, Ginkgo aggressively booked revenue and deferred revenue from related parties in which it invested. We were initially stumped.

21. Related Parties

Related party transactions included in the Consolidated Balance Sheets, excluding the Company's investments and equity method investments, are summarized below (in thousands):

	Joyn	Motif	Genomatica	Allonnia	Synlogic	Total
Balances as of December 31, 2020						
Accounts receivable, net	\$ —	\$ 2,403	\$ 1,500	\$ 1,309	\$ —	\$ 5,212
Prepaid expenses and other current assets	\$ 24	\$ 232	\$ —	\$ 13	\$ —	\$ 269
Deferred revenue, current and non-current	\$ 9,862	\$53,952	\$ 30,128	\$26,064	\$ 72	\$120,078
Balances as of December 31, 2019						
Accounts receivable, net	\$ 163	\$ 4,054	\$ —	\$ —	\$ —	\$ 4,217
Deferred revenue, current and non-current	\$17,135	\$62,513	\$ 38,059	\$24,480	\$ 144	\$142,331

Related party transactions included in the Consolidated Statements Operations and Comprehensive Loss, excluding the losses on the Company's investments and equity method investments, are summarized below (in thousands):

	Joyn	Motif	Genomatica	Allonnia	Synlogic	Glycosyn	Total
For the Year Ended December 31, 2020							
Foundry revenue	\$ 7,273	\$ 20,798	\$ 9,431	\$ 4,960	\$ 73	\$ —	\$ 42,535
Other income, net	\$ 407	\$ 314	\$ —	\$ —	\$ —	\$ —	\$ 721
For the Year Ended December 31, 2019							
Foundry revenue	\$ 9,349	\$ 18,986	\$ 6,248	\$ —	\$ 17	\$ 668	\$ 35,268
Interest income	\$ —	\$ —	\$ —	\$ —	\$ —	\$ 163	\$ 163
Other income, net	\$ 222	\$ 42	\$ —	\$ —	\$ —	\$ 1,530	\$ 1,794

We were further troubled, as the Synlogic press release on June 12, 2019, announcing the \$80MM Ginkgo investment stated that “At the closing under the foundry services agreement, Synlogic paid Ginkgo \$30MM for “synthetic biology services to be provided over an initial period of five years....” – yet Ginkgo disclosed only \$70-150K in its related party deferred revenue disclosure.

synlogic



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News Release

Synlogic and Ginkgo Bioworks Establish Transformational Platform Collaboration for the Accelerated Development of Novel Synthetic Biotic™ Medicines

June 12, 2019

Ginkgo has purchased 6,340,771 shares of Synlogic common stock as well as pre-funded warrants to purchase up to 2,548,117 shares of Synlogic common stock, both at a price of \$9.00 per share. Gross proceeds to Synlogic are approximately \$80 million. The transactions were executed and closed on June 11, 2019. At the closing, under the foundry services agreement, Synlogic paid \$30.0 million to Ginkgo for synthetic biology services to be provided over an initial period of five years which can be extended. Synlogic has exclusive rights to any Synthetic Biotic medicines that it develops as part of the collaboration and to intellectual property covering such products.

As Synlogic is a publicly-traded microcap (ticker: SYBX) with \$150MM market cap, we examined its SEC filings. We were amused, but not surprised, to discover that Synlogic’s 2020 10K, under Note 16 for Related-Party Transactions, disclosed that Synlogic “used \$13.6 million of the pre-paid research and development expenses” in 2020. This is in stark contrast to Ginkgo’s related party disclosure, which reports only \$73K of foundry revenue from Synlogic.

Synlogic 10K filed Mar 25, 2021

(16) Related-Party Transactions

In June 2019, the Company expanded its collaboration and entered into an agreement with Ginkgo for the research and development of engineered microbial therapeutic products. As of December 31, 2020, Ginkgo owns 6,340,771 shares of the Company’s outstanding common stock. See Note 10, *Ginkgo Collaboration* .

Under the agreement the Company made a prepayment to Ginkgo of \$ 30.0 million for its foundry services that will be provided to the Company over an initial term of five years . At December 31, 2020, the Company had remaining balances of \$ 4.7 million and \$ 8.9 million of current and non-current pre-paid research and development costs related to this transaction, respectively. For the year ended December 31, 2020, the Company used \$ 13.6 million of the pre-paid research and development expenses.

Synlogic's disclosures suggest that Ginkgo has misclassified Synlogic's R&D spend in order to conceal its near-total dependence on parties it finances. Had Ginkgo included Synlogic's foundry spend of \$13.6MM, related party revenues as a percent of total would have jumped from 72% to 95% - that is, Ginkgo would have had to admit that its foundry is a flop and that it can't get customers unless it gives them cash and round-trips the proceeds.

Related party revenue as reported

<i>in millions USD</i>	2020
Foundry revenue by related party	
Joyn	7,273
Motif	20,798
Genomatica	9,431
Allonnia	4,960
Synlogic	73
Glycosyn	-
Kalo	-
Total	42,535
Total foundry revenue	59,221
Related parties as % of total foundry revenue	
	72%

Adjusted to include Synlogic revenue

<i>in millions USD</i>	2020
Foundry revenue by related party	
Joyn	7,273
Motif	20,798
Genomatica	9,431
Allonnia	4,960
Synlogic	13,600
Glycosyn	-
Kalo	-
Total	56,062
Total foundry revenue	59,221
Related parties as % of total foundry revenue	
	95%

We were unable to conduct a similar analysis of Synlogic’s R&D spend with Ginkgo for periods other than 2020, given that it failed to disclose related party R&D spend under the collaboration agreement. However, we spoke with an executive of Synlogic who expressed surprise and incredulity when we asked about the negligible amounts that Ginkgo’s related party disclosures attributed to Synlogic in 2019, 2020, and Q1 2021. The executive indicated it’s not consistent with the R&D projects they have had to send their way by virtue of the \$30MM foundry credits Ginkgo gave them.

Executive indicated the paltry amounts of related-party revenue Ginkgo discloses from Synlogic does not square with the level of spending Synlogic is actually doing under the agreement

Q: “I’m actually looking at their financials, and it looks like they have only booked as far as foundry revenue from Synlogic because they disclosed revenue from related parties and classify you guys as a related party. \$72,000 in 2019, \$73,000 in 2020, \$6000 in Q1 2021. Is that really the volume of work you’re sending their way? \$17,000 a year, \$70,000 a year?”

A: “No, that doesn’t seem—I’m not looking at what you’re looking at.”

Q: “That is what is in their financials. They break it out. There’s a line for Synlogic.”

A: “Yeah, I haven’t seen that, but **that doesn’t sound accurate**...I’m probably not the right person to comment on the accounting related to the relationship.”

Q: “Does that square with the amount that you think Synlogic is spending?”

A: “I mean, we have \$30 million to spend over five years. So, it would be difficult to spend that much at those rates. I think that the agreement specified a five-year period if I’m not mistaken.”

Q: “Let me put it this way, are you aware of work that’s going on there that is more than \$100,000 a year because they’re saying it’s less than \$100,000 worth of work a year?”

A: **“Yes, we’re spending more than that.”** I don’t know how the money was captured in the original deal. I don’t know any of that stuff.”

Q: “Is there some other entity or something that is contracted with them, like do you have some other CRO that’s actually working on behalf of Synlogic?”

A: “No, it’s a direct relationship, as far as I know.” – Current Synlogic executive

4. We believe that not only is ~100% of Ginkgo's reported Foundry revenue from related parties, but that the actual truth is far more nefarious and suggests that it is a hoax for the ages. Ginkgo is opaque about the percent of its revenue and deferred revenue which is cash versus non-cash. Based on interviews with its related-party "customers," we believe that at least half of Ginkgo's reported revenue is phantom – that is, non-cash and pure accounting hocus-pocus

Ginkgo's filings are opaque and make it difficult to answer a critical question: how much of its reported revenue – already a charade given its related-party dependence - is actually cash, versus non-cash “revenue” that is just an accounting fiction? The red flag is obvious: the revenue recognition policy in the prospectus spans a preposterous four pages and is impossible to decipher. In a real business earning revenue the old-fashioned way - sell a product or service to a customer who forks over cash – the rev rec policy doesn't typically run the length of the Iliad.

Revenue Recognition

We account for revenue in accordance with ASC 606, *Revenue from Contracts with Customers* (“ASC 606”). Under ASC 606, we recognize revenue when the customer obtains control of the promised goods or services, at an amount that reflects the consideration we expect to receive in exchange for those goods or services. To determine revenue recognition for arrangements that are within the scope of ASC 606, we perform the following five steps: (i) identify the contract(s) with a customer, (ii) identify the promises and distinct performance obligations in the contract, (iii) determine the transaction price, (iv) allocate the transaction price to the performance obligations in the contract, and (v) recognize revenue when (or as) we satisfy the performance obligations.

Foundry Revenue

We generate license and service revenue through the execution of license and collaboration agreements whereby customers obtain license rights to our proprietary technology and intellectual property for use in the research, development and commercialization of engineered organisms and derived products. Under these agreements, we typically provide R&D services, which includes the provision of a license to our intellectual property. Additionally, the customer obtains license rights to the output of our services in order to commercialize the resulting output of such services. Generally, the terms of these agreements provide that we receive some combination of: (i) Foundry usage fees in the form of (i) upfront payments upon consummation of the agreement or other fixed payments, (ii) reimbursement for costs incurred for R&D services and (iii) milestone payments upon the achievement of specified technical criteria, plus (2) downstream value share payments in the form of

(iv) milestone payments upon the achievement of specified commercial criteria, (v) royalties on sales of products from or comprising engineered organisms arising from the collaboration or licensing agreement and (vi) royalties related to cost of goods sold reductions realized by our customers.

Our collaboration and licensing agreements often contain multiple promises, including (i) licenses and assignments of intellectual property and materials and (ii) R&D services, and we determine whether each of the promises is a distinct performance obligation based on the nature of each agreement. As we are generally performing R&D services that are highly integrated and interrelated to the licenses and assignments of intellectual property and materials, the promises are generally inseparable. As such, we typically combine the R&D services, licenses, and assignments into a single performance obligation. However, for certain agreements, we only grant licenses or effects such transfers and assignments upon the successful completion of the R&D services or delivery of a developed product. For these agreements, we typically consider (i) the R&D services and (ii) the licenses, transfers, and assignments as distinct performance obligations, as each is transferred separately and has a separately identifiable benefit. Options to acquire additional goods and services are evaluated to determine if such options provide a material right to the counterparty that it would not have received without entering into the contract. If so, the option is accounted for as a separate performance obligation. If not, the option is considered a marketing offer which is accounted for as a separate contract upon the counterparty's election.

At contract inception, we determine the transaction price, including fixed consideration and any estimated amounts of variable consideration. Any upfront cash payment received upon consummation of the agreement is fixed and generally nonrefundable. Variable consideration is subject to a constraint, and amounts are included in the transaction price to the extent that it is probable that a significant reversal in the amount of cumulative revenue recognized will not occur when the uncertainty associated with the variable consideration is subsequently resolved. Variable consideration may include reimbursement for costs incurred for our R&D efforts, milestone payments upon the achievement of certain technical and commercial criteria, and royalties on sales of products from or comprising engineered organisms arising from the agreement. With respect to the R&D reimbursements and milestone payments, we use the most likely amount method to estimate variable consideration. With respect to agreements that include royalties on sales or other contingent payments based on sales, we apply the royalty recognition constraint which requires a constraint until the royalty or value-sharing transaction occurs. Certain agreements contain payment in the form of equity or other non-cash consideration. Any non-cash consideration is measured at the fair value of the non-cash consideration at contract inception.

For agreements with promises that are combined into a single performance obligation, the entire transaction price is allocated to the single performance obligation. For agreements with multiple performance obligations, the transaction price is allocated to the performance obligations using the relative standalone selling price methodology. For agreements featuring variable consideration, we allocate variable consideration to one or more, but not all, performance obligations if certain conditions are met. Specifically, we assess whether the variable consideration relates solely to our efforts to satisfy the performance obligation and whether allocating such variable consideration entirely to the performance obligation is consistent with the overall allocation objective. If these conditions are not met, we allocate the variable consideration based on the relative standalone selling price methodology. The key assumptions utilized in determining the standalone selling price for each performance obligation include development timelines, estimated R&D costs, commercial markets, likelihood of exercise (in the case of options considered to be material rights), and probabilities of success.

For agreements where the licenses or assignments are considered separate performance obligations or represent the only performance obligation, we recognize revenue at the point in time that we effectively grant the license as the licenses or assignments represent functional intellectual property. For agreements where the licenses and the R&D services represent a combined performance obligation, we recognize revenue over the period of performance based on costs incurred to date as compared to total estimated costs.

We evaluate our measure of progress to recognize revenue at each reporting period and, as necessary, adjust the measure of performance and related revenue recognition. Our measure of performance and revenue recognition

involves significant judgment and assumptions, including, but not limited to, estimated costs and timelines to complete our performance obligations. We evaluate contract modifications and amendments to determine whether any changes should be accounted for prospectively or on a cumulative catch-up basis. We utilize the right to invoice practical expedient when we have a right to consideration in an amount that corresponds directly with the value of our performance to date.

Royalties received under the agreements are recognized as revenue when sales have occurred as we apply the sales or usage-based royalties recognition constraint. We have determined the application of this exception is appropriate because the license granted in the agreement is the predominant item to which the royalties relate.

As we receive upfront payments for technical services under certain of our arrangements, we evaluate whether any significant financing components exist given the term over which the fees will be earned may exceed one year. Based on the nature of our agreements, there are no significant financing components as the purpose of the upfront payment is not to provide financing, but rather to secure technical services, exclusivity rights, and Foundry capacity, or the timing of transfer of those goods or services is at the discretion of the customer.

Deferred revenue represents consideration received by us in excess of revenue recognized and primarily results from transactions where we receive upfront payments and non-cash equity consideration. In instances where we have received consideration in advance for an undefined number of technical development plans (“TDPs”) under our customer agreements, we record the advance payments as deferred revenue, net of current portion on our consolidated balance sheets. Upon the execution of a specific TDP, we reclassify the estimated consideration to be earned under that TDP within the next twelve months as current deferred revenue. We also classify unexercised material rights as deferred revenue, net of current portion on our consolidated balance sheets. When a TDP is executed, and the material right is exercised, the amount allocated to the material right, which will be earned within the next twelve months, is reclassified to current deferred revenue. All other deferred revenue is classified as current or non-current based on the timing of when we expect to earn the underlying revenue based upon the projected progress of activities under the TDP.

Biosecurity Revenue

In 2020, we launched our commercial offering of COVID-19 testing products and services for businesses, academic institutions, and other organizations. In the first quarter of 2021, we launched our pooled testing initiative which focuses on providing end-to-end COVID-19 testing services to groups of individuals, with a focus of offering pooled testing services for K-12 schools. We sell COVID-19 test kits on a standalone basis or as part of an end-to-end testing service. We record product revenue from sales of LFA diagnostic test kits. We record service revenue from sales of our end-to-end COVID-19 testing services, which consist of multiple promised goods and services including sample collection kits, physician authorizations, onsite test administration, outsourced laboratory PCR analysis, and access to results reported through a web-based portal. We recognize our product and service revenue using the five-step model under ASC 606.

Product revenue from the sale of LFA diagnostic test kits is recognized when the test kits are shipped, and risk of loss is transferred to the carrier. Our diagnostic test kits are generally not subject to a customer right of return except for product recalls under the rules and regulations of the FDA. We have elected to include shipping and handling fees billed to customers as a component of Biosecurity revenue.

Service revenue from our end-to-end COVID-19 testing services is recognized upon completion of the tests and release of the test results on the web-based portal. We have identified one performance obligation in our testing services contracts that represents a series of distinct goods or services that are substantially the same and that have the same pattern of transfer to the customer, with each test as a distinct service within the series. As the price for the testing services is fixed under each customer contract, we have elected the practical expedient to recognize revenue at the amount which we have the right to invoice for services performed. Our testing services contracts are generally one year or less in length and contain fixed unit pricing. Under typical payment terms for

We were struck by two other red flags. First, the losses and cash burn are so unusually high and margins are so shockingly negative – even for the genre of scams and promotes - that we asked, isn't there SOME cash coming in to dampen this dumpster fire? Second, we noticed discrepancies between cash flows and income, typically a sign of games being played. A Capital IQ search for healthcare companies globally >\$100MM revenue and >\$500MM market cap yielded about 500 results – and only 2 companies with margins worse than Ginkgo.

<i>in millions USD</i>	2019	2020	LTM 6/30/21
Revenue	54.2	76.7	133.1
Research & development	96.3	159.8	208.9
Operating income	(71.6)	(137.0)	(207.4)
Net income	(119.3)	(126.6)	(200.5)
Operating cash flow	(44.7)	(135.8)	(167.7)
Capital expenditure	(22.2)	(57.8)	(94.0)
Free cash flow	(66.9)	(193.6)	(261.7)
EBITDA	(60.8)	(123.2)	(187.0)
R&D as % of revenue	177.7%	208.4%	156.9%
Operating margin	-132.1%	-178.7%	-155.8%

We further noticed a handful of alarming sentences buried in the prospectus – ones suggesting that Ginkgo’s reported revenue is a mix of cash and non-cash consideration, essentially admitting that Ginkgo is using medieval alchemy to transform equity investments in related-parties into “revenue.” For example, Ginkgo reported ~\$12MM of deferred revenue as of 3/31/21 with related-party “customer” Kalo Ingredients – the prospectus states the deferred revenue is 100% non-cash.

Excerpts from prospectus

Certain agreements contain **payment in the form of equity or other non-cash consideration**. Any non-cash consideration is measured at the fair value of the non-cash consideration at contract inception.

Deferred revenue represents consideration received by the Company in excess of revenue recognized and primarily results from transactions where the Company receives upfront payments and **non-cash equity consideration**.

The Company’s collaborative arrangements with its investees and related parties typically include upfront payments consisting of cash or **non-cash consideration for future research and development services** and non-cash consideration in the form of equity securities for licenses that will be transferred in the future. The Company records the upfront cash payments and fair value of the equity securities as deferred revenue.

The common stock received under the Motif IP Agreement is considered non-cash consideration and has been recognized at fair value. The Company determined the fair value of the common stock was \$65.1 million at inception of the agreement with the assistance of **a third-party valuation specialist, which was initially recorded as non-current deferred revenue**.

As of June 30, 2021, the Company had a **deferred revenue balance of \$11.9 million with Kalo, consisting of the non-cash consideration received**.

Our worst fears were confirmed when we spoke with Genomatica, one of Ginkgo's key related-party "customers" – which provided a smoking gun leading us to conclude that most of Ginkgo's reported revenue is non-cash hocus-pocus where equity investments are round-tripped back via "R&D credits." Ginkgo reported \$9.4MM of revenue from Genomatica in 2020 and \$3.3MM in Q1 2021 – **16% percent of its total Foundry revenue.** Genomatica is also one of Ginkgo's largest sources of deferred revenue - \$38MM in 2019 and \$30MM in 2020, **26% and 23% of total deferred revenue** in each of those years, respectively. **A senior Genomatica employee told us clearly and unequivocally that they have never paid Ginkgo cash for foundry R&D services,** and are simply using R&D credits provided after large investments by Viking and Ginkgo into Genomatica.

Senior employee of Ginkgo states their foundry R&D "spend" with Ginkgo is not cash consideration

A: **"But I think what's also clear, and you'd understand this, we're not spending cash.** So, appropriately, they put GAAP revenue, but **there's no cash revenue that we're spending over at Ginkgo right now because we have credits that go back from 2018 when they got \$40 million worth of equity in Genomatica** as part of our Series B and in exchange, they gave us a bunch of R&D credits.

Q: "So, they booked \$6.2 million of revenue in 2019 and \$9.4 million in 2019 and 2020. And you're saying **you gave them zero cash?"**

A: **"Correct."**

Q: "When was the last time you gave them any cash?"

A: "I will say this, we did provide them some cash last year, which is more of a legacy the arrangement that we previously had where there was revenue sharing from deals that we had gotten."

Q: "But that wasn't R&D spend?"

A: "Foundry service—correct. You're right, exactly. That's what I was going to clarify. Just to be clear, yes, there was some cash that Genomatica had to provide to Ginkgo. **In terms of R&D services revenue that Genomatica has paid cash for, we have not paid any cash because we have a bank of credits that we have,** and we still have quite a lot of credits, like \$10 million that we need to spend." – Senior Genomatica employee

At three others points in our call, the Genomatica employee re-iterated that they have not paid Ginkgo cash for foundry R&D services, stating that they don't need to pay cash until they "run out of credits" and that they have not paid cash since 2016. When we asked what Ginkgo is booking as revenue from Genomatica then, the employee admitted that "it's like Intrexon" – the mother of synthetic biology scams, which used the same model of round-tripping from related-party to create revenue out of thin air. We corroborated the finding with a second senior employee of Genomatica, and a third who is an ex-employee.

*"There will come a time that once we run out of using those credits, to continue using the foundry services, **we would have to pay cash for that, but we're not at that point yet.**"* – Senior Genomatica employee

*"So, then it all made sense for us to do this deal. Financially, what we got was tens of millions of R&D services sort of prepaid. What did they get in exchange? They got equity, and we got our partnership opened up so that we could go pursue whatever we wanted to... **we're not in a position where we need to spend cash until we run out of credits.**"* – Senior Genomatica employee

Q: **'And since 2016, have you given them any cash for foundry R&D services?'**

A: **"No, because we never needed to,** so the relationship was set up in a different sort of way like we were collaborating.

Q: "What did they book as revenue?"

A: **"What they book as revenue from Genomatica is, yeah, so they book what I understand to be, they book GAAP revenue from—and this is what's a little bit like Intrexon. This is the part where I'll agree with you, it's like Intrexon, but I'll be clear where I think there's a difference."** – Senior Genomatica employee

"The question is, will that foundry deliver results for those companies. We have the luxury at this point, through an arrangement that we felt was a great deal for us in 2018 that gave us the luxury to say, if the Ginkgo bet pays off, meaning if they pour hundreds of millions of dollars into building this foundry and if that high-throughput capability and all that stuff ends up working out to where it is cheaper and faster for us to get that R&D done at Ginkgo as a CRO, then great, we'll go do it. And, by the way, **we don't have to spend cash to go do that, to figure that out until probably, certainly not this year and probably not next year; we don't have to spend cash.**"

– Senior Genomatica employee

The senior employee conveyed an additional point we found particularly damning: if Genomatica had to pay cash for Ginkgo's R&D services, they would not do so – all but giving away the heart of the scam, that related-party “customers” are engaging in kabuki theater, where equity investments are round-tripped back via low-value, sham projects they throw Ginkgo's way. The employee described Ginkgo as simply a CRO, a contract research organization, and one that they don't trust to pass meaningful projects to: “we don't expose the critical path stuff”; “they haven't proven themselves yet as a CRO service provider”; when it comes to paying cash: “At the moment today, we would have a hard time justifying that.”

“Would have a hard time justifying” spending cash at Ginkgo equivalent to R&D credits spend to date

A: “I think the real critical question that if I were in your shoes, **the kind of question I would wonder is - would we spend the amount of cash today at Ginkgo that we've spent on R&D credits** to get the result that we've gotten?”

Q: **“And the answer sounds like no.”**

A: **“That's right. At the moment today, we would have a hard time justifying that**... You know what? Hell, they just went and raised a billion-plus dollars. If they want to pour that into their foundry and they want to create an engine that can reduce costs and ultimately, whether it's one year, two years, three years from now, they can do something that would be a hell of a lot cheaper than having our people here do it; we're happy to pay the money. We don't need to make that decision today.” – Senior Genomatica employee

Ginkgo hasn't proven itself as a CRO

“Again, **we don't expose the critical path stuff** yet, very specifically because we don't want to yet put ourselves in a position—**they haven't proven themselves yet as a CRO service provider** to where we would put ourselves in that position. If they can deliver at the level that we need, both time, quality, all that sort of stuff, we'll be happy to give them more work. **Are we there yet? No.** Are we open-minded to see if they can get there? Sure.” – Senior employee of Genomatica

Genomatica illustrates how Ginkgo injects cash into related-party “customers,” with Ginkgo investors like Viking, Casdin, and others playing a central role. Viking appears to be the largest holder in Genomatica. The Genomatica employee indicated that Ginkgo invested \$15MM in 2016, followed by Viking leading a \$60-70MM round, and Casdin leading one after that. We have not verified these amounts but note articles that corroborate the information.

Article indicates Viking is Genomatica’s largest holder and Casdin led a round

Bioengineering export Genomatica has announced it has issued \$90 million in its latest equity offering, a move which is said to allow the company to ‘fully execute on its business plan and fuel new growth opportunities’, which include sustainable technologies.

Contributors to the funding include Casdin Capital, the new lead investor, Viking Global Investors, the company’s largest shareholder, and organism engineering partner Ginkgo Bioworks.

Ginkgo appears to have invested \$15MM in Genomatica, and Viking apparently \$60-70MM

“Ginkgo’s cash investment in Genomatica was simply the \$15 million back in 2016. **Viking has invested—let me do quick math. Let’s say somewhere between \$60 and 70 million**, something like that.” – Senior employee of Genomatica

Viking is largest holder in Genomatica

Q: **“So, Viking is your biggest holder by percentage and Ginkgo second?”**

A: **“Yes.** Ginkgo is a little bit north of Casdin; they’re close to Novo”– Senior Genomatica employee

Ginkgo introduced Genomatica to Casdin Capital, which led another round

“This is 2016. Ginkgo invested in August/September, and then it was maybe about four or five months later, then Viking came in. About two years later, we did another round where we got Casdin Capital; you may be familiar with them. They invested and led a round in Genomatica. We got introduced to them through Ginkgo. I think they had \$2 million they’d invested in Ginkgo at some point along the way, but then they led a round in Genomatica, and they invested like \$15-20 million.” – Senior Genomatica employee

A second example of Ginkgo's reliance on non-cash revenue is its related-party "customer" Allonnia, where we found a smoking gun as damning as Genomatica. We believe Allonnia is a shell with almost no employees, as we detail elsewhere in this report. Nonetheless, Ginkgo reported an absurd \$38MM of deferred revenue from Allonnia as of Mar 31, 2021, on top of \$5MM of current period revenue for 2020. Allonnia is unusual in that its financials are appended as the last few pages of the prospectus, where the critical information is buried. **We begin with the first fact to keep in mind - that Allonnia was formed in late 2019, and completed a \$33MM Series A financing that press releases suggest was led by Ginkgo, Viking, General Atlantic and other Ginkgo backers.**

Allonnia financials indicate a \$33MM financing in late 2019

Allonnia, LLC

Consolidated Statements of Cash Flows

For the Year Ended December 31, 2020 and Period from November 27, 2019 (Inception) Through December 31, 2019

	2020	2019
Cash flows from operating activities:		
Net loss	\$ (6,011,518)	\$ (24,644,497)
Adjustments to reconcile net loss to net cash used in operating activities:		
Fair value common stock issued for acquiring intellectual property	—	24,480,000
Changes in operating assets and liabilities:		
Prepaid expenses and other current assets	(3,012,433)	(3,384)
Accounts payable	704,874	23,706
Accrued expenses	783,936	117,848
Net cash used in operating activities	(7,535,141)	(26,327)
Cash flows from financing activities:		
Proceeds from issuance of Series A-1 Preferred Units, net of issuance costs	18,227,011	32,855,000
Net cash provided by financing activities	18,227,011	32,855,000
Increase in cash and cash equivalents	10,691,870	32,828,673
Cash and cash equivalents, beginning of period	32,828,673	—
Cash and cash equivalents, end of period	\$ 43,520,543	\$ 32,828,673

Simultaneous with the Allonnia financing, Ginkgo round-tripped back \$25MM as R&D spend, which it recorded as deferred revenue per a disclosure in its related-party footnote. Allonnia's financials show \$25MM in R&D expenses in 2019 – matching the entry in Ginkgo's disclosure. However, Allonnia's cash balance at the end of 2019 still remained at \$33MM – the exact amount of its initial financing in December. This proves that the \$25MM in deferred revenue that Ginkgo booked from Allonnia as of Dec 31, 2019 was non-cash. Had Allonnia's \$25MM of R&D spend with Ginkgo been cash, its cash balance would have declined by \$25MM from \$33MM to \$8MM. Instead, it didn't change at all.

Allonnia financials indicate its R&D spend with Ginkgo was non-cash

Allonnia, LLC

Consolidated Statements of Operations

For the Year Ended December 31, 2020 and Period from November 27, 2019 (Inception) Through December 31, 2019

	2020	2019
Operating expenses:		
Research and development	\$ 3,966,936	\$ 24,481,519
General and administrative	2,106,245	162,978
Total operating expenses	6,073,181	24,644,497
Net operating loss	(6,073,181)	(24,644,497)
Interest income	61,663	—
Net loss	\$ (6,011,518)	\$ (24,644,497)

Assets

	2020	2019
Current assets:		
Cash and cash equivalents	\$ 43,520,543	\$ 32,828,673
Prepaid expenses and other current assets	5,015,817	3,384
Total assets	\$ 46,536,360	\$ 32,832,057

As a third example with another smoking gun – a buried but relatively easy one to find if one paid close attention – we note the case of Kalo Ingredients, another related-party “customer.” Kalo was founded in Mar 2021 and appears to be another shell with zero employees. That of course did not prevent Ginkgo from booking \$12MM of deferred revenue from it as of 3/31/21 – a rather quick round-trip as it was only formed two weeks prior on 3/15/21 per a Massachusetts corporate registry filing. **This time, the prospectus just spits it out: the entire \$11.9 deferred revenue is comprised of “non-cash consideration.”** The key sentence is literally the second to last one in a 3 page “Summary of Arrangement” and “Accounting Analyses” of Ginkgo’s Kalo “collaboration,” buried near the very end of the prospectus. The non-cash consideration received is of course the round-tripped investment in Kalo.

Kalo Ingredients disclosure in SPAC prospectus

“As of June 30, 2021, the Company had a **deferred revenue balance of \$11.9 million with Kalo, consisting of the non-cash consideration received.**”

As a fourth example, we examined Synlogic, another key related-party “customer” in which Ginkgo invested \$80MM and where we believe the revenue is also 100% non-cash. We have already discussed our belief that Ginkgo has misclassified Synlogic revenue, in order to make it appear as third-party vs. tainted related-party revenue. We had noted that Synlogic “paid” Ginkgo \$30MM out of the \$80MM right at closing, for “synthetic biology services,” which we think has been stuffed into a non-related-party bucket and is not broken out further. The language in Synlogic’s 10K suggests the same structure as Genomatica, where Synlogic is given pre-paid R&D credits to “pay” Ginkgo vs. using cash. Synlogic indicates they “used \$13.6MM of the pre-paid research and development expenses” in 2020, which we are almost certain was booked by Ginkgo as non-cash non-related-party revenue.

Synlogic 10K filed Mar 25, 2021

(16) Related-Party Transactions

In June 2019, the Company expanded its collaboration and entered into an agreement with Ginkgo for the research and development of engineered microbial therapeutic products. As of December 31, 2020, Ginkgo owns 6,340,771 shares of the Company’s outstanding common stock. See Note 10, *Ginkgo Collaboration* .

Under the agreement the Company made a prepayment to Ginkgo of \$ 30.0 million for its foundry services that will be provided to the Company over an initial term of five years . At December 31, 2020, the Company had remaining balances of \$ 4.7 million and \$ 8.9 million of current and non-current pre-paid research and development costs related to this transaction, respectively. For the year ended December 31, 2020, the Company used \$ 13.6 million of the pre-paid research and development expenses.

If we simply add up the examples on the last few pages, we estimate that non-cash revenue is provably ~50% of Ginkgo's total 2020 foundry revenue. We suspect that at related-party “customers” Motif and Joyn, Ginkgo may be round-tripping cash, and therefore potentially booking cash-revenue.

Foundry revenue by related party, as reported

<i>in millions USD</i>	2019	2020	Q1 2021
Foundry revenue by related party			
Joyn	9,349	7,273	1,598
Motif	18,986	20,798	5,492
Genomatica	6,248	9,431	3,298
Allonnia		4,960	2,266
Synlogic	17	73	6
Glycosyn	668	-	-
Kalo	-	-	-
Total	35,268	42,535	12,660
Total foundry revenue	54,184	59,221	22,504
Related parties as % of total foundry revenue	65%	72%	56%

We estimate non-cash portion as 47%

<i>in millions USD</i>	2020
Entities where smoking guns suggest revenue is all non-cash	
Genomatica	9,431
Allonnia	4,960
Synlogic	13,600
Total non-cash revenue	27,991
Total foundry revenue	59,221
Non-cash revenue as % of total foundry revenue	47%

Our analysis on the preceding pages is consistent with our interviews with ex-executives and employees of Ginkgo, who stated that real third-party customers – versus related-party fronts - were skeptical of Ginkgo's CRO services and simply didn't want to pay, forcing them to invent a round-tripping model with equity transformed into non-cash revenue. That is, Ginkgo had to resort to paying customers to use its services, with the payments recycled back: "nobody wanted to pay an upfront payment for a technology that they really didn't understand or believe in." The color indicated that third-party partners never paid much "when it came to money" and were just hype for financing, and that the actual downstream value capture from successfully engineered strains may have only been few hundred thousand dollars per year.

"Nobody wanted to pay an up-front payment"

"Many of these deals were significantly back-end loaded because nobody wanted to pay an upfront payment for a technology that they didn't really understand or believe in." - Former Ginkgo executive

Third party collaboration partners never paid much; just hype for financing.

"Many of these companies weren't out much, if anything, when it came to money. We signed these collaborations to get the name on the marquee so that Jason could use that for the next round of funding." - Former Ginkgo executive

"Successful strains" delivered to customers may only generate ~\$500K of revenue per year; customers are unhappy with them

"The first thing is that they don't have any unique technology that other companies don't have. The second is that they don't have any product. The products that they do have or what they call their products are strains that they've delivered to customers that generate revenue. But because there aren't many of those successful strains, I believe that their actual revenue coming from those strains is something like \$500,000 a year. Customers, if they're using their strains, are using them in a very limited capacity because they're not happy with them. And they're not using them for the original purposes. So, lack of platform uniqueness, lack of product." - Former employee

5. Former employees we interviewed indicated a widespread awareness of and concern inside Ginkgo about its related-party/round-tripping business model

Former employees we interviewed indicated a widespread awareness of and concern inside Ginkgo about its related-party/round-tripping business model; fear inside the company it may not be able to go public or could “be slammed by regulators”; indicated “confusion” at what Ginkgo counts as revenue and questioned if “money actually passed hands”; and suggested that concern around the circular flows was widespread internally.

Ex-employee indicated fear inside Ginkgo that it may not be able to go public or “be slammed by regulators” for its related party business model

A: **“There was fear that Ginkgo would not be able to go public, or it would be slammed by regulators for having too many of these little companies.”**

Q: “There was fear that it would be slammed by regulators for having too many related party entities?”

A: **“Yes, that they wouldn’t be allowed to go public** or something like that. Obviously, that didn’t end up being an issue.” – Former employee

Former employees were “confused” at what Ginkgo counts as revenue and questioned if “money actually passed hands”

“I was always confused. I’ve never done accounting, but some of the ways they count revenue, I was like, okay, if you say so. I don’t know if money actually passed hands because we now own another part of a company, and then they just get platform services for free, is that revenue?” – Former Ginkgo engineer

Ex-employee suggested widespread awareness and concern among Ginkgo’s employees around circular revenue and round-tripping

“There are probably three different divisions of people. The first is the people that completely got it [the circularity] and were open about it and were getting in trouble and getting fired. ..the second group is aware of it but trying to be more careful, but getting in hot water just by virtue of knowing about it and wanting to act ethically. In the second were people that were trying to be quiet but also planning to leave as soon as their stock options vested. And then the last group were people that were just completely buying into the hype, not wanting to hear one bad word against Ginkgo. I think in that group were people that had been there for six or seven years, so they literally had \$150 million of equity, and so I think they were almost bribed to be in that group. I think they were psychologically brainwashed by their huge equity grant. And then there were people in there that were just really stupid and didn’t see it.” – Former employee

A former director-level employee stated that Ginkgo pushed for control at its related-party entities, as their model depends on “obligating” them to round-trip R&D spend back; “really needs to control these entities” or the rig breaks down; “supposedly independent companies”; described these “customers” as “child companies of Ginkgo” and that “for sure Ginkgo controls all of those.”

Ginkgo pushed for control at each entity as their revenue model was based on round-tripping; Ginkgo was allowed to do what it wanted at Joyn; “supposedly independent companies”

“Ginkgo was kind of allowed to do what they wanted to do for the first period of the company [Joyn] at least, and that was really something that Jason Kelly and the group really sort of had pushed for, especially in Motif, at Allonnia, and the subsequent ventures, Ginkgo really needs to control these entities because if they can't then—what they want is for these subsidiaries or for these joint ventures, these supposedly independent companies to be obligated to use Ginkgo services, and Ginkgo services are very expensive. It's almost prohibitively expensive in some respects. But the idea is if you comparison shop, you can go elsewhere. It's actually kind of interesting like Joyn was doing a lot of work with this group with Culture BioSciences in South City, South San Francisco. And they're basically outsourcing their fermentation to them, not to Ginkgo. I would suspect that that was because of pricing and cost savings because they actually are not entirely obligated to use Ginkgo services, but that's the whole idea, is you're obligated to basically no matter what it costs. If Ginkgo says, pay us this, that's what you pay.” – Former director-level employee

Ex-employee described the related-party “customers” as “child companies of Ginkgo” and that “for sure Ginkgo controls all of those.”

“From a business perspective, they want all the revenue, and they want to control the entity, for sure. From a longer-term perspective, it really doesn't make sense for Ginkgo to call all of the shots because you really need to be producing long-term value. It reminds me a lot of RJ because, in the short term, everything looked amazing, but they never got a single project over the finish line, as far as I can tell. They couldn't complete this. And so, Ginkgo, to be successful, has to build long-term value. I know there's short-term, and now that they're a public company, definitely revenue expectations and things like that. But I think that—it's kind of interesting because the whole Ferment Consortium, this \$300 million slush fund to start child companies of Ginkgo, I would say for sure Ginkgo controls all of those almost certainly.” – Former director-level employee

A former employee in a managerial role called circular revenue flows “a big red flag”; indicated witnessing round-tripping and stated it’s more widespread than reported; believes that explains Ginkgo’s optically better headline growth than Zymergen and other comps; pointed to Ginkgo’s “structured partnership” with related-party Synlogic as an example of Ginkgo “registering their own money as revenue.”

Former employee called circular revenue flows “a big red flag”; indicated witnessing round-tripping and that it’s more widespread than reported; believes that explains Ginkgo’s optically better headline growth than Zymergen and others in the space

“Zymergen is not doing the circular revenue thing, but when I was there [at Ginkgo], I saw it being done on several projects, even ones that aren’t completely public yet. I would say it was being done; I would say, maybe even more often than not. I don’t know for sure the exact numbers, but I wouldn’t be surprised if 50% to 60% of Ginkgo’s revenue this year is that circular revenue. That, to me, explains a lot, like why are Amyris and Zymergen and Berkeley Lights struggling with such little revenue from the platform side, and yet Ginkgo has all this revenue. I think that can help explain a lot. Personally, I would not invest in a company that has mostly circular revenue. So, that’s one example; the circular revenue is a big red flag.” – Former Ginkgo employee in a managerial role

Former employee pointed to Synlogic deal as an example of Ginkgo “registering their own money as revenue”

A: “Before I start getting into some details, have you seen the MIT Technology Review article about Ginkgo?”

Q: “I have.”

A: **“So, you’re aware of the circular revenue issue?”**

Q: “Yeah, and I’d love to understand more about that.”

A: “The issue with the circular revenue with Synlogic, basically, they provided Synlogic \$30 million, Synlogic turned around and paid it to them, **so they were registering their own money as revenue**, and then they tried to make a strain for Synlogic, and it didn’t work, and then they had to go to Zymergen and get the strain. I think just two days ago; the clinical trials showed that the Zymergen strain was effective.” – Former Ginkgo employee in a managerial role

An ex-employee expects Ginkgo to “ratchet” up round-tripping and circular flows to show growth; alleged witnessing it personally and thinks “they’re doing it left and right”; alleged and was surprised to discover a round-tripping deal had been struck with a recently announced partner, without Ginkgo disclosing its investment in the entity and that it’s therefore a related party.

Ex-employee expects Ginkgo to “ratchet” up round-tripping and circular flows to show growth; suggested desperation; alleged witnessing round-tripping personally and thinks “they’re doing it left and right”

*“I’m guessing that moving forward, because of how disappointed their customers have been, **they’re going to have to ratchet that higher** to continue their growth rate. They’re probably; they’re quite desperate. **I saw it happen on deals**. The next thing I knew, the circular thing had happened with companies and I had no say in it. I think **they’re doing it left and right.**”* – Former employee

Ex-employee was surprised to discover that a round-tripping arrangement had been struck with a recently announced partner, without Ginkgo disclosing it in their filings; stated that Ginkgo made an alleged \$5MM Series A investment and further failed to disclose it in the press release announcing the partnership

A: *“I would [redacted] a company in the space in which [redacted...and **the next thing I knew, I got told that Ginkgo had invested in that company and they were paying it back so that Ginkgo could develop a strain for them. I don’t think that’s in their filings.**”*

Q: *“Do you know what the scale this was of investment and roundtripping?”*

A: *“**This particular one was about \$5 million.**”*

Q: *“Is this one that they’ve disclosed as far as you’re aware?”*

A: *“They did make an announcement of this partnership; **they did not say that it was circular**. All they said was that they had partnered with the company, but they didn’t say anything about the way that it was being funded.”*

Q: *“So, they put out a press release, but the press release did not state that they invested in them?”*

A: *“Nope. It just said that they were partnering. They contributed \$5 million to the Series A.”*

Q: *“Were there other examples of that?”*

A: *“Of the circular revenue?”*

Q: *“Yeah, that they didn’t announce. Are you saying that was typical—there were other examples of their doing that on a smaller scale and not disclosing it?”*

A: *“Definitely, yeah. **I saw a lot of stuff, but I have a limited view. I believe that this was being done far more widely.**”* – Former employee

- 6. We spoke with several former employees who indicated that they had been terminated for refusing to play along with the scheme, and who stated they knew of others pushed out for similar reasons**

A former executive at a key related-party “customer” alleges he was terminated because he resisted Ginkgo’s pressure round-trip R&D spend from the entity; asked to do things that made him uncomfortable; “some unethical things”; alleges others had similar experiences; knows of “several others” at the entity he believes left for similar reasons.

A former executive involved in a key collaboration left because he was asked to do something that made him uncomfortable; others former employees had similar experiences

Q: “Was there ever behavior there that you observed by the management team or whatever that made you uncomfortable?”

Were you ever asked to do something that made you uncomfortable?”

A: **“Yes, absolutely. That’s why I left. I’m sure you’re speaking to other people, and I’m probably not going to be the first one who said something like this.** I mean, especially if others are being honest with you. I’m not going to say a specific issue, but I have throughout my career built a high reputation, and I was asked to step in and do something that I wasn’t willing to do because it would damage my reputation personally as what I do with the long-term relationships that I’ve had with colleagues at different companies, and I wasn’t going to do that.” - Former Ginkgo executive

Alleges he was terminated because he was trying to act in the best interests of a related-party :customer,”, which was being pressured to round-trip R&D spend to meet “Ginkgo’s revenue expectations” from it

Q: “Let’s talk about Joyn Bio. It looks like it’s co-located with Ginkgo. And Ginkgo’s on the board, and they provided the funding. How is this a separate entity?”

A: **“[Chuckles] Well, it’s a good question. That’s one of the reasons that I no longer work there is because I was trying to act in Joyn Bio’s best interest, and we weren’t meeting Ginkgo’s revenue expectations.”** – Former Ginkgo executive

Executive wouldn’t share details but suggesting it involved inappropriate and “some unethical things” that were in Ginkgo’s interest and to “the detriment” of the related-party “customer”

“I really don’t want to go into any details on it, but I think you get the gist. They were looking out for their own good, and it was at their benefit and the detriment of the partner. Since they relied on me to bring in the partner and that partner trusted me, it was putting my neck on the line with that partner, and it happened several times, and **I finally said, “No, I’m not doing this...It was after collaborations were set up, some unethical things.”** - Former Ginkgo executive

Executive knows of others who left for similar reasons

“I know there were several others that left shortly before me and shortly after me. **I know some details,** but I don’t know all. I would guess it was for similar type reasons.”- Former Ginkgo executive

A second ex-employee in a senior role at another related-party “customer” indicated he “got axed” for pushing back on Ginkgo’s attempt to exploit the entity for fake work it didn’t need: “there was real pressure to produce, to do work in their foundry that would no real value”; Ginkgo had a “certain timeline” and “certain revenue expectations” from the “customer.”

Senior employee states he was fired for pushing back on Ginkgo’s attempt to exploit one of its key related party ventures for fake work it didn’t need, given that Ginkgo needed to show fictitious revenue from the entity

A: **“There were certain revenue expectations that the company had, that Ginkgo had for [key related party name redacted], and a certain timeline to achieve those.** Those revenue expectations basically started on day one of the company, and it was clear from the start that the company wasn’t ready for prime time. There wasn’t enough thought that had been put into the materials and the infrastructure that we needed to do the work. And so, **there was real pressure to produce, to do work in the Ginkgo foundry to produce revenue in their foundry that would have no real value for [key related party name redacted]. It was sort of make work** because there was an expectation that that work needed to be done. Like I balked at that, basically, and was like, I can’t do that. I’m not working for Ginkgo; it’s not in the best interest. We have to take two steps back and think this through. **I kind of got axed for that.”**

Q: **“You got axed for that?”**

A: **“Yeah.”** –Former Ginkgo employee

A third ex-employee, in a managerial role, was afraid to bring up concerns about circular revenue flows; suggested a colleague may have been fired for doing so, based on conversations with the colleague; indicated a personal awareness of three individuals who were allegedly terminated for pushing back on circular flows with entities.

Ex-employee indicated a personal awareness of three individuals being allegedly terminated for pushing back on round-tripping

Q: **"You're aware of other people that got in trouble or terminated for pushing back on this circular mechanism?"**

A: **"Yes."**

Q: "How many of those? Was it a person that you're aware of? Were there a lot of them?"

A: **"Personally, I'm aware of about three people, but I'm sure that there are more."**

Q: "So, these are people that pushed back, and they essentially let them go?"

A: "Yes, and they blamed it on performance. But all these people had really positive reviews and were very popular."

Q: "How confident are you that that's the reason they got let go? Is that you're reading between the lines? Or that was what these people told you, that was their personal interpretation of it?"

A: "Yeah, obviously, Ginkgo didn't come out and confirm it. It's definitely some aspect of this is just what I saw and what I think. I don't have anything in writing from Ginkgo or anything like that about why. **All I know is that there were people that were trying to speak, and then they were suddenly gone. I'm just sort of putting two and two together."**

Q: "Did you talk to any of those people that were speaking up, and they conveyed to you that that's why they felt they were gone, or you're just putting two and two together?"

A: **"I did talk to one of them who confirmed her strong suspicion that that's why she was gone. And the others, I was just putting two and two together."** –Former Ginkgo in a managerial position

Former employee was afraid to bring up concerns about circular revenue flows; suggested a colleague may have been fired for doing so

Q: "Did you ever try bringing up the circularity? **Our calls have indicated that people have been pressured to direct that revenue spend back to Ginkgo, and getting terminated for pushing back."**

A: "I'm not surprised by that. There was a woman named [redacted] who was suddenly terminated from Ginkgo. She was basically the head of [redacted]. She's very talented, and now she's the [redacted] at another biotech company; less than a month later, she got another position. Yeah, I'm not surprised by that. I've seen Ginkgo fire people randomly for no good reason. Yeah, it's a very delicate thing to bring up. I did not try to bring that up, the circular thing, because I was afraid to bring that up." –Former Ginkgo employee in a managerial role

7. The related party “customers” are mostly fronts for Ginkgo and its revenue is therefore an elaborate fiction.

Ginkgo discloses revenue and/or deferred revenue from four “Platform Ventures” – its term for de novo related party companies it created. It discloses two others that were just created a few weeks ago in September, buried at the end of its prospectus, but for which related party revenue hasn’t yet been reported, though it is no doubt inevitable.

<u>Platform Venture</u>	<u>Founded</u>	<u>Purpose</u>	<u>Initial capitalization</u>
Joyn Bio LLC www.joynbio.com	2017	Engineered microbes for agricultural uses	\$25MM from Ginkgo and “a related party” investor into holdco, Bayer up to \$60MM
Motif FoodWorks Inc. www.madewithmotif.com	2018	Substitutes for animal products	Series A \$119MM including “certain of our investors”
Allonnia LLC www.allonnia.com	2019	Microbes for waste remediation	Series A \$52MM including “certain of our investors”
Kalo Ingredients LLC www.kaloingredients.com	Mar 2021	Beauty and personal care ingredients	Series A \$77MM including “certain of ours investors”
Verb Biotics LLC www.verbbiotics.com	Sept 2021	Probiotic bacteria for nutrition and wellness	Series A \$22.5MM or \$30MM from Viking and Cascade – prospectus states various #'s
Ayana Bio LLC www.ayanabio.com	Sept 2021	“Bioactives” for health and wellness	Series A \$7.5MM or \$30MM from Viking and Cascade prospectus states various #'s

We examined each of these “Platform Ventures” in detail and believe they are simply fronts that Ginkgo and its investors created and funded in order to manufacture foundry revenue via round-tripping. Ginkgo’s filings repeatedly refer to each of these related parties as a “vendor-customer relationship.” However, given that Ginkgo 1) created each of these entities, 2) financed each of them directly or via its own investors, 3) sits on their boards, and 4) simultaneously recycles back funds via prepayments and other mechanisms under R&D services agreements, their purpose is self-evident.

We note the recurring role of Viking Global, a hedge fund that is Ginkgo’s largest investor, in funding related party entities at inception or at a later stage.

In addition to the initial USD \$100 million Series A investment by Bayer, Ginkgo, and Viking Global Investors LP, Ginkgo is providing access to its organism design technology, laboratory, and office spaces, as well as outfitting a new facility specifically for Joyn Bio. Bayer is providing exclusive access to its own archive of microbial strains and the necessary

Source: <https://www.prnewswire.com/news-releases/bayer-and-ginkgo-bioworks-unveil-joint-venture-joyn-bio-and-establish-operations-in-boston-and-west-sacramento-300616544.html>

Ginkgo Bioworks Launches Motif Ingredients with \$90 Million Series A Financing

Source: <https://www.prnewswire.com/news-releases/ginkgo-bioworks-launches-motif-ingredients-with-90-million-series-a-financing-300802134.html>

Ginkgo Bioworks' Ferment Consortium Launches Ayana

Bio to Produce Plant and Fungal Bioactives for Herbal Medicines and Nutraceuticals

Source: <https://www.prnewswire.com/news-releases/ginkgo-bioworks-ferment-consortium-launches-ayana-bio-to-produce-plant-and-fungal-bioactives-for-herbal-medicines-and-nutraceuticals-301372272.html>

Source: Soaring Eagle Acquisition Corp prospectus 8/13/2021, <https://www.sec.gov/Archives/edgar/data/0001830214/000119312521246097/d177007d424b3.htm>; Scorpion Capital analysis and estimates

Allonnia Launches from Ginkgo Bioworks' Ferment Consortium with \$40M Series A to Address Critical Problems in Waste Remediation with Synthetic Biology, Appoints

Source: <https://www.prnewswire.com/news-releases/allonnia-launches-from-ginkgo-bioworks-ferment-consortium-with-40m-series-a-to-address-critical-problems-in-waste-remediation-with-synthetic-biology-appoints-nicole-richards-as-ceo-301157826.html>

Ginkgo Bioworks' Ferment Consortium Launches Verb Biotics, a Probiotics Innovation

Source: <https://www.prnewswire.com/news-releases/ginkgo-bioworks-ferment-consortium-launches-verb-biotics-a-probiotics-innovation-company-to-address-market-demands-for-effective-probiotics-301372266.html>

Beyond who created them, funded them, governs them via board seats, and locks up their R&D spend via “Foundry Services Agreements,” “Technical Development Agreements,” and other control mechanisms, the related parties share other characteristics that reveal their de facto status as captive entities and subsidiaries. Claspings collars around their necks doesn’t appear to be sufficient - Ginkgo dispenses with all pretense of legitimacy and then proceeds to chain them to the doghouse.

Ginkgo’s related party “customers” share several troubling features

- 1. The related parties are housed inside Ginkgo’s headquarters** and use the same address. They are generally located inside the office itself and share office and lab space, and in some cases are a few floors above or below.
- 2. The websites for the entities are vaporous, vague brochures** similar to superficial Instagram posts. In some cases, the sites appear to be designed and run by Ginkgo, or are simply cut and pastes of the same design template.
- 3. The related parties generally have few or zero employees on LinkedIn**, and the actual employee counts bear no plausible relationship to the related party revenue and deferred revenue that Ginkgo books from them.
- 4. The entities typically lack finance, treasury, HR, legal**, or other G&A staff on LinkedIn, indicating they are neither designed to nor actually operate independently from Ginkgo.

We begin with Allonnia LLC, which was formed in 2019 and from which Ginkgo derived a comical \$38MM of deferred revenue as of Mar 31, 2021. Allonnia's Massachusetts corporate filing lists Ginkgo's corporate headquarters as its "Principal Office." Three of the five "Managers" declared are Jason Kelly, Ginkgo's CEO; Jason Kakoyiannis, whose LinkedIn bio states is Ginkgo's Head of Ventures; and Peter Lafer, who appears to work for General Atlantic, one of Ginkgo's largest investors.

Corporations Division
Business Entity Summary

ID Number: 001425004 [Request certificate](#) [New search](#)

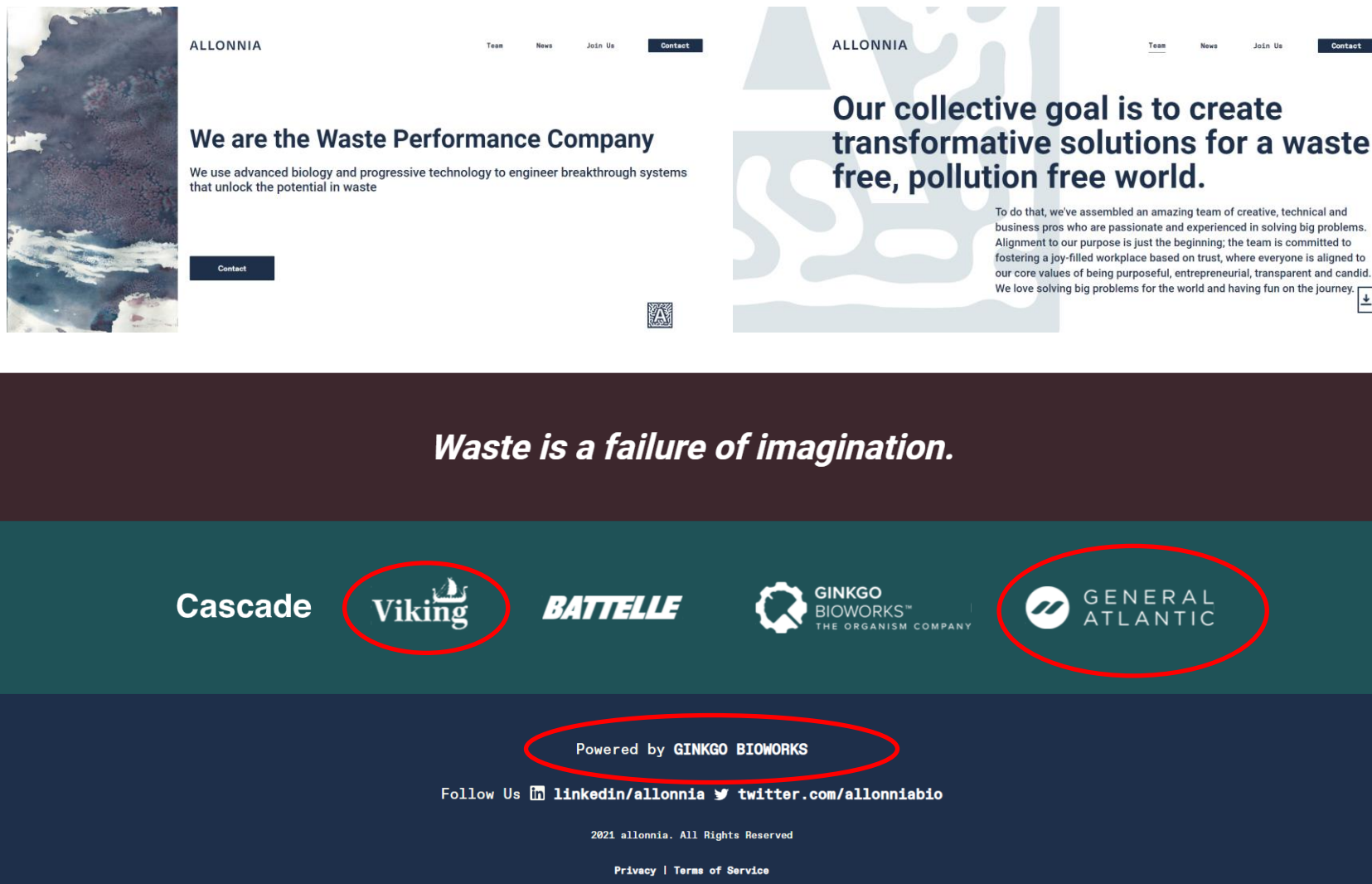
Summary for: ALLONNIA LLC

The exact name of the Foreign Limited Liability Company (LLC): ALLONNIA LLC																		
Entity type: Foreign Limited Liability Company (LLC)																		
Identification Number: 001425004																		
Date of Registration in Massachusetts: 02-10-2020																		
Last date certain:																		
Organized under the laws of: State: DE Country: USA on: 11-27-2019																		
The location of the Principal Office:																		
Address: 27 DRYDOCK AVE. FLOOR 8 C/O GINKGO BIOWORKS, INC																		
City or town, State, Zip code, Country: BOSTON, MA 02210 USA																		
The location of the Massachusetts office, if any:																		
Address: 27 DRYDOCKS AVE. FLOOR 8 C/O GINKGO BIOWORKS, INC																		
City or town, State, Zip code, Country: BOSTON, MA 02210 USA																		
The name and address of the Resident Agent:																		
Name: CORPORATION SERVICE COMPANY																		
Address: 84 STATE STREET																		
City or town, State, Zip code, Country: BOSTON, MA 02109 USA																		
The name and business address of each Manager:																		
<table border="1"> <thead> <tr> <th>Title</th> <th>Individual name</th> <th>Address</th> </tr> </thead> <tbody> <tr> <td>MANAGER</td> <td>MATTHEW L. VAUGHN</td> <td>27 DRYDOCK AVE. FLOOR 8 BOSTON, MA 02210 USA</td> </tr> <tr> <td>MANAGER</td> <td>JASON KELLY</td> <td>27 DRYDOCK AVE. FLOOR 8 BOSTON, MA 02210 USA</td> </tr> <tr> <td>MANAGER</td> <td>PETER LAFER</td> <td>27 DRYDOCK AVE. FLOOR 8 BOSTON, MA 02210 USA</td> </tr> <tr> <td>MANAGER</td> <td>NICOLE RICHARDS</td> <td>27 DRYDOCK AVE., FLOOR 8 BOSTON, MA 02210 USA</td> </tr> <tr> <td>MANAGER</td> <td>JASON KAKOYIANNIS</td> <td>27 DRYDOCK AVE. FLOOR 8 BOSTON, MA 02210 USA</td> </tr> </tbody> </table>	Title	Individual name	Address	MANAGER	MATTHEW L. VAUGHN	27 DRYDOCK AVE. FLOOR 8 BOSTON, MA 02210 USA	MANAGER	JASON KELLY	27 DRYDOCK AVE. FLOOR 8 BOSTON, MA 02210 USA	MANAGER	PETER LAFER	27 DRYDOCK AVE. FLOOR 8 BOSTON, MA 02210 USA	MANAGER	NICOLE RICHARDS	27 DRYDOCK AVE., FLOOR 8 BOSTON, MA 02210 USA	MANAGER	JASON KAKOYIANNIS	27 DRYDOCK AVE. FLOOR 8 BOSTON, MA 02210 USA
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Principal Office listed as Ginkgo's corporate headquarters:
27 DRYDOCK AVE, FLOOR 8
BOSTON, MA 02210
C/O Ginkgo BIOWORKS INC.

Managers listed include Jason Kelly, Jason Kakoyiannis, and Peter Lafer.

Allonnia's website is a vaporous, minimalist postcard with sweeping declarations of its intent to change humanity. The website appears to be hosted by Ginkgo, stating "Powered by Ginkgo Bioworks." The site has little on it, but didn't forget to credit Viking Global and General Atlantic.



A LinkedIn search turns up only 6 employees at Allonnia, plus an intern and a random Chief of Staff who lists 2 other current employers, despite being in operation since 2019 and raising \$50MM in Oct 2020. Allonnia lists a CEO, three VP's of Business Development, a VP of Research, and a CTO – we could locate no dedicated finance, treasury, HR, or legal staff.



Nicole Richards · 3rd
Chief Executive Officer at Allonnia
Ridgewood, New Jersey, United States · [Contact info](#)



Chief Executive Officer

Allonnia · Full-time
Sep 2020 – Present · 1 yr 1 mo
Boston, Massachusetts, United States

Leading a company focused on creating transformative solutions for a waste free, pollution free world through leveraging the power of biotechnology and advanced engineering systems that degrade pollutants and upcycle waste.

Source: <https://www.linkedin.com/in/nicole-richards-3268b07/>



Bryne Gramlich · 3rd
Vice President Of Business Development at Allonnia
Calgary, Alberta, Canada · [Contact info](#)



Vice President Of Business Development

Allonnia · Permanent Full-time
Jan 2021 – Present · 9 mos
Calgary, Alberta, Canada

Allonnia is a Boston, MA based bioremediation company focused on waste management tackling some of the planet's most pressing waste pollution challenges, including wastewater treatment, soil remediation, and solid waste management and upcycling.

Source: <https://www.linkedin.com/in/brynegramlich/>



Rafael Mier · 2nd
Vice President, Business Development at Allonnia
Boston, Massachusetts, United States · [Contact info](#)



Vice President, Business Development

Allonnia
Mar 2021 – Present · 7 mos
Boston, Massachusetts, United States

Our mission: leverage the power of biotechnology and engineering systems to create transformational solutions for a world free of waste and pollution

Source: <https://www.linkedin.com/in/rafael-mier-b0b1b0/>



Mack Astorga · 3rd
Vice President, Business Development at Allonnia
Atlanta, Georgia, United States · [Contact info](#)



Vice President, Business Development

Allonnia · Full-time
Mar 2021 – Present · 7 mos
Boston, Massachusetts, United States

Responsible for the development and execution of a commercial strategy that contributes to Allonnia's goal of creating transformative solutions that leverage the power of biotechnology and engineering systems to degrade pollutants and upcycle waste.

Source: <https://www.linkedin.com/in/mack-astorga-68aa512a/>



Kent Sorenson · 3rd
Chief Technology Officer at Allonnia
Boston, Massachusetts, United States · [Contact info](#)



Chief Technology Officer

Allonnia
Jan 2021 – Present · 9 mos
Boston, Massachusetts, United States

Source: <https://www.linkedin.com/in/kent-sorenson-35513827/>



Siri Gawrisch · 3rd
Founder and CEO, SiriUs Insight & Strategy, LLC
St Paul, Minnesota, United States · [Contact info](#)



Founder and CEO

SiriUs Insight & Strategy, LLC
2020 – Present · 1 yr
St Paul, Minnesota, United States

A trusted force multiplier: offering Chief of Staff services to CEOs of emerging growth companies amplifying already talented driven CEO's efforts to grow successful businesses.



Chief Of Staff

Allonnia
Dec 2020 – Present · 10 mos

Serving as Chief of Staff to the CEO for Allonnia, a company focused on creating transformative solutions for a waste-free, pollution-free world through the power of biotechnology and engineered systems.



Partner

HealthNexus
2020 – Present · 1 yr

Serving as partner and force multiplier to the CEO for HealthNexus, a company focused on helping companies adapt, innovate, decide, and grow: making companies part of the big, bold world of connected healthcare.



Dayal Saran · 3rd
Vice President and Head of Research at Allonnia
Boston, Massachusetts, United States · [Contact info](#)



Vice President Research

Allonnia · Full-time
Jul 2020 – Present · 1 yr 3 mos

Source: <https://www.linkedin.com/in/dayal-saran-43483417/>



Kristi McNary · 3rd
Strategic Marketing Analyst at Allonnia
Greater Boston · [Contact info](#)



Strategic Marketing Analyst Intern

Allonnia
Jun 2021 – Present · 4 mos

Source: <https://www.linkedin.com/in/kristimcnary/>

Nor does Allonnia appear to have any plans to grow. It lists only two open positions on its website as well as on LinkedIn – again, a striking fact for a “independent” company that raised \$50MM in Oct 2020, and one from which Ginkgo already reports a \$38MM deferred revenue balance as of Mar 31, 2021. We note Ginkgo also booked actual revenue of \$5MM from Allonnia in 2020, and \$2.3MM in Q1 2021 – an annualized \$9.2MM run rate.

2 job postings on Allonnia website

Career Opportunities

SynBio Project Manager

BOSTON, MA

[Learn more](#)

Company Description & Purpose Allonnia is a newly launched technology company focused on using biology to unlock the potential in waste. Allonnia is backed by Battelle, Ginkgo Bioworks, Viking Global, General Atlantic, Evok, and Cascade Investments. Allonnia is finding new answers to the growing challenge of waste remediation and upcycling enabled by Ginkgo Bioworks' advanced synthetic biology...

Business Development Director - Government

REMOTE - BOSTON, MA

[Learn more](#)

Company Description & Purpose Allonnia is a newly launched technology company focused on using biology to unlock the potential in waste. Allonnia is backed by Battelle, Ginkgo Bioworks, Viking Global, General Atlantic, Evok, and Cascade Investments. Allonnia is finding new answers to the growing challenge of waste remediation and upcycling enabled by Ginkgo Bioworks' advanced synthetic biology...

Source: <https://allonnia.topgradinghire.com/>

Same 2 job postings listed on LinkedIn

Allonnia has 2 job openings - find the one for you.

[Search](#)

Recently posted jobs



Business Development Director - Government

Allonnia
Boston, MA



SynBio Project Manager

Allonnia
Boston, MA

Source: <https://www.linkedin.com/company/allonnia/jobs/>

The striking absence of dedicated finance, treasury, HR, legal, or other G&A staff in LinkedIn searches makes it apparent to us that Allonnia is controlled and run by Ginkgo as a captive entity, with no purpose we believe except to create fake revenue. A few weeks ago in August, Allonnia appears to have hired - or was assigned by Ginkgo - a temp CFO, a year after raising a large Series A financing by Ginkgo, Viking Global, General Atlantic and others. The CFO appears to serve the same role for 3 entities, one of which is Motif – another Ginkgo related party. We find it unusual and troubling that sophisticated investors such Viking and General Atlantic would sponsor a \$50MM financing with an entity that appears to have lacked a finance, control, or treasury function.



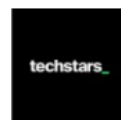
Jessica Kates · 2nd

Managing Partner at Rellevant Partners | Interim CFO at Techstars |
Interim CFO at Motif FoodWorks

New York, New York, United States · [Contact info](#)

Source: <https://www.linkedin.com/in/jesskates/>

Experience



Interim CFO

Techstars

Sep 2021 – Present · 1 mo



CFO Advisor

Allonnia

Aug 2021 – Present · 2 mos



Interim CFO

Motif FoodWorks, Inc.

Dec 2020 – Present · 10 mos

As additional evidence that Allonnia is merely a front operated by Ginkgo, we note a disclosure buried in the final pages of the prospectus regarding an unusual “G&A Services Agreement” between Ginkgo and Allonnia where Allonnia pays Ginkgo for unspecified general and administrative services, at Ginkgo’s “fully burdened FTE cost.” The disclosure is a blatant admission of Ginkgo’s day-to-day control and operation of Allonnia. We further note that Allonnia expensed an astounding \$2.1MM of G&A in 2021 – presumably siphoned by Ginkgo - despite having almost no employees. Allonnia is the only related party that filed separate footnotes, but we suspect that similar but undisclosed G&A agreements exist with other “customers” as a control mechanism.

G&A Services Agreement between Ginkgo and Allonnia

3. LICENSE, RESEARCH AND DEVELOPMENT AND CONSULTING AGREEMENTS

Ginkgo BioWorks, Inc.

On December 18, 2019, the Company and Ginkgo BioWorks, Inc. (“Ginkgo”) entered into a) an intellectual property contribution agreement (“IPCA”), b) a Common Unit Issuance Agreement (“CUIA”), c) a G&A Services Agreement for general and administrative services (“G&A Services Agreement”), and d) a Technical Development Agreement (“TDA”), collectively referred to as the Ginkgo Agreements.

Pursuant to the IPCA, Ginkgo licensed and contributed certain intellectual property to the Company in exchange for up to 9,000,000 Common Units of the Company, as specified in CUIA (see Note 4). The Company issued 3,600,000 Common Units as consideration for the license upon execution of the CUIA, the fair value of which was determined to be \$24,480,000 at inception of the agreements using the option pricing method. The Company accounted for the non-cash consideration related to the CUIA as research and development expense in the period ended December 31, 2019.

Pursuant to the G&A Services Agreement, Ginkgo will perform its obligations as an independent contractor and the Company will pay Ginkgo for services at the fully burdened full time equivalent cost, plus pass-through costs, materials and services.

“Pursuant to the G&A Services Agreement, Ginkgo will perform its obligations as an independent contractor and the Company will pay Ginkgo for services at the fully burdened full time equivalent cost, plus pass-through costs, materials and services.”

Allonnia income statement indicates \$2.1MM if G&A expenses in 2020

	2020	2019
Operating expenses:		
Research and development	\$ 3,966,936	\$ 24,481,519
General and administrative	2,106,245	162,978
Total operating expenses	6,073,181	24,644,497
Net operating loss	(6,073,181)	(24,644,497)
Interest income	61,663	—
Net loss	\$ (6,011,518)	\$ (24,644,497)

Related party “customer” Kalo Ingredients LLC exhibits the same pattern as Allonnia and is another front. Formed on March 15, 2021, its MA corporate filings suggest that it is effectively a Ginkgo subsidiary. It lists Ginkgo’s corporate headquarters as its “Principal Office” and the three “Managers” listed are Jason Kelly, Ginkgo’s CEO; Jason Kakoyiannis, Ginkgo’s Head of Ventures, who is also on the Allonnia board; and Jasmina Aganovic, an Entrepreneur In Residence at Ginkgo per LinkedIn, who may be an MIT pal of Kelly’s.

HOME DIRECTIONS CONTACT US

Corporations Division

Business Entity Summary

ID Number: 001501343 [Request certificate](#) [New search](#)

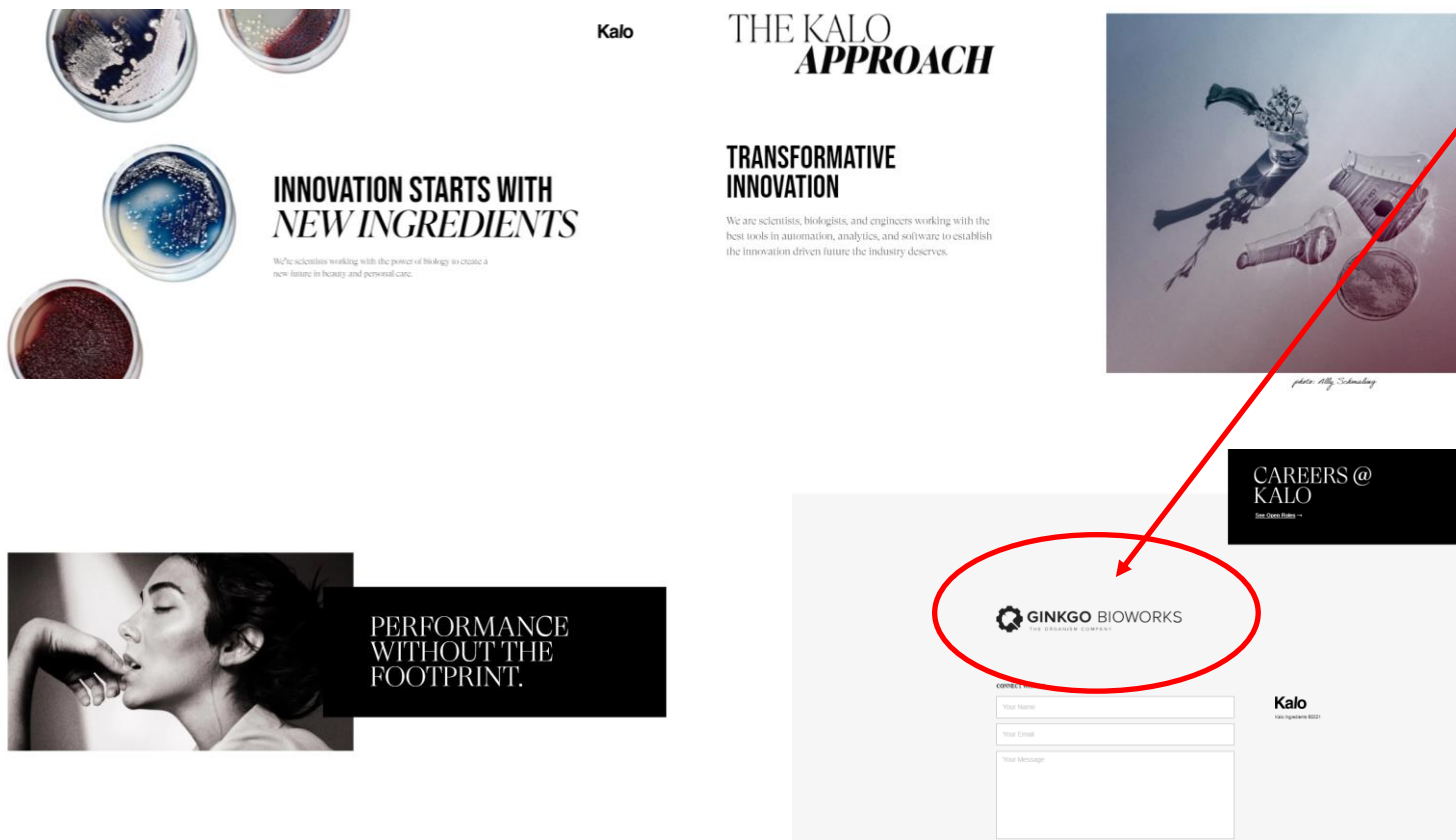
Summary for: KALO INGREDIENTS, LLC

The exact name of the Foreign Limited Liability Company (LLC):	KALO INGREDIENTS, LLC	
Entity type:	Foreign Limited Liability Company (LLC)	
Identification Number:	001501343	
Date of Registration in Massachusetts:	04-14-2021	
Last date certain:		
Organized under the laws of:	State: DE Country: USA on: 03-15-2021	
The location of the Principal Office:		
Address: C/O GINKGO BIOWORKS, INC. 27 DRYDOCK AVE., FLOOR 8		
City or town, State, Zip code, Country: BOSTON, MA 02210 USA		
The location of the Massachusetts office, if any:		
Address: C/O GINKGO BIOWORKS, INC. 27 DRYDOCK AVE., FLOOR 8		
City or town, State, Zip code, Country: BOSTON, MA 02210 USA		
The name and address of the Resident Agent:		
Name: GINKGO BIOWORKS, INC.		
Address: 27 DRYDOCK AVE., FLOOR 8		
City or town, State, Zip code, Country: BOSTON, MA 02210 USA		
The name and business address of each Manager:		
Title	Individual name	Address
MANAGER	JASMINA AGANOVIC	C/O GINKGO BIOWORKS, INC. 27 DRYDOCK AVE., FLOOR 8 BOSTON, MA 02210 USA
MANAGER	JASON KELLY	C/O GINKGO BIOWORKS, INC. 27 DRYDOCK AVE., FLOOR 8 BOSTON, MA 02210 USA
MANAGER	JASON KAKOYIANNIS	C/O GINKGO BIOWORKS, INC. 27 DRYDOCK AVE., FLOOR 8 BOSTON, MA 02210 USA

**Principal Office listed as Ginkgo’s corporate headquarters:
C/O Ginkgo BIOWORKS INC.
27 DRYDOCK AVE, FLOOR 8
BOSTON, MA 02201**

Managers listed are Jason Kelly, Jason Kakoyiannis, and Jasmina Aganovic

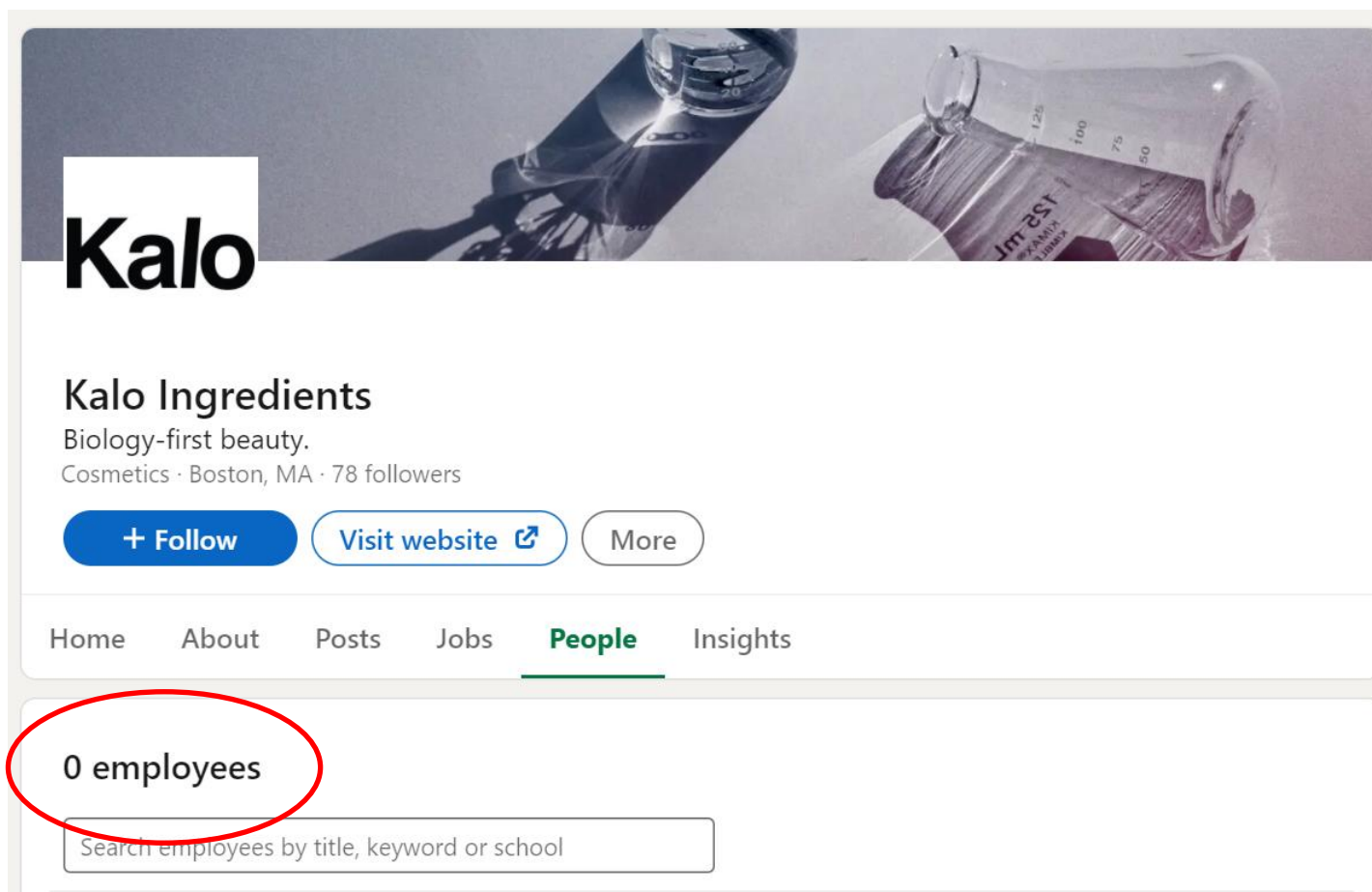
Similar to Allonnia, Kalo's website is only 2 pages, with no substantive content and only a few Instagram-type images and sweeping, meaningless declarations: “Lets have open conversations to lead the industry we want to create”; “Transformative Innovation”; “An Ecosystem to Build Together”; “Built Around Biology, Not Chemistry.” Once again, the bottom of the site suggests it is hosted by Ginkgo.



The URL on the Kalo jobs page is further evidence that Kalo is simply a front that is controlled and run by Ginkgo, as it points to Ginkgo's applicant tracking system.

<https://jobs.lever.co/ginkgobioworks?lever-origin=applied&lever-source%5B0%5D=GINKGOBIOWORKS.COM&team=Kalo%20Ingredients%20%28new%20Beauty%20%26%20Personal%20Care%20spinout%29>

Ginkgo reported \$12MM of deferred revenue from Kalo as of 3/31/21, and the SPAC prospectus states that Kalo completed a \$77MM Series A financing from Ginkgo's investors, which appears to have closed in March 2021 as well. However, we could locate no Kalo Ingredients employees on LinkedIn. The only LinkedIn profile that even mentions it is that of Jason Kakoyiannis, Ginkgo's Head of Ventures who lists Kalo as a board position.



Kalo

Kalo Ingredients
Biology-first beauty.
Cosmetics · Boston, MA · 78 followers

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[Home](#) [About](#) [Posts](#) [Jobs](#) [People](#) [Insights](#)

0 employees

Search employees by title, keyword or school

Kalo's site lists 7 job openings. The company appears to lack a CEO and is devoid of any management at all, much less employees. None of the job postings are for roles that suggest a standalone, independent company – no open roles for VP Research & Development or Chief Scientific Office, CFO, or any finance, HR, or G&A staff.

Chief Technology Officer (CTO), Kalo Ingredients

BOSTON, MA KALO INGREDIENTS (NEW BEAUTY & PERSONAL CARE SPINOUT)

APPLY

Computational Biologist

BOSTON, MA KALO INGREDIENTS (NEW BEAUTY & PERSONAL CARE SPINOUT)

APPLY

Research Associate

BOSTON, MA KALO INGREDIENTS (NEW BEAUTY & PERSONAL CARE SPINOUT)

APPLY

Senior Research Scientist – Polymers & Proteins

BOSTON, MA KALO INGREDIENTS (NEW BEAUTY & PERSONAL CARE SPINOUT)

APPLY

Senior Research Scientist Skin Microbiome

BOSTON, MA KALO INGREDIENTS (NEW BEAUTY & PERSONAL CARE SPINOUT)

APPLY

VP of Marketing, Kalo Ingredients

REMOTE/BOSTON, MA KALO INGREDIENTS (NEW BEAUTY & PERSONAL CARE SPINOUT)

APPLY

VP of New Brands, Kalo Ingredients

REMOTE/BOSTON, MA KALO INGREDIENTS (NEW BEAUTY & PERSONAL CARE SPINOUT)

APPLY

Ginkgo created two new related parties in September 2021, Verb Biotics LLC and Ayana Bio LLC. Verb raised \$22.5MM or \$30MM from Ginkgo's investors – the filings are inconsistent on the exact figure. Ayana Bio appears to have raised \$7.5MM or \$30MM from the same group. Neither has filings available yet in the MA corporate registry, though can assume they are also housed inside Ginkgo – although “they” is a misnomer if a company has no employees. The websites for both are half a screen long and cut and pastes of each other. Neither has any employees on LinkedIn nor any job postings we can locate.

verbbiotics

We identify and design new strains of probiotic bacteria for nutrition, health and wellness.

Get More Information →

<input type="submit" value="Submit"/>	



AYANABIO

Harnessing nature's bioactives for use as complementary medicine to support human health and wellness.

We are coming soon. Get More Information →

<input type="submit" value="SUBMIT"/>	



Related party Joyn Bio LLC was formed in 2017 and its MA corporate filings yet again list Ginkgo's corporate headquarters as its "Principal Office." The managers listed are Jason Kelly, Ginkgo's CEO; Reshma Shetty, Ginkgo's co-founder; the head of the venture; and two individuals from Bayer. LinkedIn and our interviews with employees of join suggest Joyn has ~60-70 employees. While it appears more substantive on the surface as an entity engaged in legitimate - albeit failed - R&D, later sections discuss why we believe that Ginkgo uses it as just another front to create fictitious revenue.

Corporations Division
Business Entity Summary

ID Number: 001289929 [Request certificate](#) [New search](#)

Summary for: JOYN BIO LLC

The exact name of the Foreign Limited Liability Company (LLC): JOYN BIO LLC																		
The name was changed from: COOKSONIA OPCO LLC on 02-02-2018																		
Entity type: Foreign Limited Liability Company (LLC)																		
Identification Number: 001289929																		
Date of Registration in Massachusetts: 09-08-2017																		
Last date certain:																		
Organized under the laws of: State: DE Country: USA on: 08-18-2017																		
The location of the Principal Office:																		
Address: 27 DRYDOCK AVE., SUITE 8																		
City or town, State, Zip code, Country: BOSTON, MA 02210 USA																		
The location of the Massachusetts office, if any:																		
Address: 27 DRYDOCK AVE., SUITE 8																		
City or town, State, Zip code, Country: BOSTON, MA 02210 USA																		
The name and address of the Resident Agent:																		
Name: C T CORPORATION SYSTEM																		
Address: 155 FEDERAL ST., STE 700																		
City or town, State, Zip code, Country: BOSTON, MA 02110 USA																		
The name and business address of each Manager:																		
<table border="1"> <thead> <tr> <th>Title</th> <th>Individual name</th> <th>Address</th> </tr> </thead> <tbody> <tr> <td>MANAGER</td> <td>MICHAEL MILLE</td> <td>27 DRYDOCK AVE., SUITE 8 BOSTON, MA 02210 USA</td> </tr> <tr> <td>MANAGER</td> <td>JASON KELLY</td> <td>27 DRYDOCK AVE., SUITE 8 BOSTON, MA 02210 USA</td> </tr> <tr> <td>MANAGER</td> <td>RESHMA SHETTY</td> <td>27 DRYDOCK AVE., SUITE 8 BOSTON, MA 02210 USA</td> </tr> <tr> <td>MANAGER</td> <td>JUERGEN ECKHARDT</td> <td>27 DRYDOCK AVE., SUITE 8 BOSTON, MA 02210 USA</td> </tr> <tr> <td>MANAGER</td> <td>ROBERT REITER</td> <td>27 DRYDOCK AVE., SUITE 8 BOSTON, MA 02210 USA</td> </tr> </tbody> </table>	Title	Individual name	Address	MANAGER	MICHAEL MILLE	27 DRYDOCK AVE., SUITE 8 BOSTON, MA 02210 USA	MANAGER	JASON KELLY	27 DRYDOCK AVE., SUITE 8 BOSTON, MA 02210 USA	MANAGER	RESHMA SHETTY	27 DRYDOCK AVE., SUITE 8 BOSTON, MA 02210 USA	MANAGER	JUERGEN ECKHARDT	27 DRYDOCK AVE., SUITE 8 BOSTON, MA 02210 USA	MANAGER	ROBERT REITER	27 DRYDOCK AVE., SUITE 8 BOSTON, MA 02210 USA
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MANAGER	ROBERT REITER	27 DRYDOCK AVE., SUITE 8 BOSTON, MA 02210 USA																

Principal Office listed as Ginkgo's corporate headquarters:
27 DRYDOCK AVE, FLOOR 8
BOSTON, MA 02201

Managers listed include Ginkgo founders Jason Kelly and Reshma Shetty.

Motif Foodworks Inc. was formed in 2018 and its MA corporate filings – you guessed it - list Ginkgo’s corporate headquarters as its “Principal Office.” The Board appears to be controlled by Ginkgo and a related party investor, as three of the five directors listed are Jason Kelly, Ginkgo’s CEO; Jason Kakoyiannis, head of Ginkgo’s Ferment Fund; and Chris Rivest, a partner at Ginkgo investor and related party Breakthrough Energy Ventures. A fourth director, Olu Beck, appears to be a figurehead board member for hire. Similar to Joyn, Motif has attempted to engage in legitimate R&D, with little to show for it, but we discuss how Ginkgo nonetheless uses it to round-trip revenue similar to other related parties.

Corporations Division
Business Entity Summary

ID Number: 001330100 [Request certificate](#) [New search](#)

Summary for: MOTIF FOODWORKS, INC.

The exact name of the Foreign Corporation:	MOTIF FOODWORKS, INC.	
The name was changed from:	MOTIF INGREDIENTS, INC. on 09-11-2019	
Entity type:	Foreign Corporation	
Identification Number:	001330100	
Date of Registration in Massachusetts:	05-30-2018	
Last date certain:		
Organized under the laws of:	State: DE Country: USA on: 04-20-2018	
Current Fiscal Month/Day:	12/31 Previous Fiscal Month/Day: 12/31	
The location of the Principal Office:		
Address:	27 DRYDOCK AVENUE, 8TH FLR	
City or town, State, Zip code, Country:	BOSTON, MA 02210 USA	
The location of the Massachusetts office, if any:		
Address:		
City or town, State, Zip code, Country:		
The name and address of the Registered Agent:		
Name:	RESHMA SHETTY	
Address:	27 DRYDOCK AVE, 8TH FLR	
City or town, State, Zip code, Country:	BOSTON, MA 02210 USA	
The Officers and Directors of the Corporation:		
Title	Individual Name	Address
PRESIDENT	JONATHAN MCINTYRE	27 DRYDOCK AVE., 8TH FLR, BOSTON, MA 02210 USA
TREASURER	PATRICK DEPRIEST	27 DRYDOCK AVENUE, 8TH FLR BOSTON, MA 02210 USA
SECRETARY	PAULA MICHELE FITE	27 DRYDOCK AVE, 8TH FLR BOSTON, MA 02210 USA
DIRECTOR	OLU BECK	27 DRYDOCK AVENUE, 8TH FLR BOSTON, MA 02210 USA
DIRECTOR	JASON KAKOYIANNIS	27 DRYDOCK AVENUE, 8TH FLR, BOSTON, MA 02210 USA
DIRECTOR	CHRIS RIVEST	27 DRYDOCK AVENUE, 8TH FLR, BOSTON, MA 02210 USA
DIRECTOR	JASON KELLY	27 DRYDOCK AVE., 8TH FLR, BOSTON, MA 02210 USA
DIRECTOR	JONATHAN MCINTYRE	27 DRYDOCK AVE., 8TH FLR, BOSTON, MA 02210 USA

**Principal Office listed as Ginkgo’s corporate headquarters:
27 DRYDOCK AVE, 8TH FLR
BOSTON, MA 02210**

Directors listed include Jason Kelly, Jason Kakoyiannis, and Chris Rivest.

- 8. Ex-Ginkgo employees as well as current employees of these related-party “customers” corroborated our findings that they are located at Ginkgo’s headquarters and are anything but arms-length vendor-customer relationships**

While Ginkgo claims that its related-party “customers” are arms-length vendor-customer relationships, former employees as well as current staff at these entities indicated the exact opposite: extensive “overlap and intermingling”; “a close association”; “specifically propped up” by Ginkgo: co-located in Ginkgo’s headquarters and “pretty much all together all the time”; murky situations where Ginkgo still paid staff even though they “worked” for the related-party “customer.”

Former direct report to CEO affirmed that the entities are intermingled and “specifically propped up”

Q: “It sounds like Joyn, Motif, and Allonnia, are they essentially pretty intermingled with Ginkgo?”

A: **“Yeah, they were all specifically propped up** because there was a very clear business opportunity that intersected with Ginkgo's core expertise.” - Former Ginkgo employee, longtime direct report to CEO Jason Kelly

The ex-employee continued that “there’s overlap and intermingling” and “a close association”

“Motif moved to the second floor of the building and built out their own food tech-specific lab. But there’s a close association and Jason's on the board of both companies, and Allonnia and stuff, too. What each subsequent company has done has been, like there's a good separation. **There's overlap and intermingling where it's really beneficial.”** - Former Ginkgo employee, longtime direct report to CEO Jason Kelly

Murky situations where Ginkgo still paying them even though they work for the related party

“What happened to prop up the companies? A few employees, like a technical person and maybe a business person who was going to eventually switch over, **would get seconded so that they keep benefits from Ginkgo, but then Joyn was paying them, or something like that.** And then, they fully switched over.... Ginkgo's PR firm does Joyn PR and Motif” - Former Ginkgo employee, longtime direct report to CEO Jason Kelly

Related parties are co-located in the same building and “pretty much all together all the time”

“Generally, they're co-located. Allonnia was just one or two floors above Ginkgo in the same Seaport building. Yeah, **they were pretty much all together all the time.**” - Former employee in a managerial role

Entities are co-located; they still conduct their research at Ginkgo

“In the building, these companies are co-located, and then eventually they're moved to their own place. So, Motif Foodworks now have their own space. **Joyn Bio still lives in Allonnia with Ginkgo...Generally, they are close by because the idea is that a lot of the research is still done at Ginkgo Bioworks.”** – Current Allonnia employee

An ex-director level employee described Ginkgo's modus operandi with related-party "customers." After Joyn Bio round-tripped less "revenue" back than it wanted, given Bayer's involvement in the venture, Ginkgo "crafted or architected" subsequent related-parties to prevent that from happening again. A current executive at Motif, a key related-party, stated that they and other related parties have had the same experience as Joyn: "Ginkgo wanted to pretty much control everything and be in charge."

Executive at related-party "customer" Motif indicates other related-parties have had the same experience, where Ginkgo attempts to exert control to divert and round-trip their R&D spend;

Q: "Do you know how any of the other platform ventures are doing, like Allonnia and Joyn? Do you talk to your colleagues there at all?"

A: "I talk to some colleagues; I talk to former colleagues, former employees of them. **They had the same experience as us. Joyn Bio's feedback is that Ginkgo wanted to pretty much control everything and be in charge.** Joyn Bio had some bio labs in there, but then they also use Ginkgo's platform. But they wanted to direct the science and the synthetic biology director back then was like, we have our own agenda and we want to do our own stuff, and we don't need Ginkgo to always be in the room. And so, they wanted to be a little more independent. And then also, **it didn't really pan so much after so many years with Joyn Bio either."**

Q: "Joyn was trying to go in a little more independent direction, and Ginkgo was trying to exert control to kind of keep their research spend, is that what you're saying?"

A: "Yes." – Current executive at Motif

After one of its related-parties – Joyn Bio – pushed back on Ginkgo control, given Bayer's involvement, Ginkgo "crafter or architected" subsequent entities to prevent that from happening

"So, I do think they [Joyn Bio] did try to push to be independent, to be fair. But **I do think that with the follow-on companies that they've announced since then have been crafted or architected in a different way to prevent that from happening because Ginkgo lost control of Joyn,** and I do think they still, I mean, they're still doing a significant chunk of work with them based on the revenue that they're getting from Joyn, but it's not actually the amount of work that they were anticipating to get." – Former director-level employee

Joyn Bio is a key related-party “customer” in which Bayer is also an investor. Multiple current employees at Joyn as well as ex-employees of Ginkgo described the day-to-operation as more like a subsidiary of Ginkgo or captive team within the company: two-thirds of Joyn’s staff is co-located inside Ginkgo’s headquarters in Boston; “fairly tight integration” and “an open space shared with Ginkgo”; “main labs are still co-located with Ginkgo”; “we’re in the foundry.”

2/3 of Joyn’s employees are in Ginkgo’s headquarters in Boston

Q: “Currently, Joyn in Boston, it’s basically cubicles or offices on a Ginkgo floor?”

A: “Yes, and **we’re in the foundry**...Right now, we’re at about 75 people; I don’t remember how many contractors, but 75 employees, and off the top of my head, **it’s about 2/3 Boston**, 1/3 in California.” – Current Joyn executive

Corroborated that 2/3 of Joyn’s employees appear to still be co-located with Ginkgo; “main labs are still co-located with Ginkgo”

Q: “Where are you based? Are you in Boston or on the west coast?”

A: **“Boston. The company’s main labs are still co-located with Ginkgo, and we’re moving to the lab space that’s not with them but in the same building.**

Q: “How big is Joyn Bio? How many people?”

A: “We’re about 70; we’ve got **about 2/3 on the east coast**, 1/3 in California.” – Current director-level employee at Joyn

Joyn had “an open space shared with Ginkgo” and had “fairly tight integration”

“It was, basically **an open space that Joyn shared with Ginkgo, and Joyn had access fairly deep into the foundry**, that is the level at which Joyn accessed, say our software, was relatively involved...**they had fairly tight integration with Ginkgo.**” - Former director-level employee

Staff of Ginkgo and Joyn could just “walk over across the land and show” each other the data

Q: “In which areas is the intermingling?”

A: “A lot of technical stuff. This stems from Joyn as a first try with this. Technically, you have people that are working in Ginkgo’s lab that are delivering work to Joyn’s technical team. **There’s something really nice and very expedient about being able to walk over across the lab and show someone the data** you just got. Like, “Is this what you’re looking for, for this experiment or not?” - Former Ginkgo employee

Comments by current Joyn employees and ex-Ginkgo employees indicates that Ginkgo and its related parties are intermingled; that the relationship is anything but arms-length; and resembles nothing like a typical vendor-customer arrangement - Ginkgo staff “got seconded over to Joyn”; “we’ve been intimately connected with them”; Joyn outsourced key functions to Ginkgo.

“We’ve been intimately connected with them”

Q: “Does the company use Ginkgo exclusively for its R&D and foundry services, or do you use other competitors to Ginkgo as well?”

A: “I don’t think we’re talking about that publicly, but **we do a lot of work with Ginkgo, I’ll say. We’ve been intimately connected with them**, and certainly, there are things that are outside their expertise, specific kinds of testing.” – Current Joyn executive

Ginkgo employees were seconded to Joyn

“For Joyn, what they did is **a number of Ginkgo’s technical people got seconded over to Joyn**, and a lot of them ended up becoming full-time employees there.” - Former Ginkgo employee, longtime direct report to CEO Jason Kelly

Outsourced key functions to Ginkgo

Q: “Why would a company that’s raised at least what appears to be \$50 million-plus - t’s kind of unusual to not have a CFO and all of those kinds of things. It’s a little bit unusual as a structure.”

A: “The goal for us is we’re running lean, so we’ve also grown pretty rapidly, so **it didn’t make sense early on to have those functions**. I think we’re going to certainly take a hard look—I should say, you can see publicly posted a job ad that we’re looking for a VP equivalent for HR. So, we are starting to fill out some of the functions. It’s a kind of a transitional point for us. **But it was a very intentional goal that we wanted to be really lean, and, in fact, if you think about the partnership with Ginkgo, it was a similar kind of thing. We didn’t set up DNA sequencing or synthesis. We didn’t set up robotics, all that. We’re leveraging them**. So, it was a consistent approach.” – Current Joyn executive

Our interviews indicated that Joyn staff have pushed back on Ginkgo's attempts at control by setting up a second office in California, but that Ginkgo made "a power move" by still keeping Joyn's engineering and synthetic biology function inside Ginkgo's headquarters – presumably the 2/3 of Joyn employees who are in Boston. Research sources indicated that while Joyn, unlike other related party "customers," wasn't "entirely run by Ginkgo" given Bayer's involvement, it still leveraged Ginkgo's staff and systems.

Joyn attempted to exert some independence by moving operations to California, but Ginkgo made "a power move" to keep all the engineering and synthetic biology in Boston

*"All of the engineering and that synthetic biology happens in Boston. So, there's that. **I do think that was a bit of a power move...**" - Former director-level employee*

Joyn "wasn't entirely run by Ginkgo" and was sort of independent given Bayer's 50/50 involvement, but still used Ginkgo staff and systems

*"Joyn was sort of set up as a, I mean, it was the first big joint venture, and it was set up to be independent. At the time, **Ginkgo HR facilitated things, and there was some outsourcing of those types of G&A functions to Ginkgo**, like they had kind of a part-time CFO, kind of independent contractor types that worked in different administration and executive functions. **So, it wasn't entirely run by Ginkgo, but it did leverage some kind of Ginkgo's people in those roles, and it did use the same software systems and things like that.**" – Former director-level employee*

We earlier quoted a current executive at Motif, a key related-party “customer,” who stated that they and other related parties have had the same experience: “Ginkgo wanted to pretty much control everything and be in charge.” He elaborated on Ginkgo’s attempts to run Motif like a front, chuckling when saying they’re “independent” of Ginkgo. He explained the role of a Ginkgo employee who was seconded to Motif as the director of R&D, and his role in funneling “R&D spend” back to Ginkgo – “The amount they charged us, there was no negotiation.”

Motif executive suggested Motif spend was controlled by a Motif “employee” who was acting in Ginkgo’s interest

Q: **“You sort of chuckled earlier when you said, yeah, we’re independent;** I think you said they try to exert control. I’m curious if you can elaborate on that - there’s always that push and pull.”

A: **“Exactly. I would say that the person that was the director of R&D, pretty much overseeing the project, was actually a Ginkgo employee at first, so he had really little say because he was an insider from Ginkgo, so he couldn’t really tell them what you’re doing is not the perfect thing.”** I came as an outsider. So, I don’t care who you are; yes, you are our founder, but I work for Motif, and I need to have the best result for Motif. And so, you can promise me everything or anything, but at the end of the day, you have to deliver commercial strains for us.”

Q: “So, you’re saying initially, the head of R&D at Motif was a Ginkgo employee?”

A: “Well, **he was a Motif employee, but he was a former Ginkgo employee, yes, that’s right.”**

Q: “For all intents and purposes, was he acting on behalf of Motif, or **was he there to make sure that Ginkgo got the right level of R&D spend from Motif?**”

A: “I would say that no matter what he was trying to do, if he was trying to be on the side of Motif, being an ex-Ginkgo employee and having worked on the Motif project, maybe it’s a little difficult to challenge them, I would say.”

Q: “Challenge them on what? The amount they wanted to charge? Or were they forcing you guys to use them?”

A: **“[Chuckles] Well, yes, all of the above. So, yes, the amount they charged us, there was no negotiation.”** I mean, the price was pretty much fixed back then by Ginkgo, I would say, very early on, and there was very little leeway and that’s changed because we are now using other syn-bio companies as well because we’re not exclusive. So, we can now negotiate a little more on the pricing. They are a stakeholder. They are an equity holder of our company; they founded us. So, their investment is really equity-based for us.” – Current Motif executive

Allonnia needs no further elaboration on its self-evident role as a front, given that it has almost no employees yet Ginkgo still reported a preposterous \$38MM of deferred revenue and \$5MM of revenue from it. Nonetheless, we find comments from ex-employees of Ginkgo to be striking, which unmistakably conveyed that Allonnia was property of Ginkgo. One said that “Allonnia was my baby” and then passed it on to Jason Kakoyiannis, Ginkgo’s Head of Ventures as well of its Ferment Fund that props up related-parties, adding that “they’re still trying to get it up off the ground.” The same person confirmed that it’s co-located inside Ginkgo – “The social stuff is nice, like a kitchen. That more the intention behind it.”

Run by Jason Kakoyiannis, head of Ginkgo’s Ferment Fund; unclear if operational

*“Allonnia was my baby, and then I passed it onto the person that’s now running Ginkgo’s investment vehicle called the Ferment Fund, which is actually just Ginkgo’s investors put money into a separate vehicle and said we’re going to keep co-investing in your spinoff, so here’s a pile of money you can pull from, so you don’t have to come back to us... **I think they’re still trying to get up off the ground.**” - Former Ginkgo employee*

Allonnia is co-located with Ginkgo; intention was to create intermingling

Q: *“So, Allonnia is co-located with Ginkgo, and they’re just trying to figure out what they want to do, basically?”*

A: *“**Yes. The co-location**, I should specify, the reason for that is not just like so-and-so is right down the hall; it’s like an incubator space. **When you start a company, you need office space; you need labs; social stuff is nice, like a kitchen. That was more the intention behind it.** instead of going to the Harvard labs, Ginkgo can just provide that.” - Former Ginkgo employee*

We spoke to a current employee of Allonnia, who stated that it “resides within Ginkgo”; that “all the research is still done at Ginkgo”; and appeared to suggest that it was part of Ginkgo irrespective of its official status “on paper.” The same employee confirmed a G&A services agreement between Allonnia and Ginkgo, but was cagey about discussing it further. We noted our belief earlier that the agreement ensures Ginkgo’s control over Allonnia’s day to day operation, while allowing it to continue the charade that it’s separate.

An Allonnia employee indicated that “Allonnia resides within Ginkgo”; “all the research is still done at Ginkgo”; appeared to suggest that it was part of Ginkgo irrespective of its official or “on paper” status

“I’m actually working with Ginkgo because all the research is still done at Ginkgo; Allonnia resides within Ginkgo. Saying that officially or on paper, I guess...Allonnia is a small company. Right now, there are six full-time employees...I think two of them were hired—the CEO was hired in October of 2020. So, officially, it’s less than a year-old company.” – Current Allonnia employee

The same employee confirmed a G&A services agreement between Ginkgo and Allonnia but was cagey about discussing it

Q: “Does Ginkgo provide any other managerial services to Allonnia, like G&A services and things like that?”

A: “In some cases, yes, but that is between Allonnia and Ginkgo. I think that may be confidential. **I don’t want to speak on that, but yes, they do provide G&A to some extent...My understanding is that in the beginning, they do help with the financial and the hiring and everything.**” – Current Allonnia employee

- 9. In the last two months, timed with its SPAC listing, Ginkgo has announced a flurry of new R&D partners, a dog whistle that the scam is about to hit overdrive. We believe these “partners” are undisclosed related parties and fronts.**

Ginkgo is making aggressive moves to expand the scale of its related-party business model, announcing a flood of new entities simultaneous with its public listing. The timing is not a coincidence. We think round-tripping is about to explode into overdrive, as Ginkgo now has a stock to support and insiders who need to dump. Without creating new “Platform Ventures” that it finances and uses as fronts, Ginkgo can’t show “growth.” We share color in a later section of the tension and conflict Ginkgo has begun to encounter with related parties as they add staff, who attempt to act as fiduciaries on behalf of the entity and push back on Ginkgo’s attempts to charge them fake or inflated R&D spend. It’s easier to simply book revenue from new entities with no apparent employees, such as Kalo – there’s no one to push back. Ginkgo’s 1H update, released Aug 18, tips off the strategy – promote topline growth and keep quiet on related parties or cash flows.

Ginkgo Bioworks Provides First Half 2021 Update

1H total revenue growth of 180%, Foundry revenue +41% - no mention of related party contribution or cash flows

First Half 2021 Unaudited Revenue Highlights & Revised 2021 Full Year Outlook

- First half 2021 Total Revenue of \$88 million, up from \$31 million in the comparable prior year period, an increase of 180%.
- First half 2021 Foundry Revenue of \$44 million, up from \$31 million in the comparable prior year period, an increase of 41%.

The number of new “partners” announced in the last few weeks just prior to the first day of trading on Sep 17th is stunning. The partnerships are for everything under the sun: GI drugs, probiotics, herbal medicines and nutraceuticals, “biomaterials,” dyes, a material inspired by spiders with “silk-like softness.” Notably, not one of the releases below discloses a deal size – and all are silent on the critical question of whether they are undisclosed related parties – that is, whether Ginkgo or its investors like Viking are investors in the entities or are providing proceeds that will be round-tripped back to Ginkgo as “revenue” in coming quarters. We shortly reveal a smoking gun that exposes the rig as exactly that.

Aug 5, 2021 - Bolt Threads

Bolt Threads to Leverage Ginkgo Bioworks' Platform for Commercial Strain Optimization of b-silk™ Protein

Aug 12, 2021 - Antheia

Antheia and Ginkgo Bioworks Announce Partnership to Accelerate Production of Essential Medicines using Synthetic Biology

Aug 19, 2021 - Huue

Huue and Ginkgo Bioworks Partner to Accelerate Production of Sustainable Dyes

Sept 2, 2021 – Cambium Biomaterials

Cambium Biomaterials Announces Partnership with Ginkgo Bioworks to Accelerate Industrial Adoption of High-Performance Biomaterials

Source: Ginkgo press release on its Investor News page, <https://investors.ginkgobioworks.com/news/default.aspx>

Sep 7, 2021 Tantu

Tantu and Ginkgo Bioworks Announce Partnership to Engineer a Living Biotherapeutic for Gastrointestinal Healing

Sep 9, 2021 – Verb Biotics

Ginkgo Bioworks' Ferment Consortium Launches Verb Biotics, a Probiotics Innovation Company to Address Market Demands for Effective Probiotics

Sep 9, 2021 – Ayana Bio

Ginkgo Bioworks' Ferment Consortium Launches Ayana Bio to Produce Plant and Fungal Bioactives for Herbal Medicines and Nutraceuticals

Sep 14, 2021 – Africa CDC

Africa CDC and Ginkgo Bioworks Announce Collaboration to Strengthen Laboratory Network Capacity Across African Union Member States

As we examined these entities, we were amused but not surprised by what we found. The press release for Ginkgo's Antheia partnership on Aug 12, 2021 makes no mention of the fact that Viking led Antheia's \$73MM financing a few weeks prior to the Ginkgo announcement.

Antheia and Ginkgo Bioworks Announce Partnership to Accelerate Production of Essential Medicines using Synthetic Biology

Source: <https://investors.ginkgobioworks.com/news/news-details/2021/Antheia-and-Ginkgo-Bioworks-Announce-Partnership-to-Accelerate-Production-of-Essential-Medicines-using-Synthetic-Biology/default.aspx>

***“Led by Viking Global Investors,
the Series B round...”***

Antheia Announces \$73M Series B to Commercialize Synthetic Biology Platform for Essential Medicines

Led by Viking Global Investors, the Series B round will be used to bring Antheia's first pharmaceutical compounds to market, expand its pipeline and scale production in global markets

NEWS PROVIDED BY
Antheia →
Jun 30, 2021, 08:00 ET

SHARE THIS ARTICLE



Source: <https://www.prnewswire.com/news-releases/antheia-announces-73m-series-b-to-commercialize-synthetic-biology-platform-for-essential-medicines-301322528.html>

The partnership with Huue was announced a week after Antheia. The release says that “Huue will leverage Ginkgo’s platform to optimize microbe strains to accelerate production of its sustainable indigo dye.” Huue appears to have only 8 employees based on a LinkedIn search, one lonely job opening, and appears to have raised a mere \$2.6MM two years ago - with presumably little left, not exactly indicative of momentum or product “acceleration” of the type that legitimate companies of Ginkgo’s size hype with releases 3 weeks prior to going public. It is self-evident that Huue could only “leverage Ginkgo’s platform” if it’s plied with cash that will be recycled back as fictitious revenue, and we question whether it’s even an active startup or a carcass being exploited as a front.

Huue appears to have only raised \$2.6MM via convertible note, prior to Ginkgo partnership

Huue and Ginkgo Bioworks Partner to Accelerate Production of Sustainable Dyes

AUGUST 19, 2021

BOSTON, Aug. 19, 2021 /PRNewswire/ -- [Ginkgo Bioworks, Inc.](#) (“Ginkgo”), which is building the leading horizontal platform for cell programming, today announced a new cell program in partnership with [Huue](#), a biotech startup that’s pioneering sustainable bio-based dyes for the fashion industry and beyond. Ginkgo, which recently announced a business combination with Soaring Eagle Acquisition Corp. (Nasdaq: [SRNG](#)), serves customers across industries seeking to develop new and better products using biology. Huue will leverage Ginkgo’s platform to optimize microbe strains to accelerate production of its sustainable indigo dye, pioneering new models for product-focused startups to collaborate with Ginkgo.

Source: <https://investors.ginkgobioworks.com/news/news-details/2021/Huue-and-Ginkgo-Bioworks-Partner-to-Accelerate-Production-of-Sustainable-Dyes-8-19-2021/default.aspx>

The screenshot shows the Crunchbase profile for Huue. The 'Highlights' section on the right lists 'Total Funding Amount' as '\$2.6M'. A red arrow points from the headline 'Huue and Ginkgo Bioworks Partner to Accelerate Production of Sustainable Dyes' to this funding amount. The 'Funding' section on the right provides more details: 'Huue has raised a total of \$2.6M in funding over 5 rounds. Their latest funding was raised on Feb 22, 2021 from a Convertible Note round. Huue is funded by 4 investors. M12 - Microsoft's Venture Fund and IndieBio are the most recent investors.'

Category	Value
Funding Rounds	5
Lead Investors	1
Investors	4
Total Funding Amount	\$2.6M

Source: https://www.crunchbase.com/organization/tinctorium/company_financials

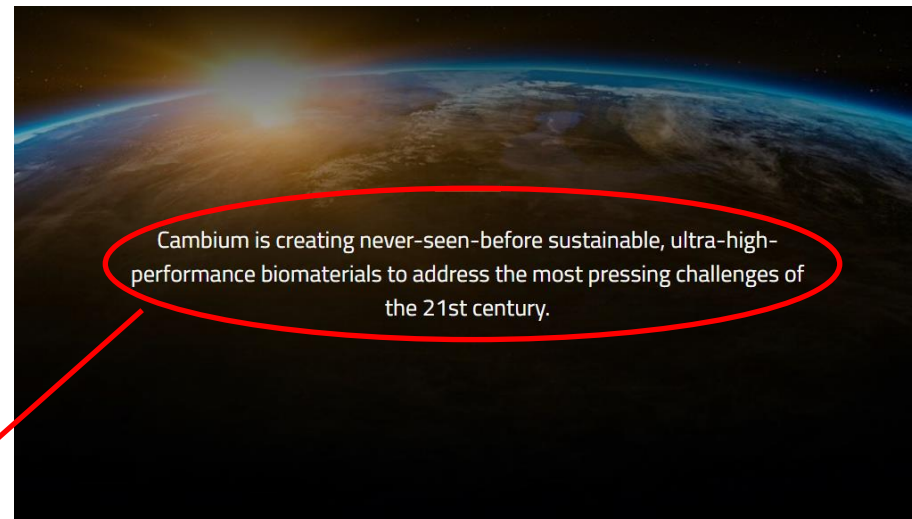
Two weeks after Huue, Ginkgo announced a partnership with Cambium Biomaterials where “the partnership will leverage Ginkgo’s Codebase and high-throughput facility” to “accelerate industrial adoption” of Cambium’s biomaterials. Small problem: Cambium appears to have never raised any funding and its site is a single page with vaporware like “Deep Tech approach, fusing biotechnology, green chemistry, computation, automated material science.” Its address appears to be a coworking space at a startup incubator.

Cambium Biomaterials Announces Partnership with Ginkgo Bioworks to Accelerate Industrial Adoption of High-Performance Biomaterials

SEPTEMBER 2, 2021

BERKELEY, Calif. and BOSTON, Sept. 2, 2021 /PRNewswire/ -- [Cambium Biomaterials](#) (“Cambium”), the company creating high-performance biomaterials, today announced a partnership with [Ginkgo Bioworks, Inc.](#) (“Ginkgo”), which is building the leading horizontal platform for cell programming. Ginkgo, which recently announced a business combination with Soaring Eagle Acquisition Corp. (Nasdaq: [SRNG](#)), serves customers across industries seeking to develop new and better products using biology. The partnership will leverage Ginkgo’s Codebase and high-throughput facility to accelerate Cambium’s discovery, development, manufacturing and commercialization of biomaterials that deliver superior performance and sustainability across a range of industries.

“Cambium is creating never-seen-before sustainable, ultra-high-performance biomaterials to address the most pressing challenges of the 21st century.”



Cambium is creating never-seen-before sustainable, ultra-high performance biomaterials to address the most pressing challenges of the 21st century.

***Address is a startup incubator:
626 Bancroft Way Suite A
Berkeley, CA 94710***

Address: 626 Bancroft Way Suite A
Berkeley, CA 94710

Cambium lists one job posting and has 3 employees on LinkedIn. The CTO appears to be the only full time employee. The CEO's bio suggests he's an operating executive at other start-ups. The third is based in Park City, UT, and is a documentary filmmaker currently working on "a character-driven, only-in-America story about the enduring consequences, politics, and polarization of school shootings."



Simon Waddington · 3rd
Entrepreneur and operating executive
Berkeley, California, United States · [Contact info](#)
500+ connections
Source: <https://www.linkedin.com/in/simon-waddington/>

Experience



Co-Founder and CEO
Cambium Biomaterials, Inc.
2019 – Present · 2 yrs
Berkeley, California

Cambium is creating never-seen-before, sustainable, ultra-high-performance biomaterials to address some of the most pressing challenges of the 21st century. We are disrupting traditional advanced materials research and development for fields such as aerospace, defense, healthcare, mobility, recreation and renewable energy. We achieve this through a Deep Tech approach, fusing biotechnology, green chemistry, computation, automated materials science, and an array of other scientific and engineering disciplines.



Operating Executive
Advising start-ups and venture capital firms in biotechnology and materials
2019 – Present · 2 yrs
San Francisco Bay Area



Stephan Herrera · 2nd
Entrepreneur, Filmmaker
Park City, Utah, United States · [Contact info](#)
Source: <https://www.linkedin.com/in/stephanherrera/>

Experience



Co-founder
Cambium Biomaterials, Inc.
Jul 2019 – Present · 2 yrs 3 mos
Berkeley, California

Cambium is creating never-seen-before, sustainable, ultra-high-performance biomaterials to address some of the most pressing challenges of the 21st century. We are disrupting traditional advanced materials research and development for fields such as aerospace, defense, healthcare, mobility, recreation and renewable energy. We achieve this through a Deep Tech approach, fusing biotechnology, green chemistry, computation, automated materials science, and an array of other scientific and engineering disciplines.



Co-founder
Two Paperboys Productions, LLC
Feb 2018 – Present · 3 yrs 8 mos
Park City, Utah

Feature-length documentary films. Current project: Executive Producer, Producer, & Co-writer of "The Student, The Survivor, The Shooter & The Sheriff" (working title) — a character-driven, only-in-America story about the enduring consequences, politics, and polarization of school shootings.



Andrew Guenther · 3rd
Chief Technology Officer at Cambium Biomaterials, Inc.
Berkeley, California, United States · [Contact info](#)
Source: <https://www.linkedin.com/in/andrewguenther/>

Experience



Chief Technology Officer
Cambium Biomaterials, Inc.
Sep 2019 – Present · 2 yrs 1 mo
Berkeley, CA

Cambium is creating never-seen-before, sustainable, ultra-high performance biomaterials to address some of the most pressing challenges of the 21st century. We are disrupting traditional advanced materials research and development for fields such as aerospace, defense, healthcare, mobility, recreation and renewable energy. We achieve this through a Deep Tech approach, fusing biotechnology, green chemistry, computation, automated materials science, and an array of other scientific and engineering disciplines.

We have already shared evidence indicating that Cambium is an undisclosed related party in which Ginkgo is an investor. As we detailed earlier, on Sep 17, 2021, Cambium filed a Form D stating that it sold \$8MM of equity on Sep 2, 2021, as part of a \$28MM round – the same day as Ginkgo’s misleading press release. The filing didn’t name the investor, so we spoke to an individual closely involved with Cambium - who stated that Ginkgo invested \$5MM as part of an \$11MM financing in conjunction with an R&D services agreement. The individual described a long-standing personal relationship between principals at both companies.

Excerpts from Cambium’s SEC Form D filing

Notice of Exempt Offering of Securities

1. Issuer's Identity

CIK (Filer ID Number) 0001883516
 Name of Issuer Cambium Biomaterials, Inc.
 Jurisdiction of Incorporation/Organization DELAWARE
 Year of Incorporation/Organization
☐ Over Five Years Ago
☒ Within Last Five Years (Specify Year) 2019
☐ Yet to Be Formed

Previous Names ☒ None

Entity Type
☒ Corporation
☐ Limited Partnership
☐ Limited Liability Company
☐ General Partnership
☐ Business Trust
☐ Other (Specify)

**Cambium Biomaterials Inc,
626 Bancroft Way Suite A,
Berkeley, CA 94710**

2. Principal Place of Business and Contact Information

Name of Issuer Cambium Biomaterials, Inc.
 Street Address 1 626 BANCROFT WAY, SUITE A
 City BERKELEY
 State/Province/Country CALIFORNIA
 ZIP/Postal Code 94710
 Phone Number of Issuer (415) 878-3752

Date of First Sale: 2021-09-02

7. Type of Filing
☒ New Notice Date of First Sale 2021-09-02 ☐ First Sale Yet to Occur
☐ Amendment

8. Duration of Offering
 Does the Issuer intend this offering to last more than one year? ☐ Yes ☒ No

9. Type(s) of Securities Offered (select all that apply)
☒ Equity
☐ Debt

13. Offering and Sales Amounts

Total Offering Amount \$28,014,379 USD or ☐ Indefinite
 Total Amount Sold \$8,012,877 USD
 Total Remaining to be Sold \$20,001,502 USD or ☐ Indefinite

\$28MM offering, \$8MM already sold

The Tantu partnership announced less than a week after Cambium is particularly grandiose: **“Tantu will leverage Ginkgo’s foundries to accelerate strain construction as a key step toward human clinical trials.”** Clinical trials presume an advanced program, typically after tens or hundreds of millions spent on R&D. Imagine our surprise after discovering that **Tantu appears to have no full-time employees. It doesn’t even appear to be a company, nor even a project – merely an idea.** Co-founder Neel Joshi is a full time associate professor at Northeastern; a second co-founder is currently CEO of another startup; a third was a post-doc in Joshi’s lab until August, when she took another job.

Tantu and Ginkgo Bioworks Announce Partnership to Engineer a Living Biotherapeutic for Gastrointestinal Healing

SEPTEMBER 7, 2021



Neel Joshi · 3rd
Co-founder at Tantu Therapeutics, Inc.
Cambridge, Massachusetts, United States · [Contact info](#)



John Tagliamonte · 3rd
Entrepreneur committed to restoring patients' lives.
Cambridge, Massachusetts, United States · [Contact info](#)

Experience

 **Co-founder**
Tantu Therapeutics, Inc.
Nov 2020 – Present · 11 mos
Boston, MA

 **Associate Professor**
Northeastern University · Full-time
Jan 2020 – Present · 1 yr 9 mos
Boston, MA

Experience

 **CEO**
Mellitus, LLC · Full-time
Dec 2020 – Present · 10 mos
Boston/Cambridge, USA
Advancing diabetes test GlycatedCD59 for better risk monitoring and treatment.

 **Co-Founder**
Tantu Therapeutics, Inc.
Nov 2020 – Present · 11 mos
Boston, Massachusetts, United States
Building company with biofilm patch to heal and locally treat gastro-intestinal lesions.



Anna Duraj-Thatte · 3rd
Assistant Professor of Biological Systems Engineering at Virginia Tech
Boston, Massachusetts, United States · [Contact info](#)

Experience

 **Assistant Professor of Biological Systems Engineering at Virginia Tech**
Virginia Tech
Sep 2021 – Present · 1 mo

 **Research Affiliate**
Massachusetts Institute of Technology
Nov 2020 – Present · 11 mos
Boston, Massachusetts, United States
Department of Nuclear Science and Engineering

 **Co-Founder**
Tantu Therapeutics, Inc.
Nov 2020 – Present · 11 mos
Boston, Massachusetts, United States

 **Associate in Bioengineering**
Harvard John A. Paulson School of Engineering and Applied Sciences
Jan 2020 – Present · 1 yr 9 mos

 **Neel Joshi Retweeted**



Anna Duraj-Thatte @AnnaDuraj · Jul 13

I am thrilled to share that I will be joining Department of Biological Systems Engineering @bse_vt at Virginia Tech @virginia_tech as an Assistant Professor in this Fall! In my lab we will strive to push the boundaries of Living Materials to new realms of Health & Environment!

Similar to Cambium, we suspected that Tantú is also an undisclosed related-party in which Ginkgo has or will make an investment which will be recycled back as “revenue.” We spoke with an individual closely involved with Tantú, who indicated that Tantú has raised \$50k to date via a university grant – “just enough to get a website and stuff like that and incorporate.” The individual stated that Tantú “doesn’t have money to pay” Ginkgo for R&D services, so “the deal ended up being like a deferred loan, essentially, for the work they are going to do.” The individual did not disclose the amount of funding Ginkgo provided, but indicated it was similar in magnitude to a Series A round.

Tantú had raised de minimis funding prior to Ginkgo

“It’s still very early stages. **Tantú has the tiniest bit of funding from a Northeastern accelerator fund**, but not really enough to meet the technical goals of a therapeutics company, **just enough to get a website and stuff like that and incorporate**. Apart from that, Tantú has some other momentum going in terms of winning a golden ticket for lab space at Lab Central in Kendall Square. There’s an incubator space.” – Individual closely involved with Tantú

The individual stated that Ginkgo is financing Tantú

I don’t think that this necessarily applies to all their partnerships, but Tantú is a special case, especially because **Tantú doesn’t have money to pay them. The deal ended up being like a deferred loan, essentially, for the work that they are going to do.**” – Individual closely involved with Tantú

10. Aside from its “customers” being related parties and effectively fake, we believe that Ginkgo takes the scam into even more aggressive territory by booking revenue from them that is simply fictitious, overstated, and/or based on overcharging.

Aside from its “customers” being related parties and effectively fake, we believe that Ginkgo takes the scam into even more aggressive territory by booking revenue from them that is simply fictitious, overstated, and/or based on overcharging. We begin with two observations: 1) the R&D and G&A services fees that Ginkgo charges these “customers” and books as revenue and/or deferred revenue bears no plausible relationship to their size, suggesting they are fraudulent; 2) the timing of when Ginkgo books revenue and/or deferred revenue is equally implausible, sometimes before the entities even appear to have any employees or be functional, which further indicates they are simply sham transactions. For example, LinkedIn indicates only 6 employees at Allonnia LLC, yet Ginkgo records a \$38MM deferred revenue balance, on top of booking \$5MM of revenue from Allonnia in 2020 and \$2.3MM in Q1 2021 – a \$9MM annual run rate.

Ginkgo revenue and deferred revenue from Allonnia LLC, per related party disclosures

<i>in millions USD</i>	2019	2020	Q1 2021
Foundry revenue from Allonnia		4,960	2,266
Deferred revenue balance from Allonnia at period end	24,480	26,064	37,959

	Joyn	Motif	Genomatica	Allonnia	Synlogic	Glycosyn	Total
For the Year Ended December 31, 2020							
Foundry revenue	\$ 7,273	\$ 20,798	\$ 9,431	\$ 4,960	\$ 73	\$ —	\$ 42,535
Other income, net	\$ 407	\$ 314	\$ —	\$ —	\$ —	\$ —	\$ 721

	Joyn	Motif	Genomatica	Allonnia	Synlogic	Total
Balances as of December 31, 2020						
Accounts receivable, net	\$ —	\$ 2,403	\$ 1,500	\$ 1,309	\$ —	\$ 5,212
Prepaid expenses and other current assets	\$ 24	\$ 232	\$ —	\$ 13	\$ —	\$ 269
Deferred revenue, current and non-current	\$ 9,862	\$ 53,952	\$ 30,128	\$ 26,064	\$ 72	\$ 120,078

Aside from the overall amount of revenue and deferred revenue being implausible – we estimate \$6.3MM of deferred revenue per Allonnia employee - the timing of when Ginkgo booked these amounts from Allonnia strains credibility and suggests they are clearly fictitious. We lay out the timeline of key events. First, in December 2019, Allonnia closed a \$33MM Series A financing, according to its financials attached at the end of SPAC prospectus. The cash was reflected on its balance sheet as of 12/31/2019.

Allonnia, LLC

Consolidated Statements of Changes in Members' Equity

For the Year Ended December 31, 2020 and Period from November 27, 2019 (Inception) Through December 31, 2019

	Units					Members'
	Series A-1 Preferred	Series A-2 Preferred	Series A-3 Preferred	Common	Incentive	Equity
Balance at November 27, 2019 (Inception)	—	—	—	—	—	—
Issuance of Common Units	—	—	—	3,600,000	—	24,480,000
Issuance of Series A-1 Preferred Units net of issuance costs of \$145,000	2,970,000	—	—	—	—	32,855,000
Net loss	—	—	—	—	—	(24,644,497)
Balance as of December 31, 2019	2,970,000	—	—	3,600,000	—	32,690,503
Issuance of Series A-1 Preferred Units net of issuance costs of \$272,000	1,664,911	—	—	—	—	18,227,011
Issuance of Series A-2 Preferred Units	—	180,000	—	—	—	—
Issuance of Series A-3 Preferred Units	—	—	180,000	—	—	—
Issuance of Incentive Units	—	—	—	—	140,000	—
Net loss	—	—	—	—	—	(6,011,518)
Balance as of December 31, 2020	4,634,911	180,000	180,000	3,600,000	140,000	\$ 44,905,996

Assets

	2020	2019
Current assets:		
Cash and cash equivalents	\$ 43,520,543	\$ 32,828,673
Prepaid expenses and other current assets	3,015,817	3,384
Total assets	\$ 46,536,360	\$ 32,832,057

Second, after Allonnia closing a \$33MM financing in December 2019, Ginkgo turned around and immediately booked \$24.5MM in deferred revenue from Allonnia as of 12/31/2019, as indicated in the financials we excerpt on slide ____. Then for 2020, Ginkgo booked \$5MM of revenue from Allonnia. **Here's the giveaway:** the start dates of the 6 Allonnia employees listed on LinkedIn indicate that the first didn't start until July 2020, meaning **Allonnia had no employees at the time Ginkgo booked deferred revenue from it.** The second employee, the CEO, didn't start until September 2020. LinkedIn profiles indicated that the others didn't start until 2021 – in other words, **Allonnia appears to have had 2 employees for 2020, both for a few months each, yet Ginkgo booked \$5MM in revenue from the entity on top of \$26MM in deferred revenue by 12/31/2020.**



Nicole Richards · 3rd
Chief Executive Officer at Allonnia
Ridgewood, New Jersey, United States · [Contact info](#)



Chief Executive Officer
Allonnia · Full-time
Sep 2020 – Present · 1 yr 1 mo
Boston, Massachusetts, United States

Leading a company focused on creating transformative solutions for a waste free, pollution free world through leveraging the power of biotechnology and advanced engineering systems that degrade pollutants and upcycle waste.

Source: <https://www.linkedin.com/in/nicole-richards-3268b07/>

CEO started as second employee in Sep 2020



Bryne Gramlich · 3rd
Vice President Of Business Development at Allonnia
Calgary, Alberta, Canada · [Contact info](#)



Vice President Of Business Development
Allonnia · Permanent Full-time
Jan 2021 – Present · 9 mos
Calgary, Alberta, Canada

Allonnia is a Boston, MA based bioremediation company focused on waste management tackling some of the planet's most pressing waste pollution challenges, including wastewater treatment, soil redemption, and solid waste management and upcycling.

Source: <https://www.linkedin.com/in/brynegramlich/>



Rafael Mier · 2nd
Vice President, Business Development at Allonnia
Boston, Massachusetts, United States · [Contact info](#)



Vice President, Business Development
Allonnia
Mar 2021 – Present · 7 mos
Boston, Massachusetts, United States

Our mission: leverage the power of biotechnology and engineering systems to create transformational solutions for a world free of waste and pollution

Source: <https://www.linkedin.com/in/rafael-mier-b0b1b0/>



Mack Astorga · 3rd
Vice President, Business Development at Allonnia
Atlanta, Georgia, United States · [Contact info](#)



Vice President, Business Development
Allonnia · Full-time
Mar 2021 – Present · 7 mos
Boston, Massachusetts, United States

Responsible for the development and execution of a commercial strategy that contributes to Allonnia's goal of creating transformative solutions that leverage the power of biotechnology and engineering systems to degrade pollutants and upcycle waste.

Source: <https://www.linkedin.com/in/mack-astorga-68aa512a/>



Kent Sorenson · 3rd
Chief Technology Officer at Allonnia
Boston, Massachusetts, United States · [Contact info](#)



Chief Technology Officer
Allonnia
Jan 2021 – Present · 9 mos
Boston, Massachusetts, United States

Source: <https://www.linkedin.com/in/kent-sorenson-35513827/>



Dayal Saran · 3rd
Vice President and Head of Research at Allonnia
Boston, Massachusetts, United States · [Contact info](#)



Vice President Research
Allonnia · Full-time
Jul 2020 – Present · 1 yr 3 mos

Source: <https://www.linkedin.com/in/dayal-saran-43483417/>

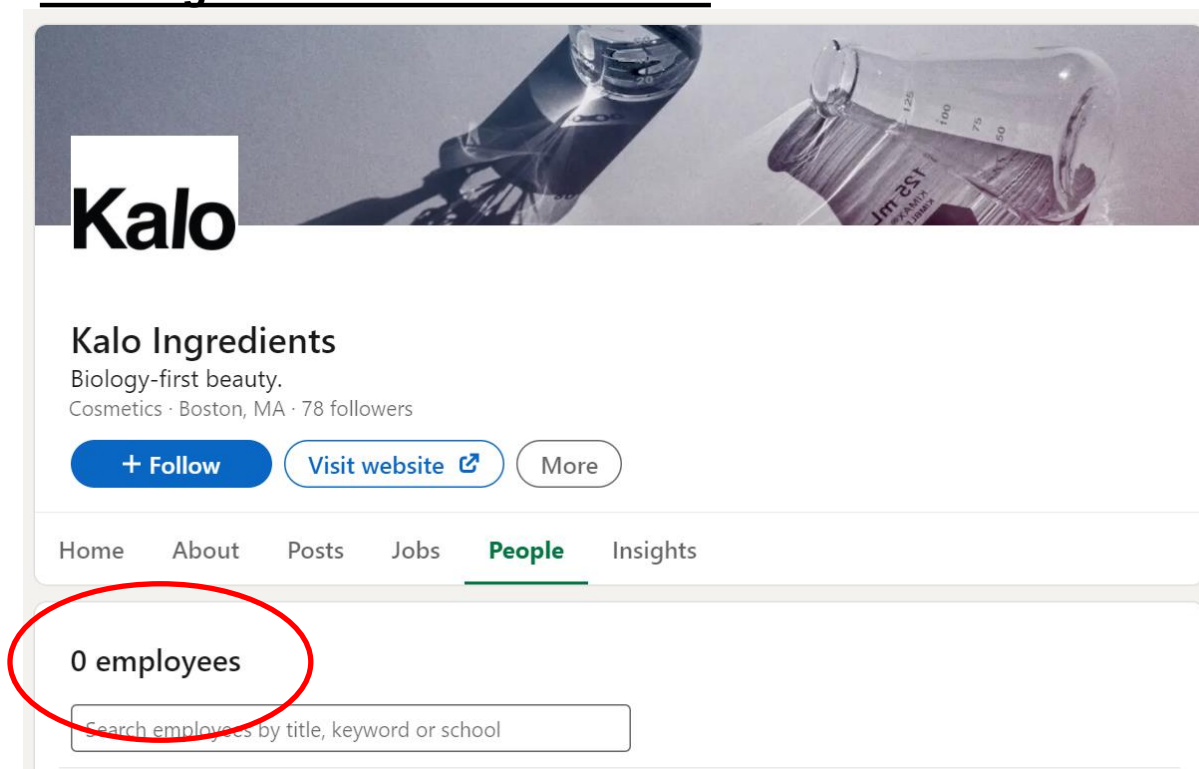
First employee started in Jul 2020

Related party “customer” Kalo Ingredients has no employees listed on LinkedIn, yet Ginkgo booked an \$11.9MM deferred revenue balance as of 3/31/2021 with the entity. We find this particularly audacious and preposterous, as Kalo was only formed 2 weeks earlier as an entity, according to its Massachusetts corporate registry filing (shown on p. ____).

SPAC Prospectus Aug 13, 2021

	Joyn	Motif	Genomatica	Allonnia	Synlogic	Kalo	Total
Balances as of March 31, 2021							
Accounts receivable, net	\$ —	\$ 3,391	\$ 1,500	\$ 2,167	\$ —	\$ —	\$ 7,058
Deferred revenue, current and non-current	\$8,281	\$53,299	\$ 26,680	\$37,959	\$ 66	\$11,897	\$138,182

Kalo Ingredients LinkedIn search



Kalo

Kalo Ingredients
Biology-first beauty.
Cosmetics · Boston, MA · 78 followers

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0 employees

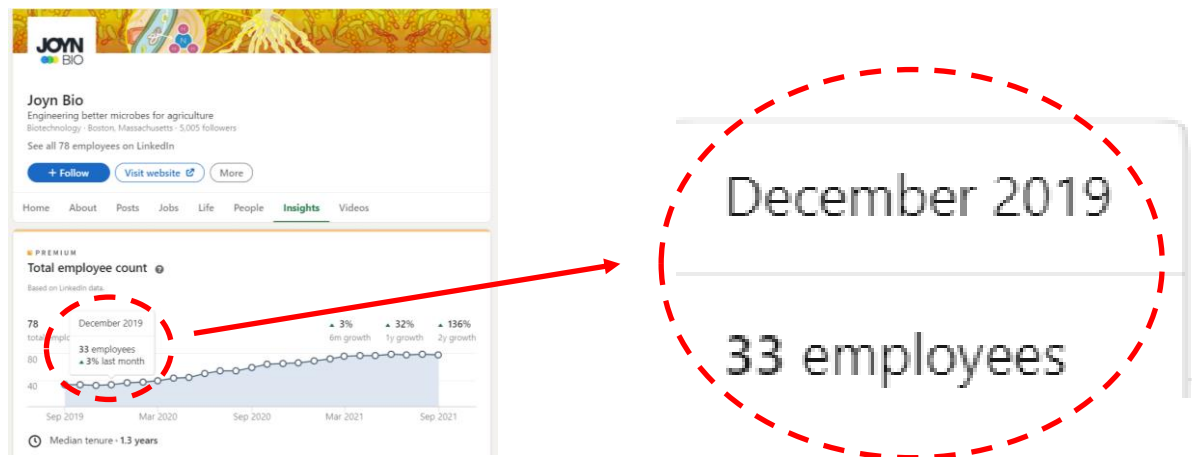
Search employees by title, keyword or school

Joyn Bio LLC was formed in 2017. By Dec 2019, Joyn had already made an astounding \$35MM of R&D “Prepayments” to Ginkgo. For the full year 2019, Ginkgo booked \$9.3MM of foundry revenue from Joyn and reported \$17MM of deferred revenue. LinkedIn’s “Employee Count Over Time” feature indicates that Joyn had a mere 33 employees as of Dec 2019 – meaning it had already made an implausible >\$1MM of R&D prepayments per employee to Ginkgo, and that Ginkgo also booked \$515K of deferred revenue and \$300K of foundry revenue per Joyn employee.

Excerpt from Prospectus, Aug 13, 2021

Joyn paid us a non-refundable \$20.0 million prepayment for services to be provided under the Joyn FSA and made **an additional \$15.0 million prepayment for services during the year ended December 31, 2019.**

LinkedIn “Total Employee Count Over Time” for Joyn Bio



Motif Foodworks launched at the end of Feb 2019 with \$90MM of Series A financing. Once again, Ginkgo went pedal to the metal to book revenue in amounts that are breathtaking and preposterous relative to Motif's small size. LinkedIn indicates Motif had only 17 employees as of Dec 2019, yet Ginkgo charged it \$19MM for foundry services in 2019 - \$1.1MM per employee - with an eye-popping deferred revenue balance of \$62.5MM on its books as of Dec 31, 2019, which works out to \$3.7MM per Motif FTE.

Excerpts from Prospectus, Aug 13, 2021

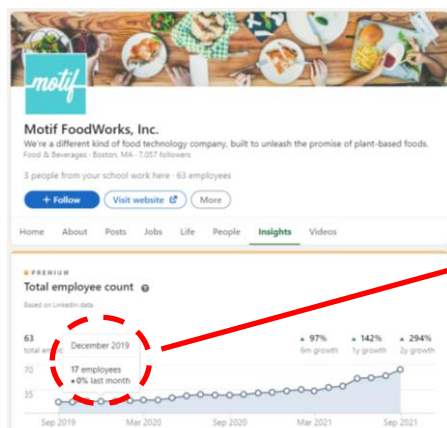
For the Year Ended December 31, 2019

	Joyn	Motif	Genomatica	Allonnia	Synlogic	Glycosyn	Total
Foundry revenue	\$ 9,349	\$ 18,986	\$ 6,248	\$ —	\$ 17	\$ 668	\$ 35,268
Interest income	\$ —	\$ —	\$ —	\$ —	\$ —	\$ 163	\$ 163
Other income, net	\$ 222	\$ 42	\$ —	\$ —	\$ —	\$ 1,530	\$ 1,794

Balances as of December 31, 2019

	Joyn	Motif	Genomatica	Allonnia	Synlogic	Total
Accounts receivable, net	\$ 163	\$ 4,054	\$ —	\$ —	\$ —	\$ 4,217
Deferred revenue, current and non-current	\$17,135	\$62,513	\$ 38,059	\$24,480	\$ 144	\$142,331

LinkedIn “Total Employee Count Over Time” for Motif



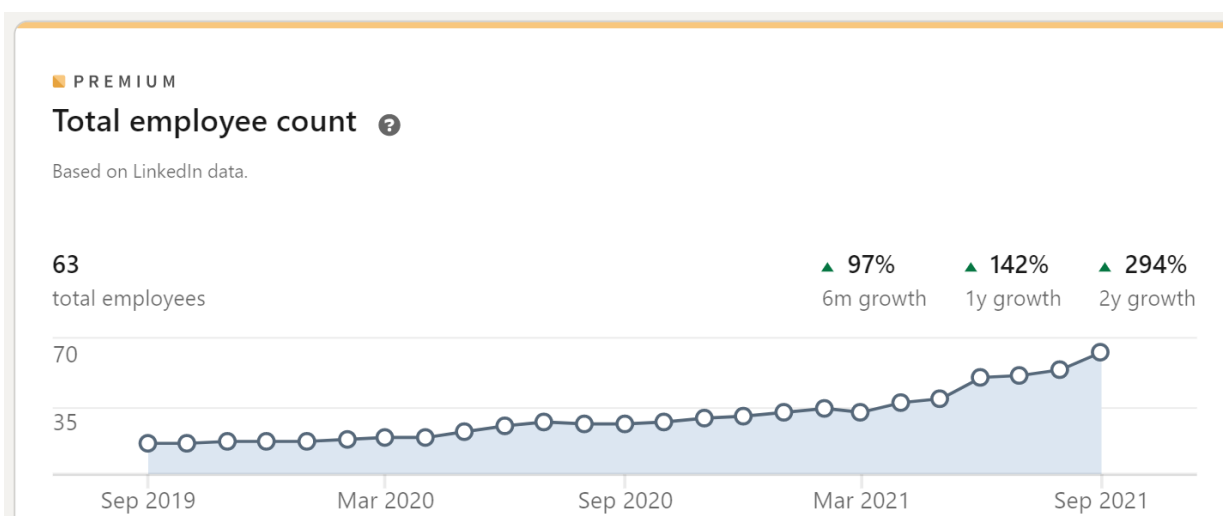
December 2019
17 employees

The “revenue” that Ginkgo continues to book from Motif remains absurd. LinkedIn indicates Motif had only 30 employees as of Dec 2020 – with the average at 23, taking the beginning and ending counts for the year - yet Ginkgo still recorded \$21MM of foundry revenue from it for 2020 – about \$1MM per Motif FTE. Ginkgo booked \$10.1MM of revenue in 1H 2021, with an average employee count of 41 (32 as of Jan 2021, 50 as of June 2021) – lower than last year, but still an annualized \$500K per head.

Excerpts from Prospectus, Aug 13, 2021

	Joyn	Motif	Genomatica	Allonnia	Synlogic	Glycosyn	Total
For the Year Ended December 31, 2020							
Foundry revenue	\$ 7,273	\$ 20,798	\$ 9,431	\$ 4,960	\$ 73	\$ —	\$ 42,535
	Joyn	Motif	Genomatica	Allonnia	Synlogic	Kalo	Total
For the Six Months Ended June 30, 2021							
Foundry revenue	\$ 2,752	\$ 10,104	\$ 6,201	\$ 3,364	\$ 10	\$ 1,191	\$ 23,622

LinkedIn “Total Employee Count Over Time” for Motif



11. Current employees at Ginkgo's related-party "customers," as well as former employees of Ginkgo, corroborated our findings that Ginkgo's model is based on overcharging captive entities.

Current employees at Ginkgo’s related-party “customers,” as well as former employees of Ginkgo, allege that Ginkgo’s model is based on overcharging captive entities: foundry fees are “prohibitively expensive; “obligated” to pay whatever Ginkgo dictates. A senior employee at key related-party Motif Foodworks provided a specific example where Ginkgo charges them 50% more than comparable strain engineering vendors.

Ginkgo’s model is basically based on overcharging captive entities; “prohibitively expensive”; “obligated” to pay whatever Ginkgo dictates

*“Ginkgo was kind of allowed to do what they wanted to do for the first period of the company [Joyn] at least, and that was really something that Jason Kelly and the group really sort of had pushed for, especially in Motif, at Allonnia, and the subsequent ventures, Ginkgo really needs to control these entities because if they can’t then—what they want is for these subsidiaries or for these joint ventures, these supposedly independent companies to be obligated to use Ginkgo services, and **Ginkgo services are very expensive. It’s almost prohibitively expensive in some respects.** But the idea is if you comparison shop, you can go elsewhere. It’s actually kind of interesting like Joyn was doing a lot of work with this group with Culture BioSciences in South City, South San Francisco. And they’re basically outsourcing their fermentation to them, not to Ginkgo. I would suspect that that was because of pricing and cost savings because they actually are not entirely obligated to use Ginkgo services, but **that’s the whole idea, is you’re obligated to basically no matter what it costs. If Ginkgo says, pay us this, that’s what you pay.**” – Former director-level employee*

Motif senior employee indicated Ginkgo charges them 50% more than comparable strain engineering vendors

Q: “How do competitors price per project? This doesn’t sound like multimillion-dollar kind of stuff. It sounds like 10 to 15-thousand dollar strain experiments.”

A: “They usually price on the lower end compared to Ginkgo. It’s not cheap.”

Q: **“How much more expensive is Ginkgo than the typical ones percentage-wise?”**

A: **“At least 50%.”** – Current Motif employee

A second employee at related-party Motif, an executive, indicated the Ginkgo's pricing was "opaque" with "very little breakdown," and chuckled when saying they're "independent" of Ginkgo; suggested Motif spend with Ginkgo was previously controlled by a Motif "employee" who was acting in Ginkgo's interest instead.

Motif executive chuckled when saying they're "independent" of Ginkgo; suggested Motif spend was controlled by a Motif "employee" who was acting in Ginkgo's interest

Q: **"You sort of chuckled earlier when you said, yeah, we're independent;** I think you said they try to exert control. I'm curious if you can elaborate on that - there's always that push and pull."

A: **"Exactly. I would say that the person that was the director of R&D, pretty much overseeing the project, was actually a Ginkgo employee at first, so he had really little say because he was an insider from Ginkgo, so he couldn't really tell them what you're doing is not the perfect thing.** I work for Motif, and I need to have the best result for Motif. And so, you can promise me everything or anything, but at the end of the day, you have to deliver commercial strains for us."

Q: "So, you're saying initially, the head of R&D at Motif was a Ginkgo employee?"

A: "Well, **he was a Motif employee, but he was a former Ginkgo employee, yes, that's right."**

Q: "For all intents and purposes, was he acting on behalf of Motif, or **was he there to make sure that Ginkgo got the right level of R&D spend from Motif?**

A: "I would say that no matter what he was trying to do, if he was trying to be on the side of Motif, being an ex-Ginkgo employee and having worked on the Motif project, maybe it's a little difficult to challenge them, I would say."

Q: "Challenge them on what? The amount they wanted to charge? Or were they forcing you guys to use them?"

A: **"[Chuckles] Well, yes, all of the above. So, yes, the amount they charged us, there was no negotiation.** I mean, the price was pretty much fixed back then by Ginkgo, I would say, very early on, and there was very little leeway and that's changed because we are now using other syn-bio companies as well because we're not exclusive. So, we can now negotiate a little more on the pricing. They are a stakeholder. They are an equity holder of our company; they founded us. So, their investment is really equity-based for us." – Current Motif executive

Motif executive indicated Ginkgo pricing was "opaque" with "very little breakdown"

"The pricing is multi-million dollars per project per year. A project lasts two or three years, so it's a significant amount... Their pricing method was very opaque, ..There was very little breakdown about what was happening behind the door for each project...But at the beginning, it was really like one pot, and you didn't know exactly what you were paying for." – Current Motif executive

A senior employee at related-party “customer” Joyn Bio corroborated and elaborated upon Ginkgo’s strategy of overcharging its entities for foundry work, suggesting that Ginkgo’s objective is “How can we make this as expensive as possible” and that they appear to stretch projects out as long as possible to inflate revenue: “they seem to be really good at making things take a long time. I don’t know how to explain it exactly.”

Senior employee at Joyn Bio commented believes Ginkgo attempts to overcharging them

Q: “There was a recent article in the MIT Technology Review, which I’m sure you can find.”

Q: “I read it. It was very skeptical. Is that your perspective?”

A: **“I do. There were several of us here who said we were afraid that they had bugged our Slack channel.”**

Q: “What were you guys saying and thinking internally that the article captured and echoed?”

A: “[Chuckles] **They talk a good game, and then six months later, they’re still saying the same things. Jason Kelly’s been giving the same pitch for eight years** or however long they’ve been around. For us, the difficulty has been, we need a result, and we need things that actually get finished, and **they seem to be really good at making things take a long time. I don’t know how to explain it exactly.** The automation is useful as long as you have a testable hypothesis and you’re looking for an answer to come out the other end, and that doesn’t seem to be a lot of what we get from them. **What we get from them is “How can we make this as expensive as possible,” to be quite frank.**”

Q: “Are they essentially trying to—is their focus to suck as much revenue out of Joyn as possible, as opposed to delivering a product? Are they just trying to stretch it out?”

A: **“That’s the way it appears to us**, and I have a colleague who claimed—so this is all rumors and hearsay at this point—but he claims that he heard someone in the Ginkgo executive team say that the best thing about the joint venture creating Joyn Bio was that they were going to get lots and lots of money from Bayer.” – Senior employee at Joyn Bio

The Joyn Bio employee described an attempt to act as a fiduciary on behalf of the entity and pushed back on overcharging by Ginkgo, but was advised to play along. The employee indicated that comparable vendors charge 40% of what Ginkgo does – which mathematically means that Ginkgo was overcharging them by 2.5 times.

Joyn Bio employee attempted to act as a fiduciary on behalf of the related-party and pushed back on overcharging by Ginkgo, but was advised to play along; indicated Ginkgo overcharged by 2.5X

A: ***"I ran into that the minute I started here where as a fiduciary of Joyn, I said, "Okay, I want to do this as cost-effectively as possible," and when I saw how much Ginkgo was charging, I went out and found another vendor who charges about 40% of what Ginkgo wants to charge us. And boy, did they pitch a fit. [Laughs] We were able to use another vendor, and we were advised to try to maintain a good relationship with them and so to buy some services from them where it made sense. **And so, we've been kind of walking that line.**"***

Q: *"And who was instructing you, out of curiosity—was it the board or management—they're telling you to play ball basically?"*

A: *"Yeah, this is was management with the message coming directly from the board."*

Q: ***"And they just said play ball?"***

A: ***"Yup."***

Q: *"It sounds to me like they were basically overcharging Joyn."*

A: *"Yeah, for a while, they had a variable pricing model, too, where they were basically charging us for all the capacity, even if we weren't using all of it. [Laughs] So, like the price went up depending on how full the lab was."*

Q: *"It sounds like this was not a standard pricing arrangement. In an arms-length relationship, this is not the kind of contract that a customer would ever sign up for."*

A: *"Right, the standard in the industry is they have a flat fee per unit of volume or per container that you rent from them or a fixed rate based on labor, for example, to do a certain number of things or to work for a certain number of hours. And so, you say, I want this many labor hours and this many containers, and it's a flat rate as long as nothing goes wrong."— Senior employee at Joyn Bio*

The Joyn Bio employee said colleagues raised red flags about Ginkgo's pricing and lack of transparency about "why everything was costing as much as it was"; indicated Ginkgo was not forthcoming with pricing and invoices for the amounts they were charged; and provided a specific example where Ginkgo charged them twice the market rate; and conveyed that Joyn has to pay up and play ball as a related-party.

Raised red flags about Ginkgo's pricing and lack of transparency about "why everything was costing as much as it was"

"What happened over the last year or so was that people like me came on, we had another person who came on, and **both of us just looked at this setup where Ginkgo wasn't being very transparent about how they were charging and what they were charging us for and why everything was costing as much as it was. And when we started to raise the red flags** about this, our management did go back to them and say, "Look, we can't carry on this way." So, they have assigned some new alliance managers who are being much better and more forthcoming about the information." – Senior employee at Joyn Bio

Ginkgo was not forthcoming with pricing and invoices for the amounts that Joyn was being charged

"Joyn would make requests for things, and they would just kind of go into a black hole. We're like, "Hey! What happened to this information? **Why is it that we haven't seen invoices? Where's the line item detail on what you're charging us for?" And it was just kind of an echo chamber**". – Senior employee at Joyn Bio

Employee provided an example where Ginkgo charges Joyn almost twice the market rate; Joyn has to play ball and pay up as a related party

"From what other people are telling me, **they are charging very high prices relative to what we could get from other vendors, and that we still have to play ball** with them to some extent and continue to buy the services from them at their prices... This is for small-scale fermentation. Let me just walk through this for you. We have a vendor who charges us about **\$840 for a small fermentation vessel. And Ginkgo charges us between \$1400 and \$1500 for the same sized fermentation...** I don't have a good frame of reference for the engineering services, the microbial engineering services but my colleagues tell me that **what Ginkgo charges is very expensive**." – Senior employee at Joyn Bio

A fourth research source, an ex-Ginkgo employee in a managerial role, suggested that overcharging and padding statements of work was part of Ginkgo's culture: "We do it all the time. They're never going to know"; "I just got the sense it happened all the time"; alleged personally witnessing an attempt to overcharge and defraud the US government on a DARPA project; "I saw them actually make it. I saw the spreadsheet."

Former employee alleges witnessing attempts to overcharge customers

A: *"On a DARPA project, they went and padded the revenue. They were like, we want to make \$7 million instead of \$4 million here, so let's figure out how we can pad the revenue as much as we can by putting in all these extra strains and all these extra fermentation processes into the SOW and into the invoice that we don't really need to do, but we're going to do it anyway. What I'm trying to say is I have seen that happen at Ginkgo, and I wouldn't be surprised if it's happening at Allonnia because both of these companies want to make money."*

Q: *"So, you saw them inflate an invoice from \$4 million to \$7 million to DARPA? Or you saw them talk about it or discuss it even if they didn't do it?"*

A: *"I saw them actually make it. I saw the spreadsheet. I saw it when it was initially formed, and then it went out. I can't remember exact numbers, it might have been 5 to 7, but it was definitely significant padding of at least \$2 million. In that specific instance, that was because they were aware that if they went above 7, the DARPA people wouldn't like it, but they were aware that they could go up to 7, and the DARPA people wouldn't know. They didn't have the expertise to know that it would be padded."*

Q: *"Was this spreadsheet of this pricing discussed among a number of people who were aware that this was going on?"*

A: *"Oh yeah, it was a team of about 25 people working on this...And everyone's like, "Oh yeah, it's fine. We do it all the time. They're never going to know." I just got the sense it happened all the time."* – Former Ginkgo employee

12. Related-party “customers” appear to be staffed with what we believe to be loyal, longtime Ginkgo plants in critical positions, who we think rotate from entity to entity to ensure that the entities play ball by round-tripping revenue back to Ginkgo via foundry R&D spend.

Ex-employees alleged that Ginkgo's co-founders plant loyalists at its "independent" related party "customers" to ensure they play ball and fire those who push back on the round-tripping scheme. A former senior level employee alleged that one such loyalist led to multiple terminations at one of Ginkgo's most significant related party "customers," and alleged that she was doing so on behalf of co-founder Reshma Shetty: "There was a Ginkgo employee that was dictating who could stay or who could go depending on whether they were yes-men or not kind of thing."

Q: **"Were there other people that were let go because of refusing to play along?"**

A: "There have been some really good people that have left [related party name redacted]. I do know that there have been people, I don't know the nature of their criticisms, but there have been a number of people that have exited [related party name redacted]. You can try and pull those out and talk to them, maybe. But I can't really speak to the true nature of why they left."

Q: **"Are there people that you suspect left for that reason?"**

A: "I'll maybe say one more thing about it. [Redacted] when I was at Ginkgo was [employee name redacted]. **She was a Ginkgo employee. [Redacted] was kind of set up at the last minute to control [redacted], I think, and the [related party name redacted]. So, that person has led to the exit of a number of different people at [related party name redacted]."**

Q: "Sorry, walk me through that. Who was this person, and what was her role?"

A: "Like the head of [redacted].. At the time, I think she now has switched over to being a [[related party name redacted] employee. **She was originally hired to temporarily help set up the company, and then she was going to go back to Ginkgo after a year because that was the nature of it, and then we were going to—she had worked at Ginkgo for five years** and knew the ins and outs of Ginkgo very well, so she was going to help do that, like rosetta stone between the new company and the old. But it **became more of a control issue, I think, where Reshma could basically—this was really the issue—they were not acting as an independent company at the time. There was a Ginkgo employee that was dictating who could stay and who could go depending on whether they were yes-men or not kind of thing."**

Q: **"This was [employee name redacted], who was that person who was doing that?"**

A: **"Yes."**

Q: **"She was doing this as a Ginkgo employee on behalf of Reshma?"**

A: **"Yeah."** – Former senior-level employee

As we examined bios of Ginkgo's executives, we noticed some that struck us as unusual. For example, the current VP of Research at related-party "customer" Allonnia appears to have played a similar, recurring role across various entities. His LinkedIn bio indicates he was previously Head of Synthetic Biology at Synlogic, another key "customer." Prior to that he was Head of R&D at Motif, another one. Each stint was short – but perhaps just long enough, we speculate, to paper up the requisite R&D agreements with Ginkgo and get the job done.



Dayal Saran · 3rd

Vice President and Head of Research at Allonnia
Boston, Massachusetts, United States · [Contact info](#)

500+ connections

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Experience



Vice President Research

Allonnia · Full-time

Jul 2020 – Present · 1 yr 4 mos



Head of Synthetic Biology

Synlogic, Inc.

Oct 2019 – Jul 2020 · 10 mos
Cambridge, MA



Head of R&D, Alliance

Motif FoodWorks, Inc. / Ginkgo Bioworks spinoff

Aug 2018 – Oct 2019 · 1 yr 3 mos
Boston MA

Led the discovery on fungal and yeast strain's platforms for a variety of novel animal and dairy proteins possessing unique properties that can improve the function (taste, texture etc.) and nutrition (PDCASS, gut health etc.) for the bulk plant proteins.



Program Director

Ginkgo Bioworks, Inc.

May 2016 – Sep 2018 · 2 yrs 5 mos
Boston, MA

Project management

We previously noted comments by a Motif executive who suggested that the entity's R&D spend with Ginkgo was previously controlled by a Motif "employee" who was acting in Ginkgo's interest instead. We speculate that the person to whom he referred is on the previous page – as he referred to "the person that was the director of R&D" at Motif.

Motif executive chuckled when saying they're "independent" of Ginkgo; suggested Motif spend was controlled by a Motif "employee" who was acting in Ginkgo's interest

Q: **"You sort of chuckled earlier when you said, yeah, we're independent;** I think you said they try to exert control. I'm curious if you can elaborate on that - there's always that push and pull."

A: **"Exactly. I would say that the person that was the director of R&D, pretty much overseeing the project, was actually a Ginkgo employee at first, so he had really little say because he was an insider from Ginkgo, so he couldn't really tell them what you're doing is not the perfect thing.** I work for Motif, and I need to have the best result for Motif. And so, you can promise me everything or anything, but at the end of the day, you have to deliver commercial strains for us."

Q: "So, you're saying initially, the head of R&D at Motif was a Ginkgo employee?"

A: "Well, **he was a Motif employee, but he was a former Ginkgo employee, yes, that's right.**"

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Q: "Challenge them on what? The amount they wanted to charge? Or were they forcing you guys to use them?"

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Part 2: Behind the synthetic biology and “Foundry” hype, Ginkgo is a glorified contract research organization that does commodity yeast strain engineering – with a decade of spectacular failure in delivering anything of value for customers.

13. No unique technology, intellectual property, or differentiated capabilities

Yeast is Ginkgo's organism of choice for strain engineering projects, with a bit of E. coli and other industry-standard hosts, same as Zymergen and every other glorified CRO. Ginkgo's 700+ page prospectus barely mentions the word "yeast," concealing that the core of its platform is commodity yeast optimization, a widespread method for 50+ years. Ex-employees and its related-party "customers" stated that the "bread and butter" of the foundry is tweaking yeast to excrete different compounds.

Ginkgo's "bread and butter" is tweaking yeast to spit out different compounds

Q: "Is this the crux of every project is there's a strain of yeast that they want to do something in particular, and they try to basically refine that strain to make it do that?"

A: **"That's the bread and butter."** I worked on 6 or 7 projects, maybe 10 if you count some of the ones I worked on briefly. In my time, I worked with various flavors of e. Coli, yeast, a weird organism that wasn't even domesticated. I know a lot of people there that worked with fungi. They'd like to be strain agnostic, meaning that they can work with whatever. In practicality, that's quite difficult. They still do not have a good way of if you bring in some grain of bacteria out of your backyard; transferring tools from one strain to another is not straightforward." – Former strain engineer

Typical strain engineering involves trial and error in manipulating yeast in different combination

"They took all of the money that they've raised, and they put it into the equipment to be able to do parallel processing of yeast cells. **It's not always yeast, but yeast is their organism of choice.**" - Former Ginkgo executive

Yeast is the organism of choice for the entire industry

"Yeast is typically the organism of choice because it's well-understood, it's fairly easy to manipulate and work with, and you can do post-translational modifications. **It's a very robust organism, and the entire industry, I would say, yeast is the organism of choice.**" -Former Ginkgo executive

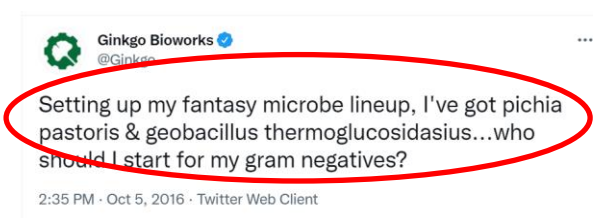
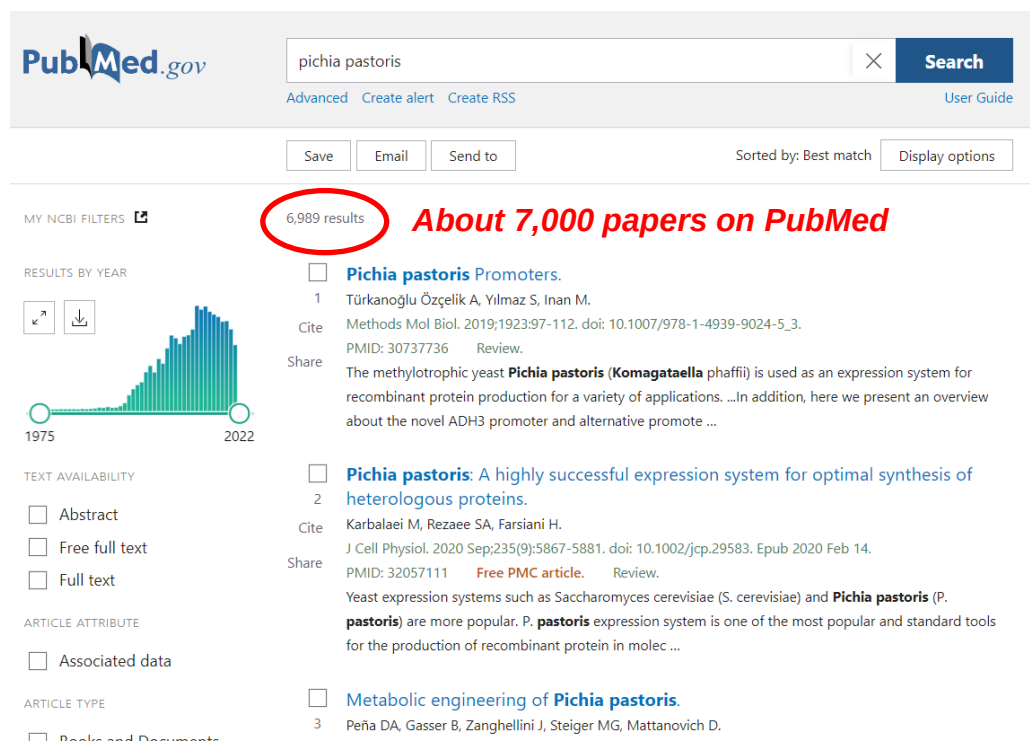
Zymergen also uses yeast; "industry standard"

"Zymergen are also using yeast." Yeast is an industry-standard, so that's just not a secret." -Former Ginkgo executive

An executive at Motif Foodworks, a key related-party “customer” of Ginkgo’s, stated that the core of Ginkgo’s platform is a strain of yeast called *Pichia pastoris*. Ginkgo indicated its preference for the strain in a 2016 tweet. A search on PubMed yields 7,000 papers that reference the strain’s use for microbial engineering and expression.

Executive at related party “customer” Motif Foodworks confirms that Ginkgo’s platform is based around yeast, mentioning a strain called *Pichia pastoris*, specifically

“The platform that Ginkgo is using is *Pichia pastoris*, which is a yeast; it’s well known to produce a lot of protein. The strain engineering that they’re doing for us is mostly yeast-based, yes.” –Current executive at Motif



Ginkgo uses fancy videos and drivel to create mystery around its commodity strain engineering lab, slapping the word “Foundry” on it. Ex-employees explained that its simply “parallel processing of yeast cells” using liquid handling machines to automate the repetitive task of using pipettes to inject samples into well plates.

Typical strain engineering involves trial and error in manipulating yeast in different combination

“Typically, when somebody talks about a fermenter, it’s a pretty big piece of equipment. What they’re doing is these micro-titer plates, which are mini-fermenters, and you can buy it. **They took all of the money that they’ve raised, and they put it into the equipment to be able to do parallel processing of yeast cells.** It’s not always yeast, but yeast is their organism of choice.” - Former Ginkgo executive

Just automated liquid handling into well plates to automate repetitive pipette-ing

Q: “They’re using large well plates, so they have hundreds of thousands of little chambers, and robots are just little machines that are able to pipette into hundreds of wells?”

A: **“Exactly. It’s a combination of a liquid handling system and analytics. It injects one sample—usually, they repeat it 10 times just to make sure. They inject it into the wells, carry out the fermentation,** and then they have analytics to look at and see the results. The ones that they like the best, they further iterate on, and if those are promising, then they’ll grow them up into larger fermenters. And these could be 10 mil or 100 mil bona fide fermenters where they’ve got agitation; they’ve got aeration, it’s a controlled environment.” - Former Ginkgo executive

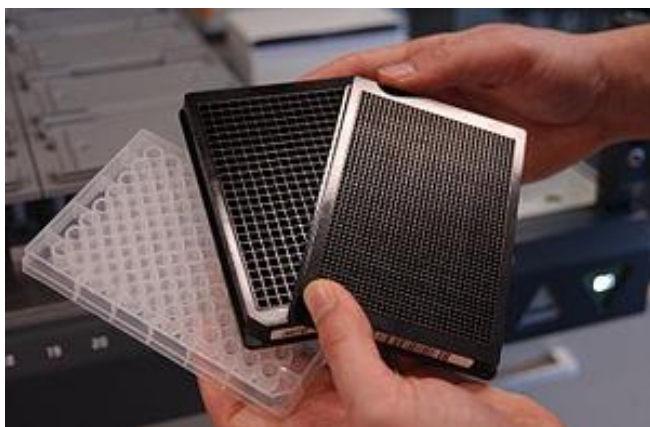
Strain engineering is a highly repetitive process

“Let’s say you want to produce an ingredient that’s used as a skin whitening agent. It exists in nature, but it’s very expensive to isolate from the tree it grows on. What you can do is go to this tree; you can say here’s a DNA sequence that encodes for it. **Take that DNA sequence and put it in a yeast cell. Grow the yeast cell up in a fermenter, and it produces it. If you put it in a yeast cell, and it doesn’t work, you look and say, what’s wrong with it? It’s a seven-step process,** and each one of these steps requires an enzyme to be able to do the manipulation. You can think of it as like you’re building a Lego model, and you need to put these pieces together, and each enzyme adds a different piece until you get the final product with all the pieces on there. You go, and you do your experiments, and you say what happened is the first four enzymes worked great, but the fifth enzyme just doesn’t work, so I need to manipulate, I need to get a different enzyme. **So, you go, and you try a different enzyme, and you put it back in the yeast cell, and you grow it up, and you try again. And then you try it again and again. It’s a repetitive process.**” - Former Ginkgo executive

Well plates sell for \$1 each and are similar to ice cube trays, except with hundreds or thousands of “test tubes” for liquid samples. Ex-employees demystified the “Foundry” hype as little more than miniaturization of well plates, using liquid handling machines from the same suppliers Zymergen and everyone else uses. The machines are a substitute for humans manually injecting samples into hundreds of wells, as part of the typical trial and error process of throwing hundreds of modified yeast strains at the wall to see which ones may excrete anything useful.

Just miniaturization of well plates

“Behind all the words and the beautiful photos and stuff, what's happening...there's miniaturization of every experiment that's happening. Traditionally, in molecular biology, one of the limiting factors was the size of the equipment that you had to run an experiment. I encourage you to google image search 96 well plate. That was the industry standard for a lot of companies where you can do 96 samples at once that are compatible with robots, but mainly not robots, mainly just a pipette in your hand. **What Ginkgo started doing early on was bringing in-house automation that would allow you to use 384 well plates.** If you google that, it's the same dimensions of a plate, but you can fit more samples in one. And the next-gen version is 1536 wells and the same thing. And so, all of the tools that they're using aren't custom, but it's the way that they're connecting everything together. So, all of these robots are part of big integrated workflows where for the same amount of time and cost, you can generate 1536 data points instead of 96.” - - Former Ginkgo employee, longtime direct report to CEO Jason Kelly



Ginkgo has no scientific innovation. It's only "innovation" is blowing hundreds of millions on third-party capital equipment – mainly fermenters and liquid handlers, stitched together with some software and engineering – to supposedly be able to process hundreds of thousands of combinations of modified yeast strains. A former executive described it as a "high throughput method of brute force," as opposed to any particular scientific insights into yeast strain biology.

Ginkgo is using "a high throughput method of brute force"

"What Ginkgo's doing differently is that they are looking at hundreds of thousands of these manipulations, and they're doing them all at the same time using robotics. What they have is the ability to have super-small fermenters that are in these microtiter plates where they use programming code to figure out. If they say it's that fifth enzyme that just doesn't work right, what they'll do is they'll try 10,000 different permutations of an enzyme that does that type of transformation they're looking for, but each one of those is different. And then they use robots to inoculate these wells in these plates, and all the analysis is done by robotics, and these things work 24/7, and they don't go on vacation. **It's a high throughput method of brute force,** trying to figure out the best set of enzymes to undertake a certain type of reaction to produce a final product. "

Former Ginkgo executive

Executive at related-party "customer" Motif Foodworks indicated throughput as Ginkgo's primary angle

"The unique about Ginkgo is their high throughput. So, they can go through a lot of iteration; they can test a lot of variants and candidates that other companies cannot **because of their automation platform,** so that's really their strong suit here...So, **putting the DNA into the yeast and then selecting the clones and then growing everything,** it's very cumbersome, very time-consuming for people, and so, they've been doing a very good job at automating these steps. So now, you can just have a robot doing 24/7" –Current executive at Motif

The fermenters and automated liquid handlers are typically purchased from two vendors - Sartorius and Hamilton, respectively - both of whom are standard suppliers to the strain engineering industry. Former employees indicated Zymergen uses the same suppliers.

Fermenters and liquid handling machines are from the same suppliers everyone in the industry uses

““The fermenters, they're buying all those from a well known manufacturer. What I think is proprietary at Ginkgo is the robotics and the software that runs the automation...They use a thing called a **Hamilton liquid handler** which is a big robot for moving around small amounts of liquid. **Those you can buy off the shelf.”** – Senior employee at related-party “customer” Joyn Bio

Same lab technology as Zymergen, both were using the same supplier

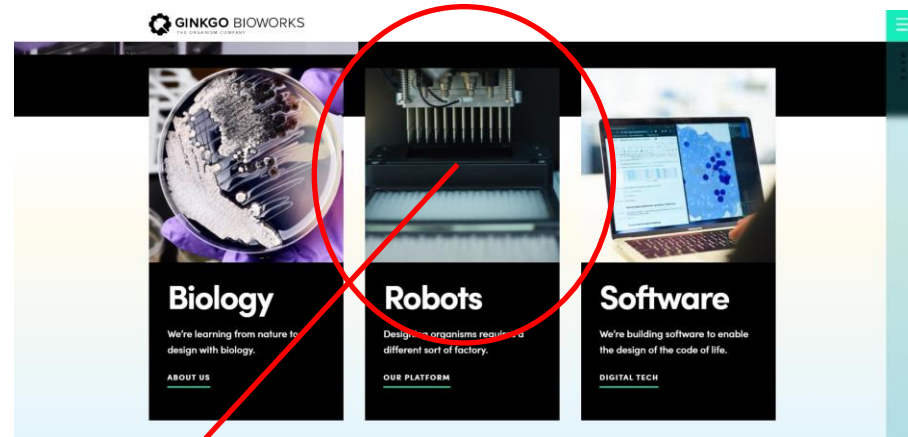
“Zymergen was doing the same thing, and Sartorius is a major company that's one of the huge suppliers in the industry. When I was at Ginkgo, Jason's biggest fear was always Zymergen. And so, he would try to corner the market by buying all the equipment so Zymergen couldn't buy any. Just stuff like that. There are very few suppliers for this type of equipment.” - Former Ginkgo executive

The fancy pictures and videos of “robots” on Ginkgo’s website and investor materials are easily apparent as liquid handling machines from Hamilton and others, as one can tell by zooming in.

Ginkgo’s website hypes its “Foundry”



...using fancy words like ‘Robots’



The same liquid handling machines for sale on Hamilton’s site

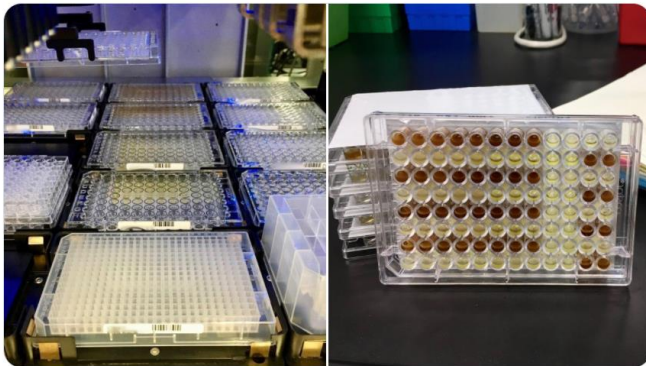


Automated Liquid Handlers | Liquid Handling Equipment

Comments by a Ginkgo reviewer on a lab equipment review site even mention the specific model of Hamilton liquid handler used, corroborated by a Ginkgo tweet in 2019 that mentions Hamilton (“And how do I prep all these samples? Hamilton Robots”), next to images of well plates. The use of “Robots” – capitalized, even – is consistent with Ginkgo’s compulsion to mystify and hype the mundane, while even Hamilton refers to them as merely “machines.”



And how do I prep all these samples? Liquid Handling Hamilton Robots.
If you've ever had to use a single channel pipette on a 384-well plate, you'll understand why I ❤️ Automation.



9:39 AM · Mar 7, 2019 · Twitter Web Client

Liquid Handling | Automated Liquid Handling Platforms Hamilton Microlab STAR Liquid Handling System

Features and Benefits Buy Your Consumables Specifications Applications Services and Support



Hamilton Microlab Star - Proven Technology

Trust in the industry-leading Hamilton Microlab STAR automated liquid handling platform for lasting confidence in results. The Microlab STAR can be changed or upgraded throughout its life to accommodate changing workflows, while robust and flexible programming through VENUS can accomplish any task, from simple to complex. As always, Hamilton's pipetting technology achieves high accuracy, precision, and repeatability from sub-microliter to large volumes.

Daniel Wen

Status: Reviewer ★

Member since: 2020

Organization: Ginkgo

“Great product for liquid handling.”

Rating: 5.0 ★★★★★

Application Area: High-throughput screening

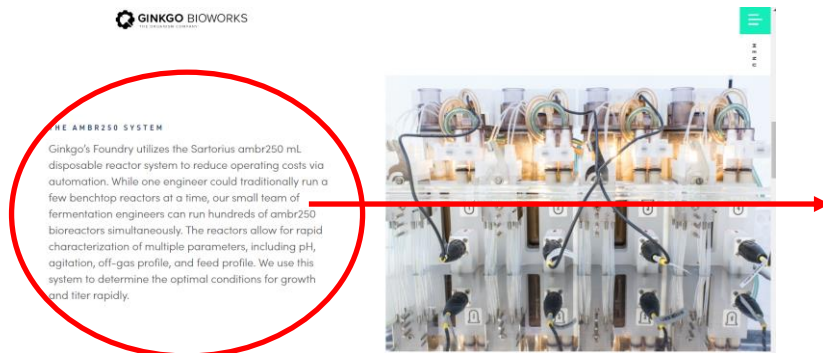
“Great product, awesome service, definitely recommend it.”

Review date: 29 Jan 2020 Microlab STAR Liquid Handling Systems

Ease of use ★★★★★
After sales service ★★★★★
Value for money ★★★★★

Combined with automated liquid handlers, fermenters from Sartorius are the second key piece of Ginkgo's "Foundry." Ginkgo admits as much, buried inside its website, bragging how its engineers have figured out Sartorius how to run **"hundreds of ambr250 bioreactors simultaneously"** – **underscoring the underwhelming nature of its "science."** The Sartorius site explains the fermenters are **simply bioreactors used to culture yeast.**

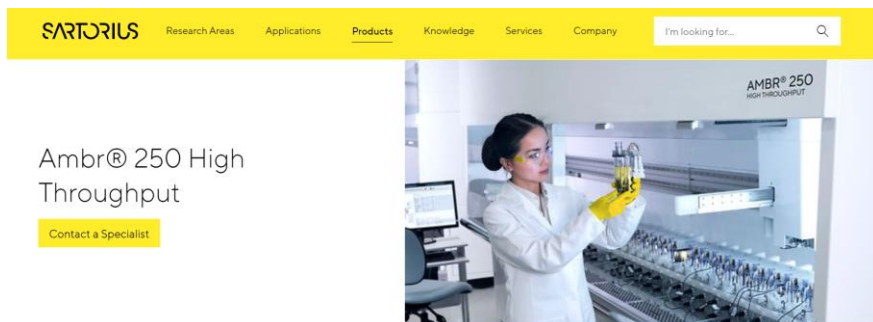
Ginkgo's website has a buried section on its use of the Sartorius Ambr250



THE AMBR250 SYSTEM

Ginkgo's Foundry utilizes the Sartorius ambr250 mL disposable reactor system to reduce operating costs via automation. While one engineer could traditionally run a few benchtop reactors at a time, our small team of fermentation engineers can run hundreds of ambr250 bioreactors simultaneously. The reactors allow for rapid

Sartorius website page for Ambr250 fermenter for "cell culture and microbial fermentation"



Single-Use Multi-Parallel Bioreactor, Fully Automated for Accelerated Process Development

The Ambr® 250 system is a high throughput, automated bioreactor system for process development with 12 or 24 fully featured single-use 100 – 250 mL mini bioreactors.

It has a wide range of applications in:

- Process development and process optimization
- Scale-down studies
- Cell culture and microbial fermentation

After the famed “Foundry,” the second part of Ginkgo’s magic and IP is it what it calls its “Codebase,” the actual substance of which is once again cloaked behind the hype: “We’re automating organism design”; “Every program, whether successful or not, generates valuable Codebase and helps inform future experimental designs and provides reusable genetic parts, making our cell program designs more efficient.”



Our Codebase



Codebase - Ginkgo’s scale data asset - is a source of long-term competitive advantage

3.4B+ unique gene sequences pulled from all public databases

440M proprietary gene sequences acquired

And growing... codebase accumulates as we run new experiments in the foundry and build new:



Organisms



Genetic Code



Biological Tools

MOVE OVER, JONY IVE—
BIOLOGISTS ARE THE NEXT
ROCK STAR DESIGNERS



“Ginkgo will organize the world’s biological code and make it useful.”

- GINKGO HEAD OF CODEBASE, PATRICK BOYLE

GINKGO BIOWORKS INVESTOR PRESENTATION | MAY 2021



27

Former employees explained that codebase “is just a fancy word for a genetic sequencing database.” One indicated that “Zymergen has way more codebase than Ginkgo.” An executive at Synlogic, a key related-party “customer” of Ginkgo also described it as just another “sequenced database,” and added that Ginkgo’s isn’t even functional: “we can go on the internet right now and search sequences”; “I couldn’t tell you how unique it really is, to be honest.”

Half of Ginkgo’s project fail at front-end strain engineering

“I mean, yeah, **codebase is just a fancy word for a genetic sequencing database**, and they’re constantly trying to get more codebase and get more diversity. **Another interesting thing is that Zymergen has way more codebase than Ginkgo. Their diversity is way higher, and that’s part of why they were able to engineer that Synlogic strain and Ginkgo wasn’t. Ginkgo is way outcompeted by Zymergen in that regard.** Anybody who really wants a strain engineered should probably go to Zymergen. They’re obviously having a bunch of trouble, they have a bunch of executional and management issues, but they fundamentally have better stuff to be able to make these strains. And I’ve heard that from several folks that are there now as well.” - Former Ginkgo employee

Ginkgo “Codebase” is just another DNA sequences databases

“The codebase is effectively a sequenced database...It’s like accessing sequences; it’s not functional. It’s not like they’ve expressed proteins from all that sequencing; they’re screening those proteins. It wasn’t for sequences that you think might have activity... I think most of it is DNA sequences. So, if we want to find a certain kind of enzyme, we could search those sequences with different computer algorithms to try to pull out things that we think might be there.” – Synlogic executive

Nothing unique; lots of public sequencing databases online

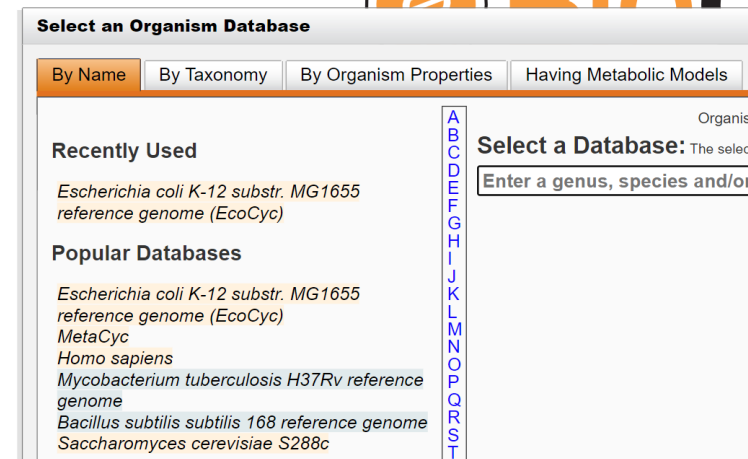
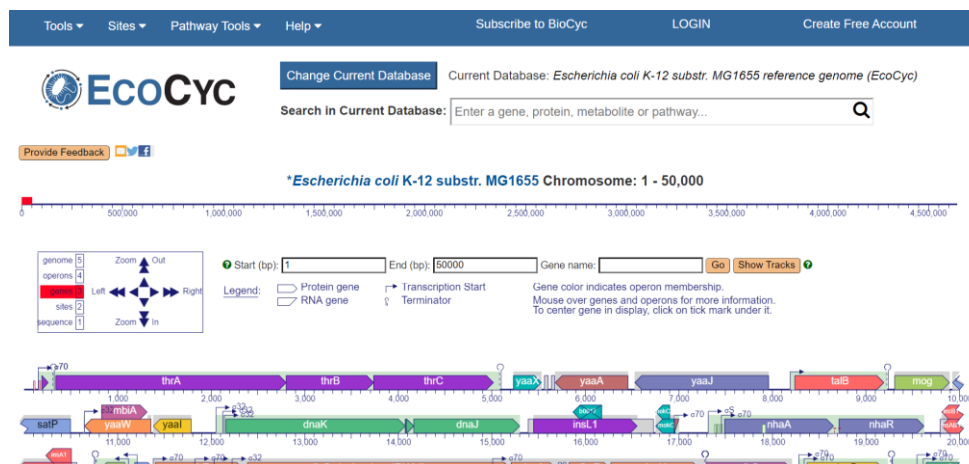
“We can go on the internet right now and search sequences. There are public databases for sure. So, there are lots of sequencing databases. I couldn’t tell you how unique it really is, to be honest. ...there are lots of public repositories of sequence data. And every time you publish a paper in the scientific literature that have sequences in it, you’re required to deposit those sequences in databases, so those are growing all the time.” – Synlogic executive

Numerous sequencing databases are available online, including various open-source, industry-wide repositories, whether for yeast – Ginkgo’s platform of choice – or other industry-standard organisms like E. Coli. For example, BioCyc contains “curated data from 130,000 publications” and is a “collection of 19.534 Pathway Genome Databases (PGDB’s) for model eukaryotes and for thousands of microbes, plus software tools for exploring them.”

YeastGenome.org sequence database



BioCyc.org repository of ~20,000 sequence databases



14. Ginkgo's related-party "customers" slammed Ginkgo's foundry services.

Given that Ginkgo is simply engineering yeast and doing so with the same third-party equipment others use, we were not surprised that former employees indicated the “Foundry” is nothing special: “not unique”; just standard small molecule screens people have done for decades; robotics are nothing innovative, just a “schtick” around automated liquid handlings; no unique equipment or IP; nothing “that other companies don’t have.”

Foundry is “not unique”; just standard small molecule screens people have done for years, with higher throughput through miniaturization

“The foundry itself is not unique. People for many years, whether that’s Amyris or Intrexon or name your favorite pharma company, people have been doing small molecule screens and high throughput and doing similar types of work for many years. And so, what they’ve done really well is miniaturized experimentation. They’ve invested heavily in very cutting-edge robotics that allows you to miniaturize experimentation down to pico-liters.” - Former director level executive

Robotics are nothing special, just a “schtick” to hype automated liquid handling from well to well

“They use these instruments called the Echo that basically moves around very tiny volumes of material. And their whole schtick about we’re three times cheaper than a human pipetting at the bench and things like that is really a function of an acoustic liquid-handling robot that transfers a very tiny amount of material from well to well and that allows them to achieve those types of numbers. It’s just a robot, basically.” - Former director level executive

Ginkgo has no unique equipment or “any really unique IP”

“On a high level, I don’t think they’re unique and special at all...it doesn’t have any unique equipment; I don’t think it has any really unique IP.” - Former employee in a managerial role

Ginkgo has no particularly unique technology “that other companies don’t have”

“The first thing is that they don’t have any unique technology that other companies don’t have. The second is that they don’t have any product. The products that they do have or what they call their products are strains that they’ve delivered to customers that generate revenue. But because there aren’t many of those successful strains, I believe that their actual revenue coming from those strains is something like \$500,000 a year. Customers, if they’re using their strains, are using them in a very limited capacity because they’re not happy with them. And they’re not using them for the original purposes. So, lack of platform uniqueness, lack of product.” - Former employee in a managerial role

A current executive at key related-party “customer” Joyn Bio – co-located in Ginkgo’s Boston offices and foundry – corroborated comments by ex-employees that Ginkgo’s foundry services are nothing differentiated: wouldn’t personally invest in Ginkgo; “I can’t understand the basis for the hype”; “synthetic biology” is a commodity; “hard for me to point to something that they do that is totally different than anybody else in the industry.”

Joyn executive wouldn’t invest in Ginkgo; can’t understand the “hype”; not differentiated and syntenic biology is becoming a commodity

Q: **“Would you ever personally invest in Ginkgo?”**

A: **“I would say not**, mostly because of what I was talking about before. I think Amyris is actually a more interesting company...I think Ginkgo is overvalued. **I can’t understand the basis for the hype**...Like I said before, you have to be able to—I think **you have to be differentiated** in some way to be competing against...Let’s take a step back. Think about DNA sequencing. There was a point where that was something special, and people would do it in-house.. And now it’s become effectively commoditized. So, what’s differentiating about DNA sequencing? Maybe you have enormous capacity, or maybe you have super, super-fast turnaround, or maybe you’re the backend, and your informatics is really good. But it’s hard to be special as a sequencing company. And I worry that some of the other elements of **synthetic biology, as people talk about it industrially, are a little bit that they are ripe to be commoditized. What I don’t have a vision of for Ginkgo is what is it about them that’s going to be in that context is going to be differentiating?”** – Current Joyn executive

Joyn Bio can’t point to anything different at Ginkgo’s foundry vs. other strain engineering service providers

Q: “The foundry services that Ginkgo provides, are these are basically services you can get from a lot of other people?”

A: “Yeah, I would say there’s probably—it would be **hard for me to point to something that they do that is totally different than anybody else in the industry.**”

Q: “That doesn’t bode well in the valuation.”

A: “No comment.” – Current Joyn executive

A second employee at Joyn Bio referred to Ginkgo as “a kind of a glorified CRO” – a contract research organization. An executive at another key related-party, Motif Foodworks – also co-located with Ginkgo’s offices and foundry in Boston – indicated that similar strain engineering services are widely available from other vendors.

Senior employee at Joyn calls them “a kind of a glorified CRO”

*“They can do strain engineering and DNA synthesis and sequencing at a very high level, high throughput quickly. You can run a lot of tests through their program very quickly. They can engineer a strain and test them with very high throughput. **But how we are working with them now is as a kind of a glorified CRO.**”* – Senior employee at Joyn

Current executive at related party “customer” Motif indicated similar strain engineering services are widely available from other vendors

*“As far as the other people that one might go-to for this kind of work, you can think of that from different from angles, but **Zymergen is one of them. You can think about Amyris**; you can almost think about Genomatica; they could do the same thing. There’s also a privately-owned company locally. And then there are other smaller ones that have expression platforms on different bugs, e. Coli bacteria, yeast, fungi. **So, there are different levels of players; there are other people out there.**”* – Current Motif executive

Motif Foodworks is a key related-party “customer” and the one with whom Ginkgo reports the largest revenue and deferred revenue - \$21MM in 2020, and \$54MM as of 12/31/21, respectively – yet a current executive slammed Ginkgo’s foundry services. He indicated Ginkgo’s foundry staff can’t deliver strains without babysitting and micromanaging; that staff are junior and inexperienced, resorting to simplistic high-throughput screens as a crutch, which doesn’t result in successful yeast strains; and that Ginkgo lacks “senior people” who can oversee the science.

Current executive at related party “customer” Motif indicated Ginkgo’s staff can’t deliver without babysitting and micromanaging

*“As a customer, I know how much input I have in our success in helping them be successful. If I am another company and I don’t have a head of synthetic biology that’s going to be really working closely with them and monitoring their progress, it’s going to be risky...It’s going to be risky, yes, because **I see how Ginkgo works and I see how much guidance I give them, and they’re not at the level of, I could just forget about them and come back in a year, and they have a strain**, and they have the best thing they can do. I’m going to be paying a lot of money without overseeing anything, and it’s going to be—**their chance of success without me guiding them is going to be very limited.**” – Current Motif executive*

The executive indicated that Ginkgo’s staff is junior and inexperienced, and resorts to simplistic high throughput screening as a crutch, which doesn’t result in successful strains; “no senior people” that can oversee the science
“I would say they are too junior, yes. They’ve been growing very fast; they have amazing people, don’t get me wrong, there are very smart people out there, but they’re very young, and **there are no senior people that really have the time to look and spend time on the science** and understanding what’s going on and make the right decisions. And so, they always revert to their high throughput platform, and **it’s just like their solution is always to go and test 1000 mutants.** It’s good to get all those mutants, and at the same time, sometimes you need to do and add those mutants together, and you don’t need a robot to do that, you just need a little time to do it, and you need to select the right ones, and move forward. **So, it’s not like, magically, by screening a lot of mutants, you’re going to be able to move forward.** If you have a strain development that requires you to improve five different things, they’ll be very good at focusing on each of these individual steps and find the best one of each of those steps. But then, when it comes to merging them together, they’re going to be lumped into how many best mutants they do and things, and **they need guidance. They get sidetracked sometimes,** I would say.” – Current Motif executive

He indicated that Ginkgo can't deliver and lacks capabilities to manage strain engineering projects – the entire purpose of the foundry – and that projects are bound to fail unless a customer is sophisticated enough to micromanage them. He stated that foundry staff are generally straight out of school and in their first jobs. A senior employee at Joyn provided identical color: Ginkgo is unable to execute on foundry projects; team is young, inexperienced, and has no “mindset around customer service.”

Current executive at related party “customer” Motif indicated Ginkgo’s staff needs to get their act together; customers have to be sophisticated enough to project manage Ginkgo’s projects themselves, as they fail if Ginkgo oversees them

“They’re going to have to get their act together to make sure that they understand what the customer needs and what they need to deliver...Honestly, I don't see how—they might get lucky, but in my experience right now, if you don't have, from the customer side of things, if they don't have someone that understands synthetic biology and understands the process and can really review the data, review the project, the advancement of the project is going to be difficult to be successful, unless Ginkgo hires really amazing program directors.” – Current Motif executive

Ginkgo staff is generally straight out of school and in their first job; Ginkgo can't get senior people

“A lot of them are pretty much fresh out of Ph.D. So, this is their first job...at the speed they're growing right now, it's difficult for them to recruit senior people.” – Current Motif executive

Ginkgo was unable to execute; young, inexperienced team; no “mindset around customer service”

“I think it just really comes back to having a **very young, inexperienced team**. We would give them a statement of work and a proposal for what we wanted to have done, and they would just seem to **not be able to execute on it. Or we had them try to develop an assay for us, and we kept getting no results back from it. The assay just wasn't done; it wasn't developed. They kept dragging on,** or they would give us something that we thought should be done, and we would test it and find that it didn't work the way it was supposed or they hadn't used the proper controls...**the mindset around customer service didn't seem to be there. It was very casual about,** oh well, we can just keep fiddling around with this and charging you for it.” – Senior employee at Joyn

Engineering east and other standard hosts involves two phases: 1) the front-end where strains are identified via trial and error, where previous pages indicate that Ginkgo offers nothing unique; and 2) the fermentation and scale-up phase, where strains of interest are grown in quantities that are industrially relevant, versus the microscopic quantities at the front-end. Scaling and cost remain an ongoing, unsolved challenge for the “synthetic biology” industry. Many have coaxed microbes to excrete biofuels, but 1/100th of a teaspoon doesn’t power a car, not is it feasible if it costs \$100,000 to fill up a tank with precious microbial juice. An ex-Ginkgo executive stated that half of their projects fail at the front end and the other half at fermentation/scale-up, given Ginkgo’s inability achieve any cost metrics: “over-promised and under-delivered”; stated Zymergen “imploded” for a similar inability to manufacture their film.

Half of Ginkgo’s project fail at front-end strain engineering

*“The conversion rate that I spoke of, like how many projects actually succeed, I think a lot of that has to do with, I’d say **half of them fail on the very front-end**, where it’s just very hard to make that product, it’s a very complex molecule, it’s difficult. Biology is hard, and there’s a lot of crosstalk, and you might be able to make a trace amount of something, but you can’t make grams per liter of it the way you could from a barrel of oil or something like that.” - Former Ginkgo executive*

Other at back-end fermentation and scale-up; Ginkgo can achieve any cost of goods metrics; “over-promised and under-delivered”; same reason Zymergen “imploded” – couldn’t manufacture their film

***“Half of them probably fail on the backend** where you’re trying to meet a certain cost of goods. Ginkgo historically has done very little work in pharma, and so, a lot of the extracts from plants and naturally-occurring vitamins and things like that, there are naturally-occurring alternatives, and so you really do need to provide a low-cost, uniform, high-quality product. **They can’t really achieve those metrics. It’s sort of over-promised and under-delivered in the backend. That’s something like Zymergen just went through that and kind of imploded in part because they couldn’t figure out how to scale up their proprietary film.**” - Former Ginkgo executive*

The ex-executive elaborated that Ginkgo has “no real differentiating technology” on the back-end either, and little ability to scale fermentation and strain production. As a result, Ginkgo just outsources to third-party fermentation facilities and “hopes for the best.” He described their woes as a reflection of the space, where companies can’t make products cost-competitively: every synthetic biology “ever started” has tried to make vanilla and other “marquee ingredients” but failed, as it’s cheaper to buy natural vanilla from third-world countries; “market doesn’t want what Ginkgo’s able to make”; “problem with the business model.”

Ginkgo has “no real differentiating technology” and little ability to scale fermentation and strain production; just outsourcing to third-party fermentation facilities and “hope for the best”

*“The backend of it is the fermentation and scale-up. I think that Ginkgo has no real differentiating technology when it comes to that. They just sort of test their strains and fermenters to try to scale them at the University of Pennsylvania or another kind of commercial fermentation facility in Mexico City or elsewhere, and they hope for the best. They try a bunch of strains and hope that one of them scales. **That’s the foundry in a nutshell.**” – Ex-Ginkgo executive*

Ginkgo isn’t cost competitive, similar to every other synthetic biology company

*“I think it really just comes down to; you can have very sophisticated technology that makes very elaborate assembly lines inside of cells, but **they are not able to compete on cost. Some of this is basic economics. You have to pay for glucose or your sugar feedstock and all of your other ingredients.** Agriculture is pretty efficient when it comes down to it. I mean, some of it is heavily subsidized, too. But if you want rose oil, you can go plant roses in a third-world country and produce some for pretty darn cheap, and vanilla and some of these other ingredients. **Every synthetic biology company that’s ever started has tried to make vanilla and a few other marquee ingredients, but they don’t have meaningful revenue, or those products were never commercialized.**” – Ex-Ginkgo executive*

“Market doesn’t want what Ginkgo’s able to make”; “problem with the business model”

*“But **it’s really the market doesn’t want what Ginkgo’s able to make. I think it’s as simple as that.** Maybe that’s because the **cost of producing all of these ingredients in a global economy is just very low,** and they should actually just focus on low volume, high-value products and not do these kinds of commodity-like ingredients because anyone can make those commodity ingredients. **I think there’s a problem with the business model and what they have focused on historically.**” – Ex-Ginkgo executive*

15. Long history of failure with little to nothing delivered in the way of successful strains and products.

Former employees and executives painted a consistent and damning profile of Ginkgo as a company with a long history of failure with little to nothing delivered in the way of successful strains and products: “What went wrong is probably the \$15 billion question”; all hype with little substance – “What successful products have come out of the back end? I have not seen anything that I would call a material success”; Ginkgo has “way over-promised and way under-delivered”; company has failed, even if measured by delivering minor product successes vs. major ones.

What went wrong is “the \$15B question”

“What went wrong is probably the \$15 billion question. Who knows?...I'm surprised, and I'm disappointed myself that it just hasn't happened yet.” - Former Ginkgo executive

All hype with little substance and no materially successful products to date

*“They have announced a lot of deals, and they've got a ton of fanfare because they bring investors through, and they've got very beautiful, shiny equipment, and Jason tells an incredible story. They've convinced a lot of people to invest in the company. But **you hit the nail on the head. What successful products have come out of the back end? I have not seen anything that I would call a material success** coming out the back end. ”* - Former Ginkgo executive

Ginkgo “has way over-promised and way under-delivered”

*“One of the things that I think has gone wrong is Ginkgo has **way over-promised and way under-delivered**, and they should never have made that many promises. I think Jason did that so he could raise the money.”* - Former Ginkgo executive

Ginkgo has failed, even if one lowers the bar to more modest successes vs. billion dollar products

*“Because not every product needs to be a billion-dollar product and succeed. But **you need to have a reasonable conversion, and I don't think they have a reasonable conversion..**”* - Former Ginkgo executive

We quote three different ex-employees whose commentary can only be described as devastating: “There’s no real business there”; “Where are those products? For a company that’s been around for 13 years”; “Who has been successful using your technology that has led to a meaningful product? And the answer is there isn’t any.” One was surprised at Ginkgo’s inability to deliver on even easy products like flavors and fragrances that are the bread and butter of strain engineering CRO’s, and noted the lack of announcements indicating success in the product pipeline when he was there.

“There’s no real business there”; nothing tangible; lack of any product success after 13 years is red flag

“I think the big hiccup or red flag right now is you could say, **where are those products? For a company that's been around for 13 years**, it seems to me, if I was going public, I'd have this long list of amazing products—they don't all have to be billion-dollar products, but **something that looks tangible** that says, we have succeeded and provided value to our partners and provided a good return on investment. **And they don't have that story. They don't have those products.** And so, do you believe that they are going to build those products and are going to have some success? Or do you believe that it's just a smokescreen? **There's no real business there.**” - Former director-level employee

No meaningful product has come out Ginkgo

“Maybe Ginkgo will be successful at some point, but when the rubber meets the road, you look at what products are out there. **Who has been successful using your technology that has led to a meaningful product? And the answer is there isn't any...**I would have expected more success for the amount of money that they've raised at this point in time.” - Former Ginkgo executive

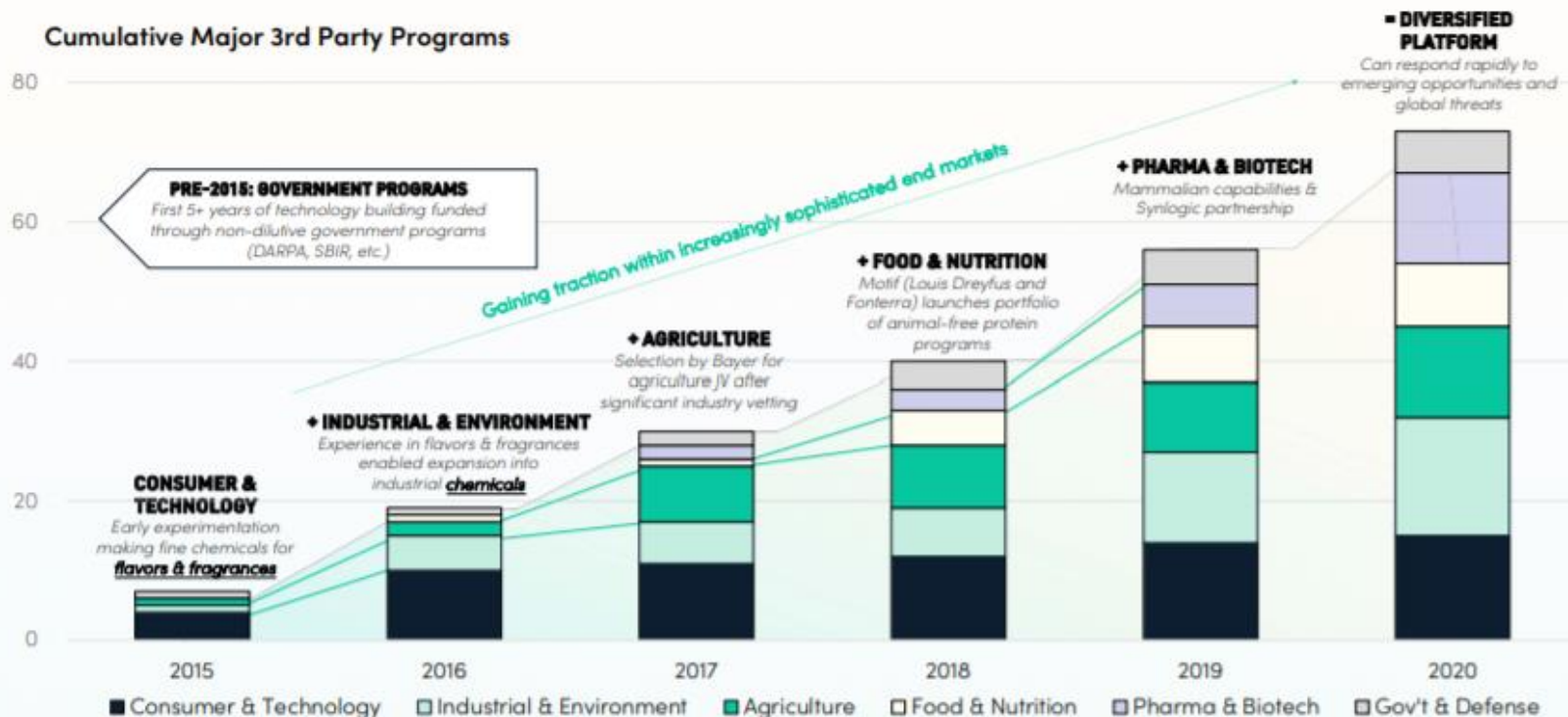
Former director surprised at lack of product success even in easy products like flavors and fragrances that many companies synthesize; hasn't seen product announcements for pipeline projects he worked on

“I'm actually surprised that there haven't been more product announcements, things that succeeded that were in the pipeline while I was there because you have manufacturers, like these large multinational agricultural companies that make ingredients for all sorts of things. There are a lot of manufactured good ingredients and flavors and fragrances and things like that. **You'd think it shouldn't be too difficult to do to succeed on some of them to have a story to tell.**” - Former Ginkgo employee

Ginkgo hypes the number of “collaborations” it has signed – an elaborate word for serving as a contract research organization (CRO) that attempts to engineer strains of yeast to excrete various compounds. Ginkgo states it has run 70 “Cumulative Major 3rd Party Programs” on its platform through 2020.

From May 2021 Investor Presentation

Over 70 major customer programs across diverse industries
have run on our platform through 2020



Note: "Major" programs exclude proof of concept work and ancillary projects and typically have at least \$500K actual / expected development costs on behalf of a customer

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Given Ginkgo's abject, decade-long failure to deliver strains that work – a devastating indictment of the secret sauce behind its “Foundry” and “Codebase” - the color from an ex-business development executive and others was revealing. The executive indicated he and his team personally signed 20 collaborations and “zero percent panned out.” He stated the syntenic customers were generally major, well-known names, and stated the actual number of failed collaborations is higher, as Ginkgo fails to list many of them: “Those collaborations are just gone. They just ended them.”

All hype with little substance and no materially successful products to date

*“One thing that Jason’s really good at is publicizing success. So, if any of those collaborations were successful, he would have publicized it, and I haven’t seen any of those publicized. So, as far as I’m concerned, **they’re likely zero percent that panned out. When I was there, between my team and me, we signed 20**...I mean, every single one of these companies were—if I named the name of the company, you would have heard of them. They may not all have been Fortune 500, but a bunch of them were.” - Former Ginkgo executive*

Failed collaborations aren’t even listed – “just gone,” “just ended them”

*“I was involved in setting up and bringing in Joyn, that collaboration, and also Genomatica. There were many others on top of that they didn’t list. In many cases, **those collaborations are just gone. They just ended them**. They also—and I’m pretty sure it’s buried somewhere in there, and if not, you can find it elsewhere—they also have a collaboration with Amyris, and that ended as well.” - Former Ginkgo executive*

Three other ex-employees provided color identical to that of the business development executive. A former strain engineer stated that “9 out of 10” projects didn’t succeed, out of a cumulative 200+ projects he was aware of since inception. A second ex-employee stated “Lot of projects fail. I don’t know if it’s 90% or 70%. But yes, it is high.” A third stated that the few strain engineering projects that work - the 1 out of 10 or so - are only minor successes that generate almost no revenue and are ones that customers are still “not happy with.”

Former strain engineer who led projects says 90% of Ginkgo 200+ projects have failed

A: “Their big success stories are projects like Cronos, but **for every project like Cronos, there are nine others that don’t make money, probably don’t pay for themselves, and don’t result in a final product.** When I was leaving, they were past 200 projects; they were up in the 220’s.”

Q: “So, you’re saying 215 projects, you’re saying about 9 out of 10 of those didn’t really work?”

A: “Hard to say. I would say that resulted in a strain that the customer turned around, put in a vat, and then did all of the processing, sold it on the market, and made a profit. **Yeah. Probably 9 out of 10 of those did not succeed** at that level. The other thing is that because it’s so expensive to do the fermentation and processing, if the strain does not hit a certain threshold of efficiency, it’s not necessarily economically viable to even go to the hundred-thousand-liter fermenters.” - Former strain engineer

Ex-business development employee confirmed that most projects fail

“I would say certainly at the level of individual projects; yeah, **lots of projects fail. I don’t know if it’s 90% or 70%. But yes, it is high. And certainly, some customer collaborations have come and gone.**” - Former Ginkgo employee

“Successful strains” delivered to customers may only generate ~\$500K of revenue per year; customers are unhappy with them

“The first thing is that they don’t have any unique technology that other companies don’t have. The second is that they don’t have any product. The products that they do have or what they call their products are strains that they’ve delivered to customers that generate revenue. **But because there aren’t many of those successful strains, I believe that their actual revenue coming from those strains is something like \$500,000 a year. Customers, if they’re using their strains, are using them in a very limited capacity because they’re not happy with them.** And they’re not using them for the original purposes. So, lack of platform uniqueness, lack of product.” - Former employee

One ex-employee provided commentary we found particularly withering - **“I continually was asking during my entire tenure there, can you give me a list of successful delivery examples?” – and indicated an inability to find “even one or two examples of customers” who are “actually super happy with the strains that Ginkgo is producing. The individual stated “there’s a huge amount of hype,” and conveyed discomfort from a personal reputational and “ethical perspective” in bringing on customers via personal relationships if Ginkgo couldn’t deliver for them.**

Former employee couldn’t find examples internally of customers “that are actually super-happy with the strains that Ginkgo is producing,” despite asking multiple teams inside Ginkgo for the information; the ex-employee was concerned about personal reputational consequences and “didn’t feel comfortable” in “bringing on customers” if Ginkgo couldn’t deliver

*“The other thing was there were huge issues with delivery. I think there’s a huge amount of hype,, and as far as I’m aware, **there isn’t even, maybe even one or two examples of customers that have been paid, whether it’s circular or not, that are actually super happy with the strains that Ginkgo is producing.** The Synlogic thing is one example of customers who aren’t happy. But I continually was asking during my entire tenure there, can you give me a list of successful delivery examples? When have we executed on a promise to a customer? **And nobody could give me a single example.** I thought that was very concerning because in my role, I was bringing on customers left and right, and I didn’t feel comfortable bringing on customers to a company that couldn’t deliver because my reputation was on the line... **I asked people in all the different teams I was intersecting with and nobody gave me an example,** so I started going to the technical team, with the folks who are actually executing on the customer projects... **For me, from an ethical perspective, I just needed to know.** I didn’t want, not only my reputation, but I didn’t want to engage my customers’ time that they could never get back in something that wasn’t going to work for them.” - Former Ginkgo employee*

A current executive at key related-party “customer” Joyn Bio substantiated the color provided by former employees of Ginkgo: “I’m just worried that I’m not seeing Ginkgo delivering on that”; “I can’t tell if that is really playing out”; “I haven’t seen the substance for their other partners or other projects.”

Current Joyn executive is “worried that I’m not seeing Ginkgo” deliver on its collaborations; hasn’t “seen the substance for their other partners or other projects”

“That’s the challenge for me. **I can’t tell if that is really playing out.** What to me is more interesting about Amyris and I think Zymergen if you look at the DOD or DOE, this thing where they ask them can you use synthetic biology to produce 100 arbitrary novel compounds that there’s no synthetic route to? And if they were able to do that at scale relatively rapidly, that to me is proof is in the pudding there. **I’m just worried that I’m not seeing Ginkgo delivering on that. I’m not saying that they don’t or that they haven’t; I haven’t seen the substance for their other partners or other projects.**” – Current Joyn executive

16. Each of Ginkgo's key related-party collaborations has been a flop. We present five case studies

Ginkgo has six key collaborations, each of which is a related party in which Ginkgo and/or its investors invested. We spoke with current and/or former executives at five. The sixth appears to be a shell with no employees. Our interviews indicate that each of the six is a flop where Ginkgo has failed to deliver strains for the vertical the collaboration is in. The revenue and deferred revenue Ginkgo attributes to each is shown below. Two of the six are “Structured Partnerships” where equity is provided to an existing entity that agrees to “purchase” R&D from Ginkgo’s foundry: Synlogic (a microcap drug discovery company) and Genomatica (private, focused on chemicals). The rest are “Platform Ventures” where Ginkgo creates a new entity that it funds: Joyn Bio (nitrogen-based fertilizer); Motif (substitutes for animal products); Allonnia (waste remediation); and Kalo (beauty and personal care).

<i>in millions USD</i>	2019	2020	Q1 2021
Foundry revenue by related party			
Joyn	9,349	7,273	1,598
Motif	18,986	20,798	5,492
Genomatica	6,248	9,431	3,298
Allonnia		4,960	2,266
Synlogic	17	73	6
Glycosyn	668	-	-
Kalo	-	-	-
Total	35,268	42,535	12,660
Total foundry revenue	54,184	59,221	22,504
Related parties as % of total foundry revenue	65%	72%	56%

<i>in millions USD</i>	12/31/2019	12/31/2020	3/31/2021
Deferred revenue by related party			
Joyn	17,135	9,862	8,281
Motif	62,513	53,952	53,299
Genomatica	38,059	30,128	26,680
Allonnia	24,480	26,064	37,959
Synlogic	144	72	66
Kalo	-	-	11,897
Total	142,331	120,078	138,182
Total deferred revenue	147,898	128,475	147,408
Related parties as % of total deferred revenue	96%	93%	94%

Case study 1: Synlogic collaboration. We cover each key collaboration and share our findings, beginning with Synlogic, a \$200MM market cap early-stage biotech focused “synthetic biotic” drugs. **A current executive of Synlogic spoke at length about their experience with Ginkgo and indicated Ginkgo failed to deliver successful strains: Synlogic had to switch to Zymergen instead, which successfully engineered a strain after Ginkgo flopped; stated Ginkgo has “not impacted” their clinical programs “at all”; and suggested at Ginkgo’s “SPAC pitch deck” is promotional and was dismissive of their focus on trivial areas like fragrances: “we’re not trying to do science experiments, we’re trying to develop drugs.”**

Synlogic had to turn to Zymergen instead for the strain it needed, after Ginkgo failed to deliver

*“In the PKU program, we have two programs for PKU: one is a first-generation strain that we developed ourselves, homegrown. The second-generation program has an enzyme that we did work with Zymergen on, and **Zymergen was able to identify an enzyme with improved activity. And that’s one of the enzymes that we ended up engineering into our next-generation strain**”* – Synlogic executive

Ginkgo has made no impact on Synlogic’s clinical programs

Q: “Have they made any major impact? Or is this all incremental-level stuff that they’re doing here?”

A: “For the two clinical-stage assets for the PKU [phenylketonuria] program and the HOX [enteric hyperoxaluria] program, **they have not impacted those programs at all...We’re not trying to do science experiments; we’re trying to develop drugs.**” – Current Synlogic executive

Executive suggested that Ginkgo is promotional and has had little meaningful product success to date

*“The whole SPAC pitch deck, it’s very aspirational and beautiful, and it looks great and all that stuff. If you like that kind of stuff, it’s an impressive example of how to do that, I guess. **I hope that they can make a bigger impact than some of the stuff that has been their successes so far, things like the fragrances.**”* – Synlogic executive

Case study 1: Synlogic (cont'd): Synlogic executive indicated they never would have “spent” \$30MM with Ginkgo if Ginkgo hadn’t financed them; agreed that it was basically just a round-tripping agreement; and was diplomatic but conveyed that they struggled to find strain engineering work that Ginkgo could do well, and that Synlogic still does most strain engineering in house.

Suggested Synlogic is only doing R&D with Ginkgo because of the \$30MM prepaid research deal; will “have a higher level of scrutiny” thereafter

“We had that \$30 million to spend. We want to spend it eventually, and we are. I would go to them for certain things or at least consider them for certain things. **It’s expensive to do work there** without the prepaid component of it; it would be hard to do everything there because it would get expensive pretty quick. But there would be some things, I can say for sure that we would consider doing work with them, and we are, and we will. Will that change when we get through all the money that we’ve pre-allocated? **We will probably have a higher level of scrutiny for what we do for sure.”** – Current Synlogic executive

Synlogic never would have spent \$30MM with Ginkgo if Ginkgo didn’t finance them; agreed that it was basically just a round-tripping arrangement

Q: “It just seems an odd deal, more on Ginkgo’s part. If I was Synlogic, I’d be like, okay, you’re going to basically finance us to be your customer, sure. My car dealer wants me to buy a car, and he’s happy to put \$35,000 in my account, and he wants me to give that to him in exchange for the car, sure. I’ll buy whatever he has.”

A: **“We did not have \$30 million to spend on it for sure without the financing that came with it.... What I would say, I think you’re reading the agreement correctly in terms of what the deal was.** Yeah, I don’t know how sustainable that is. I’ll just say that.” – Current Synlogic executive

The executive was diplomatic but suggested that hey struggled to find project work that Ginkgo could actually do well; still do most of the strain engineering in-house

Q: “What are you having them do? Is it re-engineering e. Coli?”

A: “I would say it’s changed over time; like with anything, **it probably took us a little time to figure out what makes a good project between both companies, just what we needed from them, what worked for us...** **We do most of the engineering of the strains** and combining the parts into the clinical candidates and all that kind of stuff ourselves at Synlogic. Ginkgo was really helping us at the early stages—you can think about it as helping us find parts to use in the strains to satisfy different types of criteria.” – Current Synlogic executive

Case study 1: Synlogic (cont'd): Synlogic executive continued that Ginkgo is just another commodity contract research organization (CRO) that thinks it has “a unique business model”; and that Synlogic has sent less than 10 projects to Ginkgo in two years, despite a \$30MM R&D prepayment – “It’s like working with a CRO”; “there’s nothing really that unique”; can get “the same kind of thing at any other CRO for a piece of work.”

Ginkgo is just another commodity CRO that thinks it has “a unique business model”

“The PR is definitely amazing; they’re good at that for sure...I think if you break down the different parts of the business and what they offer, you certainly can find other places that can do aspects of that. To me, they’re a CRO, that they’re providing a service, and they may be thinking they have a unique business model behind that in some way, but we would look at a project with them just like we would look at a project with Zymergen or anyone else if we were scoping out what the question was and what the right person would do and we definitely get competitive bids for things when we have to.”

– Synlogic executive

Executive elaborated that one can get “the same kind of thing at any other CRO for a piece of work”

“It’s like working with a CRO. We decide that we want to work on enzyme x; we want to improve that enzyme's activity. We get a proposal from them. We work on the proposal together, and we negotiate aspects of it, we take stuff out and put stuff in, and that proposal has a cost associated with it, and it has milestones associated with it, and as the project progresses, we make payments accordingly. **As far as I know, or I can tell, there's nothing really that unique like I would expect to do the same kind of thing at any other CRO for a piece of work.”** – Current Synlogic executive

Synlogic has sent less than 10 projects to Ginkgo in two years, despite a \$30MM R&D prepayment

Q: “How many projects have you done with them in these two years?”

A: “Many. Not two and not 20. Many, **less than 10.**” – Current Synlogic executive

Case study 1: Synlogic (cont'd): Two former employees of Ginkgo corroborated the Synlogic executive's account, one in a more roundabout manner and another more specifically indicating that Synlogic had to switch to Zymergen after Ginkgo failed.

Former employee indicated that Ginkgo failed to deliver a strain for Synlogic, who then had to switch to Zymergen who delivered successfully

*"The issue with the circular revenue with Synlogic, basically, they provided Synlogic \$30 million, Synlogic turned around and paid it to them, so they were registering their own money as revenue, **and then they tried to make a strain for Synlogic, and it didn't work, and then they had to go to Zymergen and get the strain. I think just two days ago; the clinical trials showed that the Zymergen strain was effective.**"* – Former employee in a managerial role

Former Ginkgo employee's verbal gymnastics suggested to us that the Ginkgo project was a failure

Q: "How did that project go with Synlogic? Was it successful? Was it a failure?"

A: "I wouldn't be able to comment on that... This was between Ginkgo and Synlogic **when I was there; they were still working on it. I can't comment beyond that.**"

Q: "Are you able to offer an opinion as to whether you think it will be successful in the future?"

A: "I hope so."

Q: "You hope so, but do you think it will?"

A: "It's biology. **Nobody even at Ginkgo can say whether it will or not**" – Former Ginkgo employee

Case study 2: Genomatica, a privately-held company focusing on synthetic biology applications in the chemicals space. We earlier detailed color from a senior employee who indicated that they have never paid Ginkgo cash for foundry r&d services. The employee conveyed an additional point we found particularly damning: if Genomatica had to pay cash for Ginkgo's R&D services, they would not do so. The employee described Ginkgo as simply a CRO, a contract research organization, and one that they don't trust to pass meaningful projects to: "we don't expose the critical path stuff"; "they haven't proven themselves yet as a CRO service provider"; when it comes to paying cash: "At the moment today, we would have a hard time justifying that."

"Would have a hard time justifying" spending cash at Ginkgo equivalent to R&D credits spend to date

A: "I think the real critical question that if I were in your shoes, **the kind of question I would wonder is - would we spend the amount of cash today at Ginkgo that we've spent on R&D credits** to get the result that we've gotten?"

Q: **"And the answer sounds like no."**

A: **"That's right. At the moment today, we would have a hard time justifying that"**... You know what? Hell, they just went and raised a billion-plus dollars. If they want to pour that into their foundry and they want to create an engine that can reduce costs and ultimately, whether it's one year, two years, three years from now, they can do something that would be a hell of a lot cheaper than having our people here do it; we're happy to pay the money. We don't need to make that decision today." – Senior Genomatica employee

Ginkgo hasn't proven itself as a CRO

"Again, **we don't expose the critical path stuff** yet, very specifically because we don't want to yet put ourselves in a position—**they haven't proven themselves yet as a CRO service provider** to where we would put ourselves in that position. If they can deliver at the level that we need, both time, quality, all that sort of stuff, we'll be happy to give them more work. **Are we there yet? No.** Are we open-minded to see if they can get there? Sure." – Senior employee of Genomatica

Case study 3: Joyn Bio, is a key related-party “customer” with Bayer involved in the venture, which is attempting to develop a nitrogen-fixing organism that can replace conventional nitrogen fertilizers. An ex-executive involved in setting up the venture called it “a moonshot” with “pretty small” chances, suggesting he’s heard nothing indicating success. An ex-strain engineer stated “there is not a lot of success coming out of that.” A current Joyn executive called them “a glorified CRO” and said that while of their R&D spend is with Ginkgo today, by virtue of it being a captive entity, they’d slash that to 15-20% of their R&D budget if it was up to them.

Executive involved in setting it up hasn’t heard of any progress, changes of success are “pretty small”

“I haven’t heard anything. Honestly, I just know that they keep saying they’re making progress, but they don’t say what that progress is. I monitor it as well. Nothing’s been announced. Nitrogen fixation is extremely difficult. It truly is a moonshot. If they get this to work, it’s huge. But **what are their chances? I think it’s pretty small.**” - Former Ginkgo executive

Senior Joyn employee would only spend 15-20% with Ginkgo if given the freedom to choose other vendors

Q: “It sounds like you tried to use other vendors. What percentage of your R&D spend would you give to the foundry if you were just doing what’s in the best interest of Joyn?”

A: “The high throughput sequencing, the DNA synthesis, all of the highly automated processing of samples that they are really good at—all of that stuff is fine. **I’d probably leave it at the 15% to 20% level. The percentage of our budget that goes to them today is** probably 60%, 70%, or more right now. I don’t actually have visibility on the whole R&D budget, unfortunately... **how we are working with them now is as a kind of a glorified CRO.**” – Senior employee at Joyn

Former strain engineer suggested Joyn is a failed project

“Joyn, it’s hard to tell how far they’ve gotten. I know that testing things in the soil with plants takes a long time, but it seems like **there is not a lot of success coming out of that**... They are testing a strain in the field, from what I understand. But no, they’ve spent a long time working on it, and they haven’t announced anything that was a big wow moment.” – Former Ginkgo strain engineer

Case studies 4 and 5: Motif and Allonnia. Motif is a key related-party “customer” in the food space. A senior employee at Joyn who speaks with colleagues at Motif – both are “platform ventures’ co-located in Ginkgo’s building – told us that **Motif “are very spicy about Ginkgo. They did not have a good experience with them,”** and added that **“All I know is that the CEO of Motif is barely on speaking terms with the Ginkgo folks anymore.”** An ex-employee of Ginkgo in a managerial role alluded that **Allonnia is “extremely unhappy with Ginkgo,”** implying Ginkgo had failed to deliver.

Former employee alluded Allonnia is “extremely unhappy with Ginkgo”

*“[Redacted] on the technical team was assigned to execute on a **customer project in the environmental remediation space** where they needed to design enzymes to help clean the environment in a specific way. **That customer had raised a lot of money to be able to do this and was paying all of it to Ginkgo. I think a huge amount of it was Ginkgo's own investment.** [Redacted stated] that in any other company, it wouldn't have been impossible to do this project if Ginkgo had been approaching it the right way...**That specific company is extremely unhappy with Ginkgo, and it's very tragic because they have to work very closely with Ginkgo**” - Former Ginkgo employee in a managerial role*

Senior employee at another related-party “customer” believes that Motif “did not have a good experience” with Ginkgo and that Motif’s CEO is “barely on speaking terms” with Ginkgo anymore

*“If you are lucky enough to get a hold of someone from Motif, **they are very spicy about Ginkgo. They did not have a good experience with them** and separated themselves as completely from Ginkgo as they could at the earliest possible date. I've not been told any of the details. **All I know is that the CEO of Motif is barely on speaking terms with the Ginkgo folks anymore**...” – Director-level employee of Joyn Bio*

17. Ginkgo is virtually identical to its closest synthetic biology competitor Zymergen (ticker: ZY), which fell >70% in a single day two months ago. We believe a similar outcome is more likely than not at Ginkgo

One ex-employee of Ginkgo after another stated that Zymergen and Ginkgo are virtually identical animals. A former executive indicated they both had the same “Foundry” technology, even using the same suppliers to equip their labs; that **Ginkgo CEO’s “biggest fear was always Zymergen”**; and that Ginkgo was worried after Zymergen’s stock imploded in August, and “announced all these deals to try to be able to show that they’re different.” An ex-employee who was a longtime direct report to the CEO stated **“they’re very similar”** with the only difference being that Zymergen wanted to specialize in a few verticals while Ginkgo wanted to be a generalist strain engineering CRO.

Same lab technology as Zymergen, both were using the same supplier; CEO’s “biggest fear was always Zymergen”

“Zymergen was doing the same thing, and Sartorius is a major company that’s one of the huge suppliers in the industry. When I was at Ginkgo, **Jason’s biggest fear was always Zymergen**. And so, he would try to corner the market by buying all the equipment so Zymergen couldn’t buy any. Just stuff like that. There are very few suppliers for this type of equipment.” - Former Ginkgo executive

Ginkgo was worried and tried to downplay its similarity to Zymergen after its stock collapsed

“Ginkgo was quickly in defensive mode [after Zymergen’s stock collapsed] because they didn’t want people to make that comparison, and so they announced all these deals to try to be able to show that they’re different. Maybe Ginkgo will be successful at some point, but when the rubber meets the road, you look at what products are out there. **Who has been successful using your technology that has led to a meaningful product? And the answer is there isn’t any.**” - Former Ginkgo executive

“They’re very similar”; some minor differences in vertical targeted

“I think that in certain ways, they’re very similar. They competed on a number of very large-dollar number projects...I would say where Ginkgo and Zymergen diverged was a few years ago, Zymergen started to specialize in a few application areas: materials, ag, pharma, a few other one-offs. Whereas Ginkgo said, we’re not going to specialize. We want to be able to engineer everything. And so, the types of projects that the company took on varied from cannabis projects to materials to biodefense.” - Former Ginkgo employee, longtime direct report to CEO Jason Kelly

A former director-level employee of Ginkgo re-iterated comments by his colleagues, stating that Zymergen and Ginkgo are very similar, and that if anything Zymergen's science was lower-risk. He indicated that both companies "were always looking jealously at each other"; that there's little differentiation between the two; and that they're "more similar to each other than they are different."

Zymergen and Ginkgo are very similar, except that Zymergen focused on "lower-risk science"

"I mean, they had very similar beginnings, Zymergen and Ginkgo. What was different about the two companies, really, and this maybe seems like I'm splitting hairs a little bit, but Zymergen was always focused on lower-risk science. Their bread and butter was really taking an industrial strain that made an antibiotic or a biopesticide or something like that, and they would improve the titer of that strain, the cost of goods. And the whole idea was, we get a percentage of the upside. They were doing strain improvement at Zymergen for most of their existence. The problem was that they needed like 900 people to do that work, and they were losing \$200 million-plus a year doing that work. It didn't exactly work out, and that's why they had to pivot." - Former Ginkgo director level employee

Whereas Ginkgo focused on "higher risk but more sexy" de novo products

"Ginkgo has always been more of the let's not do strain improvement per se, but let's actually build novel products or start from scratch, and then, here's a very elaborate fragrance, sandalwood or something elaborate that exists in nature that nobody's ever been able to make as a product, a permutation, and we will piece together a pathway. Totally de novo, totally from scratch using our own scientific intuition and technology and informatics. That's how we're going to do things. And so, Ginkgo was always a higher risk but more sexy projects. Zymergen was just kind of ho-hum. Let's just improve the strains." - Former Ginkgo director level employee

Ginkgo and Zymergen "were always looking jealously at each other"; not much differentiation between the two; "more similar to each other than they are different"

"I think both companies were always looking jealously at each other or trying to one-up each other. But I don't think there's a large differentiation between the two companies per se besides that historical philosophy. I think they've worked on very similar projects and ideas. Like, for Joyn Bio, Zymergen was a runner-up, and you can see in their S1. They kept working on a nitrogen-fixing microbe because they thought it was such a valuable idea or useful thing to do. There are a lot of parallels; they're more similar to each other than they are different." - Former director level employee

An ex-Ginkgo employee in a managerial role, with numerous contacts at Zymergen, elaborated on the risk of a similar disaster; why Zymergen was better positioned than Ginkgo, if anything; and indicated that the culture at both companies is similar in that both are “like Theranos.” The ex-employee shared the example of Ginkgo related-party customer Synlogic, who switched to Zymergen for a strain engineering project, which they delivered successfully after Ginkgo failed.

Ginkgo could blow up like Zymergen

“In short, I think that the company is over-hyped, overvalued, and set up to have its Zymergen-like moment. You saw how the Zymergen stock crashed 80% in a day, and I would not be very surprised if that happened to Ginkgo sometime in the next year. I would not be surprised. In fact, I have several friends at Ginkgo who agree with that and several friends who are planning to leave Ginkgo as soon as a good fraction of their stock options vest because of this reason. I can give examples, very high level, very non-confidential about **things that I saw that made me extremely concerned.**” -Former Ginkgo employee in a managerial role

If anything, Zymergen was better positioned than Ginkgo: their genetic sequencing database (“codebase”) was larger; Ginkgo failed to deliver for its key related-party customer Synlogic who turned to Zymergen instead

“I mean, yeah, **codebase is just a fancy word for a genetic sequencing database**, and they're constantly trying to get more codebase and get more diversity. **Another interesting thing is that Zymergen has way more codebase than Ginkgo. Their diversity is way higher, and that's part of why they were able to engineer that Synlogic strain and Ginkgo wasn't. Ginkgo is way outcompeted by Zymergen in that regard.** Anybody who really wants a strain engineered should probably go to Zymergen. They're obviously having a bunch of trouble, they have a bunch of executional and management issues, but they fundamentally have better stuff to be able to make these strains. And I've heard that from several folks that are there now as well.” - Former employee in a managerial role

Ex-employee indicated culture of both Zymergen and Ginkgo is both similar and that both are “like Theranos”

“The culture is very toxic. What's going on with firing people who speak up? I mean, that's like Theranos. Theranos started firing people who spoke up. We used to call Zymergen Theranos because it was lying so much about the Hyaline film, and all the stuff that came out about Hyaline being not good for customers and stuff, we all knew that for a whole year, we just couldn't talk about it. **I just think that it's lying about stuff that's creating an extremely toxic culture, silencing, it's basically killing off naysayers by firing them... The culture is pretty toxic. There are some really terrible things going on.**” –Former employee in a managerial role

An ex-Ginkgo employee shared comments from a colleague at Ginkgo who worked at Zymergen: “Wow, it’s like Zymergen too here. All the same stuff is happening, except it’s happening even faster at Ginkgo.” A current executive at key related-party “customer” Joyn Bio offered color identical to what we heard from ex-employees: “can’t tell what’s different” between Zymergen and Ginkgo; stated that founders at both “have their little magical thing they spin to get deals”; and can’t figure out the purported “secret sauce” at either company.

Ginkgo employee who worked at Zymergen stated it’s “all the same” but the decline is even faster at Ginkgo

“[A colleague who worked at Zymergen] was telling me, “Wow, it’s like Zymergen too here. All the same stuff is happening, except it’s happening even faster at Ginkgo.” The decline in the culture.” There was a very fast decline when it got really close to IPO; the culture became terrible, I would say, in a matter of a few months.” –Former Ginkgo employee

Current executive at related-party “customer” Joyn Bio “can’t tell what’s different” between Zymergen and Ginkgo; founders at both “have their little magical thing they spin to get deals”; can’t figure out the purported “secret sauce” at either company

“As I understand it I think Zymergen actually has products, I mean, they’re in partnerships, but I think that they actually have some products that they’re working on either themselves or in collaboration with somebody. That’s one difference. I can’t assess under the hood because when we talked to Zymergen a couple of years ago, they were in the same thing. Zach Serber, the CSO, was the same kind of thing as Jason. They each have their little magical thing that they spin to get deals. You could never really get to what was the secret sauce. I can’t tell what’s different about them.” – Current Joyn executive